

Practical Design and Production of Optical Thin Films

Second Edition, Revised and Expanded

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Practical Design and Production of Optical Thin Films

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Preface to the Second Edition

The field of optical thin films continues to expand at a rapid rate in both commercial application and technology. As individuals and as a technical community, we continue to gain new understanding as new tools, techniques, processes, and experimental results become available. Since the publication of the first edition, enough new material has become available to warrant a new edition with expanded usefulness for the reader.

Some of the areas of expansion and addition include: new design visualization and optimization techniques, coating equipment updates, a review of spectral measuring equipment and techniques, a significantly expanded materials section, further ion-assisted and energetic process experience, more detail on and examples of design of experiments methodology, expanded understanding of optical monitoring sensitivity, the use of constrained optimization to promote more reproducible designs, consideration of the rapidly expanding dense wavelength division multiplexing applications of optical coatings, and new simulations of the error compensation effects available in optical monitoring with associated tolerance estimations.

This book continues to serve as a tutorial text for those new to the field and as a reference for those with long experience.

Ronald R. Willey

Preface to the First Edition

This book deals with the basic understanding, design, and practical production of optical thin films, or interference coatings. It focuses on two main subjects that are critical to meeting the practical challenges of producing optical coatings. The first is the design of coatings, an understanding of which allows the practitioner to know the possibilities and limitations involved in reducing, enhancing, or otherwise controlling the reflection, transmission, and absorption of light (visible or otherwise). The second deals with the practical elements needed to actually produce optical coatings. These include equipment, materials, process development and know-how, and monitoring and control techniques.

Emphasis is placed on gaining insight and understanding through new viewpoints not found in existing books. The approaches are primarily empirical and graphical. Extensive mathematical derivations have been well covered in other texts, but there remains a need for additional guidance for coating engineers who must design and produce coatings. As optical coatings have become more sophisticated and demanding over the past few decades, optical engineers have had to better understand the limitations and possibilities of design and production. The widespread availability of computers now makes the optimization of detailed designs easy for anyone in the field. A broadened insight will lead the designer to better starting designs that will converge reliably to workable final results. One aim of this text is to aid the reader by adding global views that enhance the creative thought process and lead to new and improved design solutions. Another aim is to share results and information from practical experience in the areas of equipment, materials, and processes which may shorten the reader's path to the production of practical optical coatings.

This book offers the reader practical concepts, understanding, and approaches to use modern computer tools to get better design results. Some insight and tools are provided for knowing where certain boundaries or limitations lie in thin film design. Additional understanding is provided in the monitoring and control of some of the more complex and difficult-to-manufacture modern optical thin films.

Among the book's unique features is an empirically discovered family of functions of index of refraction versus thickness that are discussed here which provide a selection of various antireflection coatings that can be used as needed. The field of Fourier analysis and synthesis of thin film designs is explored from a new viewpoint. Optical monitoring sensitivity concepts and error correction techniques that can enhance production results are elaborated upon in ways not found in other texts.

This book should be of benefit to every optical and thin film engineer and scientist, whether experienced or new to the field. It should serve as a tutorial text and also as a reference for even masters in the field.

Ronald R. Willey

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1

Fundamentals of Thin Film Optics and the Use of Graphical Methods in Thin Film Design

1.1. INTRODUCTION

Getting started is often the most difficult part of a new task, and it is always the first thing to do. The goal supported by this book is to produce a practical thin film optical coating which meets whatever particular requirements are presented to the reader within the limitations of the technology. We will give a brief overview of what is needed to do this and how one might proceed.

The first thing needed is a clear statement of the requirements and/or goals of the coating such as spectral reflectance versus wavelength, spectral range of concern, substrate characteristics, environment to be encountered (and survived), etc. Often it is necessary to work with the end user or customer to establish these requirements and desires within the framework of what is possible and practical. Chapter 2 provides some assistance in estimating what can be done at this first stage.

The second thing needed by the coating developer is a basic understanding of the underlying principles of optical thin film performance and design. These are set forth in Chapters 1 and 3 from several viewpoints to give a more global perspective of the optical thin film design task. The possibilities, limitations, and options in various types of thin film coating designs are discussed in Chapter 2. This leads to a choice of the type of design to use to meet the requirements and an estimate of the number of layers and materials needed.

The third stage is to select materials which will have the desired properties over the required spectral range and which will perform in the required environmental conditions. These conditions often include temperature, humidity, abrasion, adhesion, salt fog, cleaning solutions, etc. Chapter 5 discusses the more commonly used materials and the processes by which they may be deposited. The equipment to be used in the production of a coating will affect the design and material choices. Chapter 4 reviews the typical equipment currently used for thin film production.

The fourth step is to perform the detail design of the coating to meet the optical performance requirements. If the index of refraction versus wavelength in the region of concern of the chosen materials is well known and stable, and the control of film thicknesses in the deposition process is adequate, the final optical coating product can be expected to be the same as predicted by the detail design. A computer evaluation and optimization program is used for the detail design process. The preliminary or starting design, derived typically from the choices made from Chapters 2 and 5, is evaluated and then the layer thicknesses are optimized with respect to the spectral requirements of the product. Sometimes a few more layers may need to be added to meet some requirement. It is our practice also to try to remove layers from a near-final design and reoptimize to determine if fewer layers might still meet the requirements. The fewest number of layers is generally desirable as long as there is enough margin in the design to allow for expected process variability and still meet the requirements.

The fifth phase of the process is to test the design in actual deposition. The first part of this should be to verify that the indices of the deposited materials are the same as expected and used in the design. If this is true, this phase is simple and short. If significant differences are found, the material processes may need refinement and greater control. This may require extensive effort. Chapter 6 discusses process development to reduce variability and characterize material properties. This can feed back improved information for the material and process know-how covered in Chapter 5.

The sixth stage which leads to the successful production of the required optical thin film is to gain adequate control of the layer thicknesses. The actual strategy for thickness control should be considered at the fourth or design step and influenced by the equipment to be used (Chapter 4). Chapter 7 deals with the many ways that film thickness might be monitored and controlled. It also discusses monitoring strategies, where the different strategies might best be used, and their strengths and weaknesses. When a monitoring process is established with sufficient repeatability, then the only task is to set the film thicknesses to produce the required results. This is typically a “Kentucky windage” process. That is to say that the actual resulting film thickness is usually somewhat different from that given as a parameter to the monitoring system. As long as the difference is the same from run to run, the difference can be subtracted from what is given as a parameter to the system so that the resulting thickness deposited meets the requirements. Section 7.7.3 gives a detailed example of how this was done in a specific case.

The application of the above six steps and the information in the chapters mentioned should lead to a successful coating in the great majority of optical thin films used in the world today.

We would like to mention a few key things which may be helpful to keep in mind when starting a new coating development. These are discussed in more detail in the chapters. They are usually true, but we will not state that there are no exceptions.

It is best to view all optical thin film coatings as affecting the reflectance of a surface as discussed in Chapters 1, 2, and 3. This may be to reduce the reflectance in a given spectral range or to increase it. Transmittance, optical density, etc., are all derivatives of reflectance.

More layers and thereby a thicker coating in a design give more control over fine details of the spectral reflectance profile as shown in Chapter 2. However, the

improvements with increasing thickness seem to be asymptotic. For example, more than eight (8) layers in a broadband antireflection coating, where the optical thickness is about one wavelength in the band, typically gives limited reduction in the reflection over the band. It requires an order of magnitude thicker coating to reduce the reflectance to half that of eight layer.

We also show in Chapter 2 that it is desirable to use high and low index materials in a design where the difference in index is as great as practical. There is more reflectance produced at each interface, and therefore fewer layers are needed to produce a given result:

It has further been shown in Chapter 2 that the lowest practical index should be used for the last layer in an antireflection coating.

The fewest layers practical in a design is most desirable from a production point of view. It also has been shown in broadband antireflection coatings that a minimum number of layers is optimum for given designs and more layers cannot produce as good a result.

The simplest solutions which can meet the requirements are usually the most elegant and the most successfully produced.

Absolute values of deposition parameters such as pressure, temperature, etc., are difficult to obtain and may be unnecessary. The most beneficial characteristic of almost any process is its stability or reproducibility. With stable indices of refraction and material distribution in a process, the parameters of a design and process can be adjusted by "Kentucky windage" to meet the requirements.

The design of experiments methodology discussed in Chapter 6 can often minimize the number of experiments necessary to optimize and stabilize a deposition process.

In recent years, "Concurrent Engineering" has been a topic of some interest in the industrial world. Our interpretation of the topic is that it is nothing new, but is a rediscovery of a truth that had been lost by some cultures. The best results are obtained when the engineers/designers are in close communication with the fabricators/producers of the product that they develop and when they understand the details of the production processes needed to produce their designs. Anyone who has worked in a design and production field will have examples where a design was submitted to be fabricated without any prior communication between the designer and the fabricator. This usually leads to problems. The design might call for things which cannot be achieved or might not take advantage of processes or capabilities which exist. The engineers/designers should be in adequate communication with the producers/fabricators to optimize the results of their combined efforts. It has been our practice for decades as engineers/designers to gain as much hands-on experience as practical with the processes for which we design. This allow us to understand the capabilities and limitations of the processes so that our designs are achievable and properly utilize the existing capabilities. We further attempt to discuss design and fabrication ideas at the formative stages with all of the pertinent parties. This often takes the form of going into the shop and saying, "We need to achieve this result. What choices might we have as to how to fabricate it, and how might the details and tolerances best be arranged?" The producers/fabricators should also be included in all of the formal design reviews from preliminary to final review. The essence of Concurrent Engineering in our opinion is communication and the inclusion of the knowledge and experience of all of the appropriate parties involved in a development

at all stages of a project. In general, if the designer is not also the producer, the designer and producer need to coordinate as a unit to attain the best result from the application of their combined resources and experience.

The application of Concurrent Engineering to the optical thin film development process is no different from the development of a machine, a building, or a spacecraft.

With these few suggestions in mind, we will now go into the details behind the preceding overview.

This chapter reviews the fundamentals of thin film optics and elaborates on the use of graphical methods to gain understanding and insight into some of the basic principles of optical thin film design. Reflectance diagrams and admittance plots are described in some detail. The insight gained from the admittance plots of antireflection (AR) coatings with ideal inhomogeneous index profiles connect these results to Fourier thin film design techniques discussed in Chapter 3. The limited choice of low index materials for broadband antireflection coatings necessitates a discussion of the possibilities and limitations of equivalent index approximations. A specific example in section 1.7.1 illustrates the design process to achieve the goal of a very broad band and low reflectance AR coating on crown glass.

The emphasis of this chapter is on graphics, concepts, insight, and reduction to practice in optical thin film design.

1.2. REVIEW OF THIN FILM OPTICS PRINCIPLES

Some years ago, I was privileged to hear a presentation by Prof. Robert Greenler (then president of the Optical Society of America) wherein he aptly demonstrated the interference colors in a soap bubble. A mixture of equal parts of dish soap and glycerine (which can be purchased in a pharmacy) is diluted with water until bubbles can be blown which persist for many seconds without breaking. An ordinary flower pot of about 6 inch diameter is dipped upside down in a pie tin of the solution and withdrawn from the solution with a bubble formed on its top lip. With the pot still upside down, one blows into the hole in the bottom of the pot until the bubble becomes hyperhemispheric. The hole is then sealed by wiping some of the solution across the hole with the thumb. The pot is then gently turned right side up and set on a table. Figure 1.1 illustrates the configuration.

If the solution is homogeneous and the air currents around the bubble are

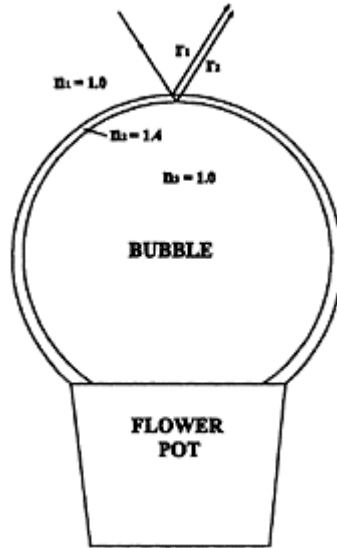


Fig. 1.1 Soap bubble on a flower pot illustrating interference colors.

negligible, the solution will gradually drain from the top of the bubble to the bottom causing the top to become thinner and the bottom sides to become thicker. The colors of the top will change from pale “pastels” to increasingly saturated colors. The top will eventually become white and then black just as the bubble breaks (because it has gotten too thin). The colors will form horizontal bands. The bottom sides will become less saturated colors as they get thicker. These colors illustrate various principles related to our interests, but we will here focus on the top of the bubble. The physics of the situation is explained below.

We will first review some of the underlying principles and the mathematical foundations of thin film design briefly from a fundamental point of view. The more experienced reader might choose to skip ahead to Sect. 1.5.

The Fresnel amplitude reflection coefficient (r) for an interface between two nonabsorbing media at normal incidence can be represented by the following equation:

$$r = \frac{n_1 - n_2}{n_1 + n_2} \quad (1.1)$$

where n_1 and n_2 are the (real) indices of refraction of the two media. For the more general case of absorbing media with complex refractive indices $n_j = n_j - ik_j$ ($j = 1, 2, \dots$); the reflection intensity coefficient (R) is the amplitude reflection coefficient (r) times its complex conjugate (r^*),

$$R = rr^* \quad (1.2)$$

For most of the common applications of absorption-free thin films at near-normal incidence of light it is sufficient to treat the intensity as the square of the amplitude.

The soap bubble is a mixture of water with an index of refraction of about 1.333 plus soap and glycerine of higher indices. For the sake of simplicity, we will assume that the index of the mixture is 1.40. This would make $r_1 = (1.0 - 1.4)/(1.0 + 1.4) = -0.1667$ while $r_2 = +0.1667$. Thus, as we will examine in more detail below, if the soap bubble film were very thin with respect to (WRT) a wavelength of light, the two reflections would cancel each other out. There would be no reflection, i.e., it would look black! This is because they are of equal magnitude and opposite sign. If, on the other hand, the film were thick enough to delay r_2 by one half wavelength WRT r_1 , then the two would be in phase and add to give constructive interference of $r = -.3333$ or $R = .1111 = 11.1\%$. This would look white in reflection.

In a rigorous sense, the amplitude of the residual reflection resulting from the interaction of light with two interfaces with a phase separation of ϕ (ϕ) is:

$$r = \frac{r_1 + r_2 e^{-i\phi}}{1 + r_1 r_2 e^{-i\phi}} \quad (1.3)$$

for a more extensive discussion, see Apfel¹.

Let us look at a more general “real life” example. Fig. 1.2 represents a single layer of magnesium fluoride ($n_1=1.38$) on crown glass ($n_2=1.52$) in a medium of air (or vacuum $n_0=1.0$). Equation (1.4) gives the amplitude reflection coefficient at the interface between the air and magnesium fluoride at normal incidence of light:

$$r_1 = \frac{n_0 - n_1}{n_0 + n_1} = \frac{-.38}{2.38} = -.1597 \quad (1.4)$$

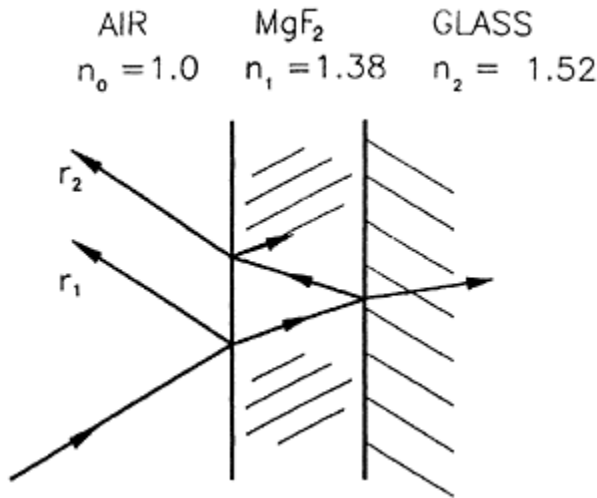


Fig. 1.2 Reflections at a single layer coating of MgF₂ on crown glass.

Equation (1.5) gives the amplitude reflection coefficient at the fluoride-to-glass interface.

$$r_2 = \frac{n_1 - n_2}{n_1 + n_2} = \frac{-.14}{2.9} = -.0483 \quad (1.5)$$

When these two reflections are in phase, they add to give -0.2080 or an intensity of 0.0433 (4.33% reflectance). However, when we apply Eqn. 1.3 with $e^{-i\phi} = 1$, we find $r=.2063$ and $R=4.26\%$ which is the same as an uncoated substrate of index 1.52. When the reflections are 180 degrees out of phase ($e^{-i\phi} = -1$), they add destructively (subtract) to give -0.1123 amplitude or 0.0126 intensity. This single beam interference consideration yields approximately the typical residual reflectance of 1.26% for a single layer antireflection coating (SLAR) on crown glass at the design wavelength. Thus, for the exact reflectance value, multiple beam interference was taken into account. This was done using Eqns. 1.3. The derivation of this is also found in Macleod², page 51.

1.3. REFLECTANCE DIAGRAMS

Figure 1.3 illustrates the interaction of the reflections as a function of phase angle for a single thin film. When the layer is very thin, the two reflections are nearly in phase. As the layer increases to one quarter wave optical thickness (QWOT) at the reference (or design) wavelength, the phase delay of r_2 with respect to r_1 approaches 180 degrees which gives a minimum residual reflectance. If the layer thickness increases further from this point, the reflection vector in Fig. 1.3 will return to its starting value at a phase of 360 degrees or one half wave of optical thickness. This situation is often referred to as an “absentee layer” which has the same reflectance (at the design wavelength) as the bare substrate surface, i.e., as if it were not there at all. The path that the tip of the residual reflectance vector travels is the circle seen in Fig. 1.3. This is the underlying principle of the reflectance or circle diagram which was expounded by Apfel¹.

For a coating of a different index on this same substrate, we get, of course, different reflectance. Figure 1.4 shows various examples as a function of optical thickness. Layers of higher index than the substrate will increase the reflectance to a maximum at the QWOT point and then decrease again to the reflectance of the bare substrate at the half wave point, which is true for all half waves, high or low index. We see that the simplest low reflector or AR coating is a QWOT of low index on the substrate. The lowest reflectance ($R=0$) would be obtained by a layer whose index is the geometric mean between the substrate and the medium ($n_1=(n_0 n_2)^{0.5}$ as in Fig. 1.2); for this crown glass, that would be 1.233.

Fig. 1.4 Reflectance intensity as a function of the optical thickness for a single layer of the various indices on a 1.52 index substrate.

The reflectance amplitude from each interface would be equal because: $(n_1 - n_0)/(n_1 + n_0) = (n_2 - n_1)/(n_2 + n_1)$. The simplest high reflector would be a QWOT layer of the highest available index.

The amplitude reflection coefficients in the reflectance diagram are represented by the distance from the center where the real and imaginary axes intersect. The extreme of this “circle” diagram is the unit radius circle which would represent an amplitude reflection coefficient of 1.0 or 100%. Since the reflection intensity (reflectance R) is effectively the square of the amplitude, the circles of equal reflectance are not equally spaced as is shown in Fig. 1.5. Note how far from the center the 1% reflectance boundary lies.

1.3.1. Low Reflectors, Antireflection Coatings

The single layer antireflection coating (SLAR) illustrated in Figs. 1.2 and 1.3 is shown on a reflectance diagram in Fig. 1.6. This represents a change of reflectance from the bare substrate at $R=4.26\%$ to $R=1.26\%$ for the SLAR. If one were interested in using two layers of high and low index materials to achieve a zero reflectance coating at one wavelength, Fig. 1.7 shows how it might be done. The last layer (toward the incident medium) would have to have the low index,

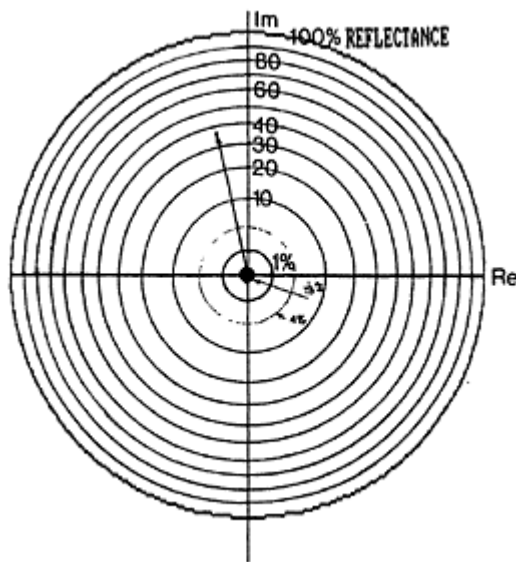


Fig. 1.5 Equal percent reflectance intensity contours on a reflectance circle diagram.

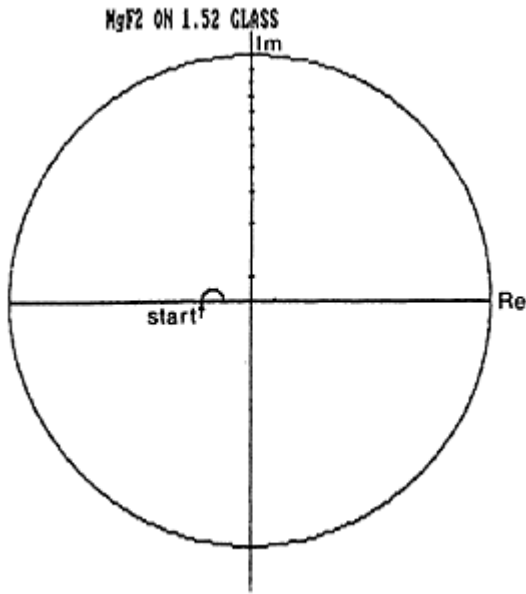


Fig. 1.6 Reflectance locus of the coating in Figs. 1.2 and 1.3 with increasing thickness on a reflectance circle diagram.

and it would have to be represented by a circle passing through the origin. The high index first layer (on the substrate) would have to start at the substrate index and intersect the low index circle (generally at two points).

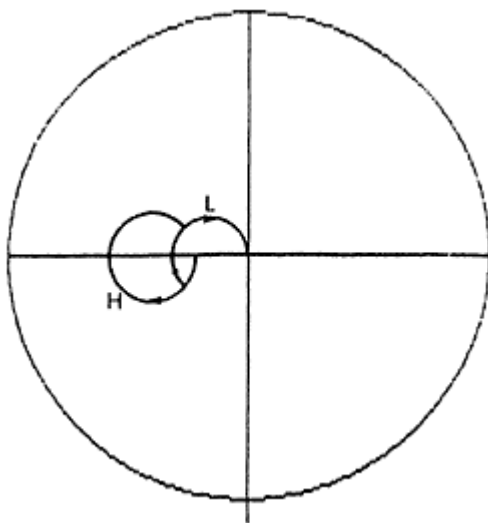


Fig. 1.7 Reflectance diagram of the two solutions to a two layer AR coating (V-coats).

There are two solutions to this classical “V” coating. The first has a short high index layer which terminates at the first intersection with the low index locus, and the low index layer completes the path to the zero reflectance point. The second solution has a longer high index layer which terminates at the second intersection with the low index circle, and a shorter low index layer to stop at zero. This is like traveling between two points on a subway in New York City. There could be an “H” train which leaves from your office and crosses paths in two places with the “L” train that will get you home. You then have a choice to transfer at the first crossing or wait until the second. The ride might be longer in the latter case! Curves V1 (with the shorter first layer) and V2 in Fig. 1.8 are the reflectance versus wavelength of these two V-coats, their shape shows where the name V-coat originates. V1 is generally preferred because it has less material (shorter ride) and is somewhat more tolerant to shifts in wavelength and/or thickness errors. Curve A in Fig. 1.8 is the reflectance of the SLAR of Fig. 1.6. Note that the minimum reflectance wavelength is where the layer is one QWOT. As illustrated in Fig. 1.9 (on an expanded scale), for a longer wavelength (700 nm), the layer is not a full QWOT and therefore the reflectance is higher. For a shorter wavelength (400 nm), the layer is more than a QWOT, and therefore the reflectance is starting to increase again as the optical thickness increases toward a half wave. This same behavior can be seen in Fig. 1.10 to cause the reflectance of either V-coat to rapidly move away from zero as the wavelength changes from the design wavelength.

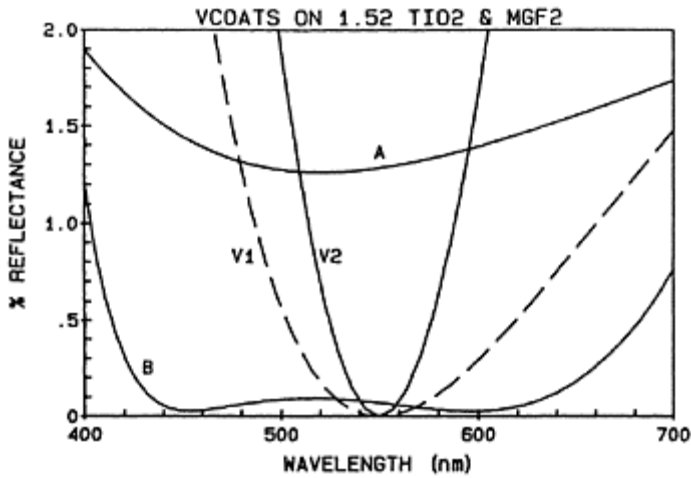


Fig. 1.8 Spectral reflectance: A is a single layer AR coating of MgF₂ on glass, B is a three layer broadband AR from Fig. 1.11, and V1 and V2 are V-coats from Fig. 1.7.

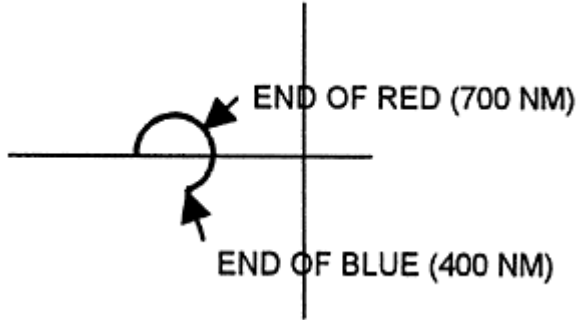


Fig. 1.9 Change in the locus of reflectance curve as in Fig. 1.6 with wavelength.

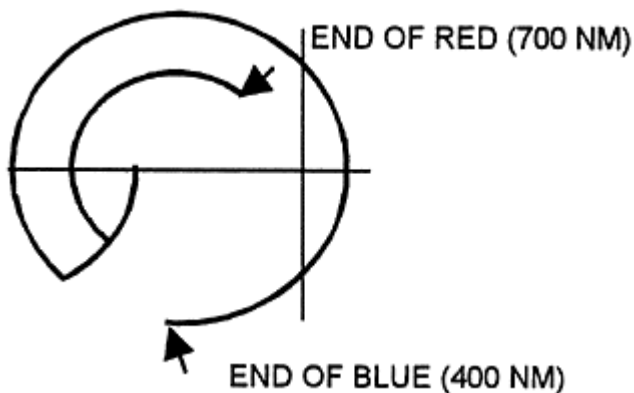


Fig. 1.10 Change in the locus of reflectance curve as in Fig. 1.7 with wavelength.

A special case of the V-coat as seen in Fig. 1.11 can be constructed for a 1.52 substrate with a medium index layer M and low index layer L which are each a QWOT (Fig. 1.11). This happens if $n_M=1.7$ and $n_L=1.38$. This two layer coating would have similar sensitivity to wavelength or “dispersion” as the other V-coats. However, the addition of a half wave layer of high index (about $n_H=2.25$) between the M and L layers will “achromatize” the design or make it insensitive to wavelength changes over a fairly broad range. Curve B in Fig. 1.8 shows the result of this broad band AR of the classical quarter-half-quarter wave (QHJ) design. Figure 1.12 illustrates this compensation effect on the reflectance diagram for the red (650 nm) and Fig. 1.13 shows the blue (450 nm) which comes quite close to zero reflectance at the origin. These three cases discussed so far (1, 2, and 3 layers) form the basis of most of the AR coatings produced in the world today. We will later discuss a modification of the three layer to a four layer for practical reasons of producibility when the required index (of $n_M=1.7$) is not available.

Apfel¹ showed how to generate reflectance diagrams for any materials of interest. Figs. 1.14 and 1.15, for example, are for materials of index 1.38 and 4.0 in air or vacuum. Figure 1.16 is an aid to using these graphical techniques which shows the starting points on the real axis of several representative substrate materials. The one with the “dumbbell” marks is ordinary glass at $n=1.52$. The others range from fused silica at about 1.45 to germanium at about 4.0.

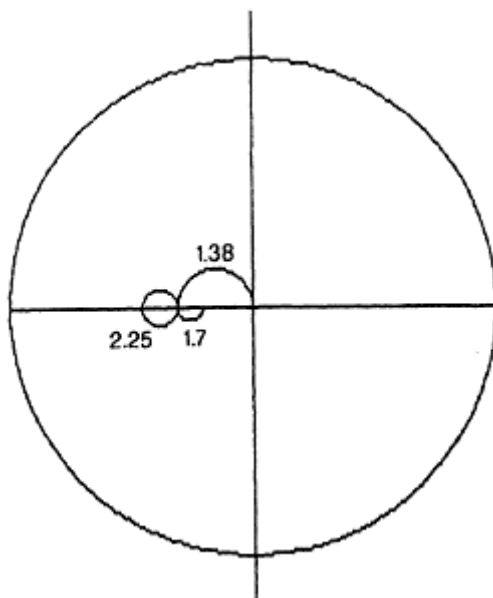


Fig. 1.11 Reflectance diagram of a three layer AR on crown glass.

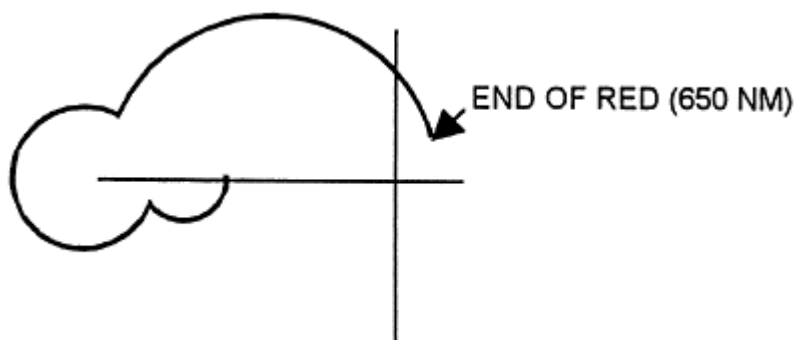


Fig. 1.12 Change in the locus of reflectance curve as in Fig. 1.11 at 650 nm.

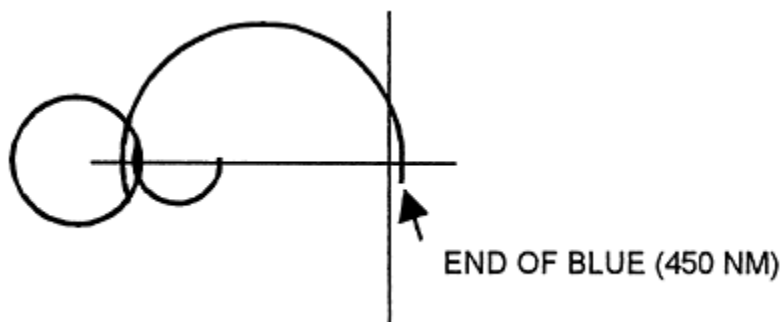


Fig. 1.13 Change in the locus of reflectance curve as in Fig. 1.11 at 450 run.

These diagrams can be readily used for graphical overlay construction of the reflectance of a thin film design following these steps:

(1) Lay tracing paper over the Fig. 1.16 diagram and mark the substrate starting point and the real and imaginary axes for alignment with the other diagrams.

(2) Now align the axes of the tracing paper over the first layer material "spider web." Using the circle of which starts from that substrate point, trace the path from that point on the reflectance diagram moving clockwise to the termination of the first layer (1/2 circle is a QWOT).

(3) Now align the axes of the tracing paper over the second layer material spider web. Trace the second layer from that new point in the spider web to its termination.

(4) Repeat these steps with the remaining layers. The reflectance of the complete coating at this particular wavelength can then be found using Fig. 1.5 which shows the square of the magnitude of the vector from the origin (intersection of the axes) to the endpoint of the last layer ($R = r r^*$). A dozen or so such spider diagrams are all that the author has needed to represent common materials for preliminary (or back of the envelope) type of coating design.

Note that the "radial" spokes on the spider webs have been plotted to represent 1/20th of a QWOT or 1/80th of a wave of optical (phase) thickness.

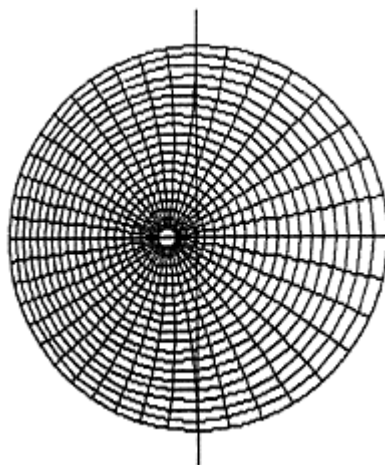


Fig. 1.14 Reflectance loci and equal optical thickness lines for 1.38 index material.

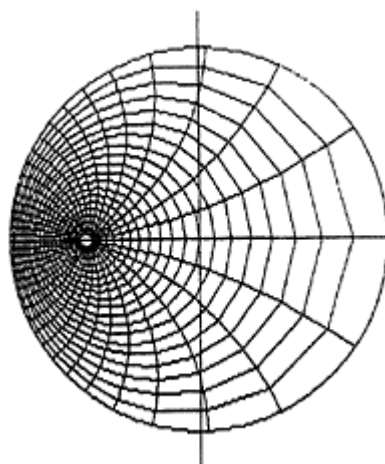


Fig. 1.15 Reflectance loci and equal optical thickness lines for 4.0 index material.

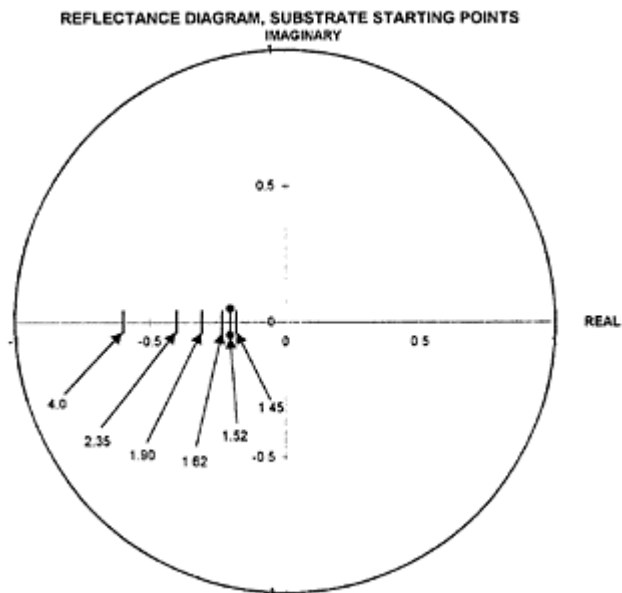


Fig. 1.16 Graphic aid of substrate starting points from indices 1.45 to 4.0.

We can use these to count the “spokes” (and fractions thereof) for each individual layer and thereby obtain an adequate estimate of the thickness of the layer in QWOTs. Such results can then be entered into an appropriate thin film design program as a starting design. The first results are usually a good preliminary design which can be quickly optimized to the best solution of its type. However, we have found the greatest utility of these graphical techniques to be in gaining insight and as an aid to understanding of the behavior of a given coating.

1.3.1.1 Behavior of a High Index Slab

The narrow bandpass behavior of a high index slab seen in Fig. 1.4 can be visualized also on a reflectance circle or spider web diagram. Figure 1.17 is a spider diagram for a hypothetical index of 10.8 as in Figs. 1.14 and 1.15. This index was chosen just for illustration such that the highlighted circle passes through the origin. Here we imagine a substrate of index 1.0 ($r=0$) on which we start to deposit material of index 10.8. The locus of the total reflection follows the bold circle from the origin in a clockwise direction. At one QWOT, it will have a maximum reflection near $r=-1$ (at the left on the diagram). An additional QWOT, to make a total of one half wave, will bring the reflection back to zero at the origin.

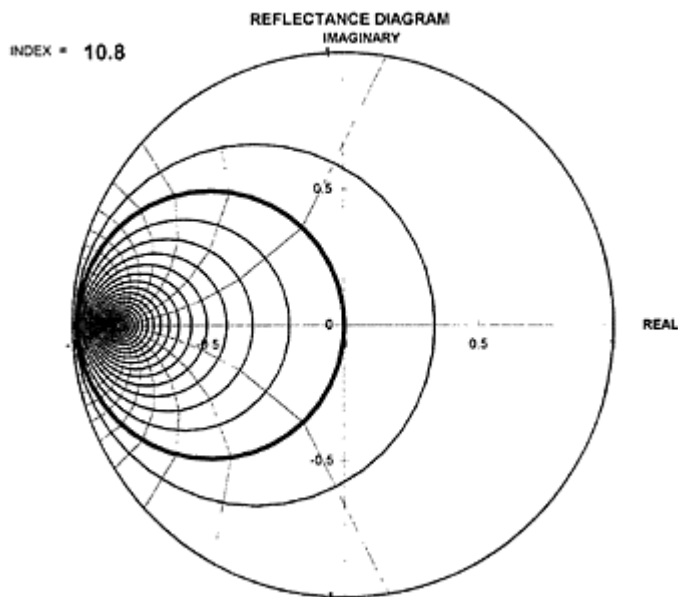


Fig. 1.17 Locus on a reflectance amplitude diagram for a 10.8 index material freestanding in air, illustrating the principle of the NBP filters in Fig. 1.4.

The point to note here is that the lines of equal phase thickness (spokes) are very widely spaced near the origin for high index materials, and they are very closely “packed” at the end of the $-r$ axis on the left. The result is that the reflectance grows very quickly from zero to a high r with only a small addition of phase thickness on the substrate. For a very high index like 99, the first phase thickness line representing $1/20$ th of a QWOT on such a spider diagram would have an $r = .93$. Almost all of the action in such a case is clustered near the left end of the $-r$ axis. This means that the reflection is high for almost all optical phase thicknesses except those near one half wave, or multiples of a half wave.

With reference to Eqn. 1.3, r_1 and r_2 are equal in magnitude and opposite in sign and have large values close to unity. In such a case, the denominator of Eqn. 1.3 becomes a major factor in the shape of the locus. Without it, the locus of r would start at the origin and circle out of the $r=1$ boundary as it pivots clockwise about the high index point. However, the denominator tempers this so that as $e^{-i\phi}$ approaches -1 , the r approaches $-2/2$. Thus, the behavior seen in Fig. 1.4 can also be visualized on a reflectance/spider diagram such as Fig. 1.17.

1.3.2. High Reflectors

We next address the subject of producing a high reflector and how it would appear on a reflectance diagram. Figure 1.18 shows the reflectance diagram of a stack of QWOT layers of the form (HL)⁴. This means a pair of high and low index QWOTs is repeated four times for a total of eight layers. The reflectance diagram starts with the first H at the substrate and progresses clockwise until it reaches a maximum reflectance at one QWOT where it intersects the real axis. The next low layer continues clockwise to the real axis again, etc. Figure 1.19 shows the reflectance versus wavenumber (10,000/wavelength in micrometers) after each pair is applied. By the time eight layers have been added to the substrate, the reflectance at the design wavelength has become high. More pairs will make it correspondingly higher and the transmittance correspondingly less. The edges of this “stop” band become steeper as the number of layers increase. On either side of the stop band are pass bands. As illustrated in Fig. 1.20 after Thelen⁴, the edges of the stop band can be used to create a long-wave-pass (LWP) filter on one side of the center wavelength of the QWOT stack or a short-wave-pass (SWP) filter on the other. This high reflector stack is the building block of mirrors, LWP and SWP filters, and certain types of bandpass filters. The latter are made by bringing separate edges of a LWP and a SWP filter toward each other in the design until the desired pass band is achieved. The AR class of coatings of Fig. 1.20 have been

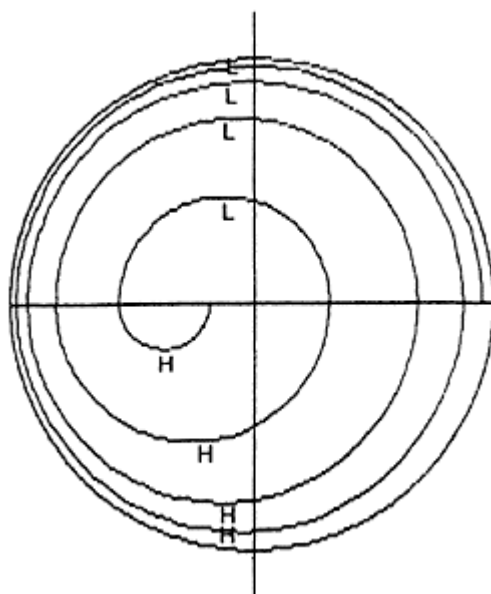


Fig. 1.18 Locus of the reflectance of a high/low QWOT layer stack spiraling outward from the substrate to high reflectance as deposited.

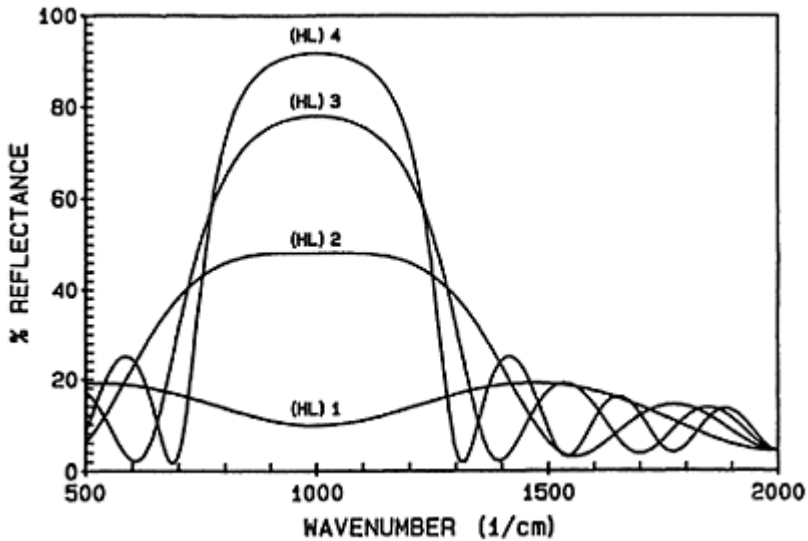


Fig. 1.19 Spectral reflectance as H/L pairs are added to the stack of Fig. 1.18.

mentioned already and the high reflectors in this section. Neutral beamsplitters are related to the high reflectors with only a few layers. The basis of the narrow band pass filter is seen in Fig. 1.4 and will be discussed in more detail below. The minus filter is discussed extensively by Thelen⁴ and would be created by a QWOT stack where the difference in indices from H to L was small. The principles of the polarizing beamsplitter will be discussed below in connection with angular effects.

A stack of many periods of (HL) will be the same as the same number of periods of (LH) except at the start and end of the stacks. It is seen in Fig. 1.19 that there are ripples on either side of the stop band. These ripples can cause problems in some applications if they are in a band where high transmittance is required. The cause of the ripples is the lack of a proper “impedance match” or admittance matching layers (AR coatings) between the effective admittance of the stack and the admittances of the substrate and the medium. This subject is treated extensively in Thelen’s early paper³ and more recent book⁴. Basically, the last few layers on both sides of the stack can be adjusted in thickness to simulate a desired admittance and to create an AR coating for the stack over some (usually limited) wavelength band. The region of good “impedance match” can be shifted in wavelength, broadened, and can often be made to reduce the ripple to an acceptable level for the application. We will cover this in later chapters.

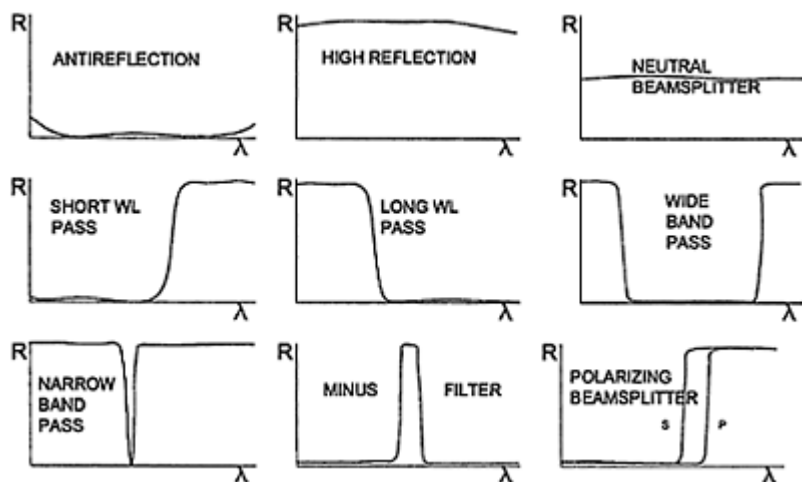


Fig. 1.20 Various types of filters made with only dielectric materials.

1.3.3. Narrow Bandpass Pass Filters

Narrow bandpass (NBP) filters are usually of a somewhat different design than the bandpass constructed of a LWP and SWP filter combination. If we examine Fig. 1.4 and Sect. 1.3.1.1 again, we see that a very high index single layer would create a filter with a narrow passband at $1/2$ wave and multiples of $1/2$ wave optical thickness. If such very high refractive indices existed, the high Fresnel reflectance at the interfaces with the spacer medium would have high transmission (low reflectance) at all multiples of one halfwave of optical thickness and a high reflectance for all other wavelengths. The passband width gets narrower as the index gets higher. This is in effect a Fabry-Perot interferometer with two mirrors (the interfaces) spaced by a dielectric layer. The early applications of this principle used partially transparent metal coatings such as silver for the mirrors on each side of the spacer layer. Such metal mirrors, however, come with some unavoidable absorption losses that are not inherent in all dielectric designs. Indices higher than 4.2 (germanium) are not commonly available or used, so we have to use some other method to get the same effect. Figure 1.19 shows the way. A high reflector can be made by a stack of high/low layers to give the same reflectance as any desired index interface (at a given wavelength). This mirror stack can then be spaced from a similar symmetric stack by a layer which is a multiple of one or more half waves at the design wavelength of the passband. We can visualize this by reference to Fig. 1.18. It would be like a clock spring unwinding or winding tighter. The fixed point would be the starting point of the substrate, and the moving or “free” end would be the end of the last layer as seen on the reflectance diagram. In the case of a single cavity NBP filter, the first “spring” at the design wavelength would coil out from the start to an intersection with the real axis near the outer bounds of the reflectance diagram. A halfwave spacer would move from the termination of the mirror reflectance locus for one full circle and then

back to its starting point. The second mirror stack would be like a spring which is symmetrical to the first and start from that point and progress back to the substrate starting point. Since it returned to the starting reflectance, it is the same as a simple absentee layer at that wavelength. However, since there are so many semicircles spiraling around the reflectance diagram, it is easy to imagine that a small change in wavelength might cause the balance of the “half wave” effect to be upset and the reflectance to become high. This is in effect what happens to cause the passband to be narrow about the design wavelength. Multiple cavity NBP filters are where two or more of these resonant cavities are placed in properly coupled sequence (by a one QWOT layer) to create steeper sides, squarer tops, and greater reflection or rejection outside the passband. Figure 1.21 shows an example of a spectrophotometer trace of an actual three-cavity NBP filter in the infrared (IR). The three humps at the top of the passband are a clue to the fact that this is a three-cavity filter. Two humps would imply a two-cavity filter. However, with more than three cavities, counting humps is not a reliable clue.

1.3.3.1 NBP Wavelength Effects as Seen on Reflectance Diagrams

In connection with Figs. 1.9–1.10 and 1.12–1.13, we discussed how the reflectance diagram changes with wavelength. Similarly, the “springs” mentioned above will wind and unwind with changes of wavelength from the design wavelength. Figure 1.22 shows the locus or path of the last reflectance point of a coating design as a function of wavelength. The partial circle toward the lower left of Fig. 1.22 is for a single cavity NBP filter centered at 1550 nm. As the wavelength is changed from the design wavelength, the reflectance (r) moves away from the origin of the diagram toward the high reflectance boundary (along this circular locus). The two-cavity design, showing a cardioid-shaped locus in Fig. 1.22, moves more rapidly away from the origin. Its reflectance will therefore increase more rapidly with wavelength than does the one-cavity filter (steeper skirts or sides). Similarly, the three-cavity design moves even more rapidly toward higher reflectance after having gyrated about the origin over a small wavelength range. This gyration is associated with the broader flat top on the three-cavity transmittance spectrum. The points on each locus for ± 0.4 nm are indicated by arrows. The design wavelength of 1550.0 nm is centered on the origin in each case. The reflectance

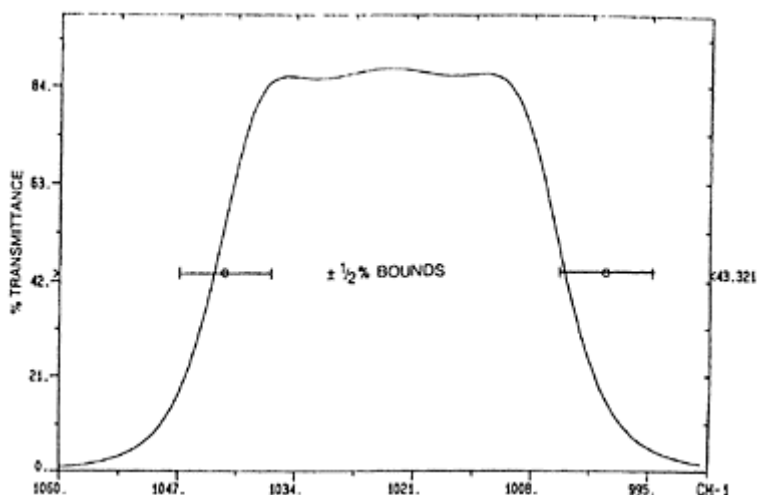


Fig. 1.21 Actual spectral transmittance of a three-cavity NBP filter at about $9.77 \mu\text{m}$.

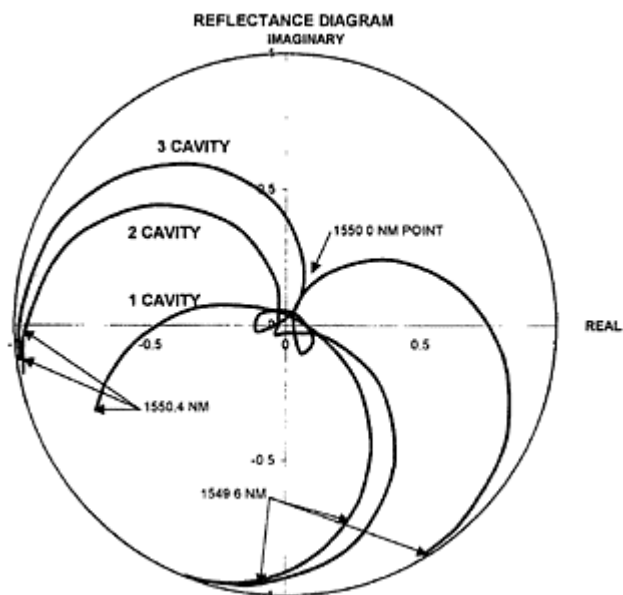


Fig. 1.22 Locus of the final reflectance point of a coating design as a function of wavelength for 1-, 2-, and 3-cavity filters.

is higher still in the blocked regions as the wavelengths move further from the design than we have plotted here. This is another way to view what is happening in NBP filters.

1.3.3.2 Dense Wavelength Division Multiplexing (DWDM) Filters

At the time of this writing (2001), DWDM is a topic of great interest in the fiber optics communications field. Very narrow bandpass filters are required to separate out the many wavelengths which are traveling in a single fiber. This seems to one of the most challenging optical coating production problems of this period.

The design of these filters is relatively straightforward, which we will address in this section. In Chap. 2 we will give aids for estimating the design details needed to achieve specific bandwidth requirements. The control or monitoring during the production of such designs is the most significant issue and will be discussed in Chap. 7. It will be seen that DWDM filter production contains extreme contrasts between the error forgiveness in some areas and requirements for rigid adherence to stable control in others.

Figure 1.23 shows the case of effective index 32 from Fig. 1.4 on a decibel (dB) transmittance scale as used by the communications industry. The decibel scale in this context is 10 times the optical density (OD), where $OD = \log_{10}(1/\text{Transmittance})$.

The communications industry also tends to work with a frequency scale rather than wavelength. The L-band is at a range of frequencies from 185 to 192 teraHertz (THz= 10^{12} Hz) while the C-band is from 192.2 to 196 THz. This covers the contiguous wavelength range from 1528 to 1620 nm. The bandwidth is expressed in gigaHertz (GHz= 10^9 Hz). The bandwidth $\Delta\lambda$ in nanometers can be found from the bandwidth Δf in GHz by: $\Delta\lambda = -C \times \Delta f / f^2$, where 0.2997925×10^{18} nm/second is C (the velocity of light) and f is the frequency. Therefore, 100 GHz represents 0.8 nm bandwidth at 1549 nm and 50 GHz is 0.4 nm.

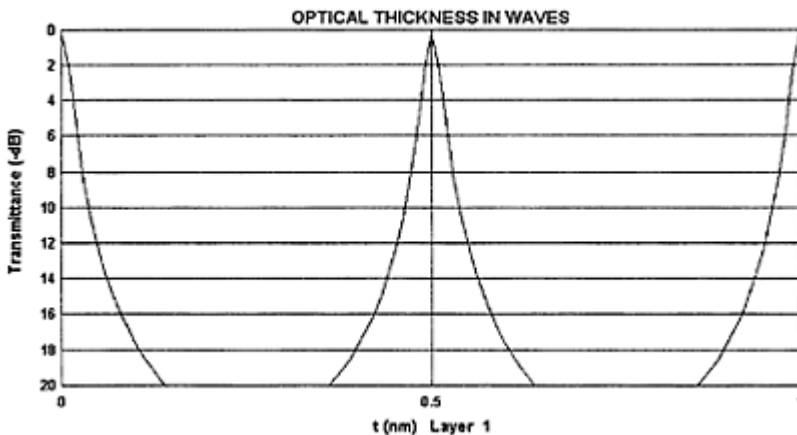


Fig. 1.23 Spectral trace of effective index 32 NBP filter from Fig. 1.4 on a decibel (dB) transmittance scale.

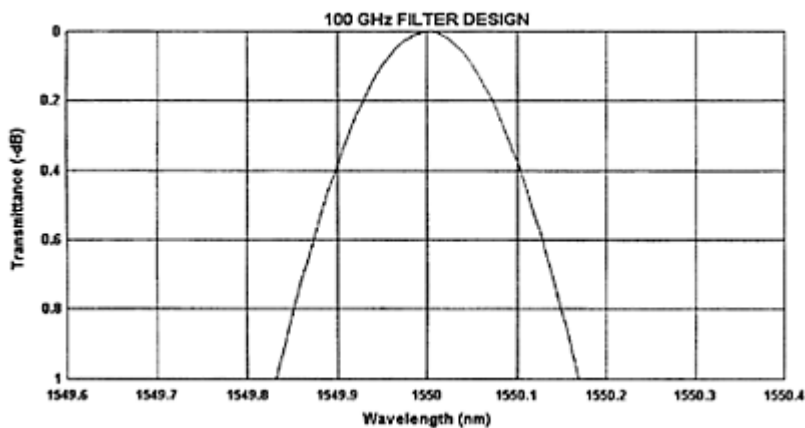


Fig. 1.24 Spectral trace of a “rear” one-cavity design to similar Fig. 1.23.

Figure 1.24 shows the peak of a similar “real” design to Fig. 1.23 using materials such as SiO_2 and Ta_2O_5 for the low and high index materials in the mirrors and spacer. This design at 1550 nm is:

(1H 1L)9 4H (1L 1H)8 1L .59917H .76524L

where there are nine (9) pairs of high and low index QWOTs in each mirror and two half waves or four (4) QWOTs in the spacer. The last two layers are an antireflection coating or “impedance matching layers” to the air or vacuum beyond the filter designed to maximize the peak transmittance.

The filter in Fig. 1.24 is a “single cavity” filter. The top of the transmittance peak is rounded and the “skirts” do not drop very precipitously from the peak. If two such filters were concatenated and the phase between them properly maintained, then the peak of such a “two-cavity” filter would be broader and the skirts steeper as shown in Fig. 1.25. The proper phase between cavities is maintained by a QWOT coupler layer, which in this case is of low index. A three-cavity filter would look like Fig. 1.26, and the extended skirts for each of the designs is shown in Fig. 1.27.

Figure 1.27 also shows typical specification limits for a 100 GHz filter wherein the adjacent channels must have greater than 20 dB blocking of one channel to the other on 0.8 nm (100 GHz) spacing, and each must have better than 0.3 dB transmittance in a 0.34 nm (42.5 GHz) pass band. Of the designs shown, only the three-cavity design meets these requirements.

A simple extension of this concatenation of cavities to more than three will create steeper sides and a broader top. However, another problem appears as seen

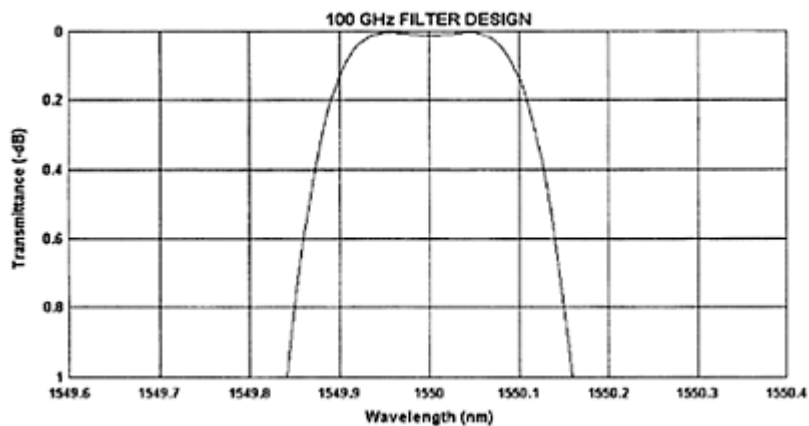


Fig. 1.25 Two-cavity NBP filter for 100 GHz DWDM applications.

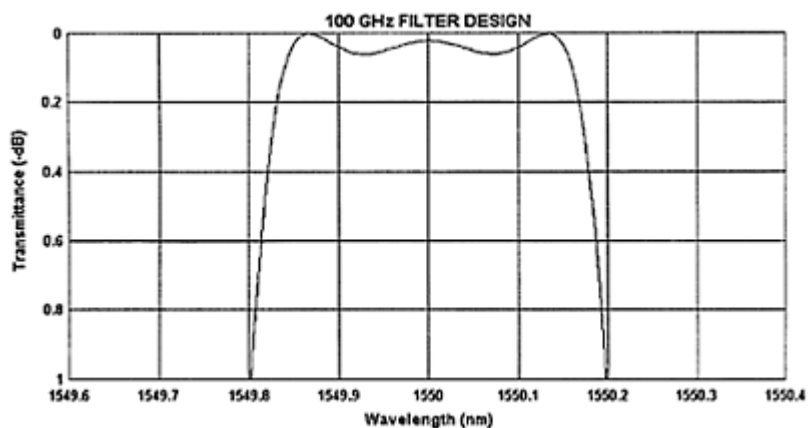


Fig. 1.26 Three-cavity NBP filter for 100 GHz DWDM applications.

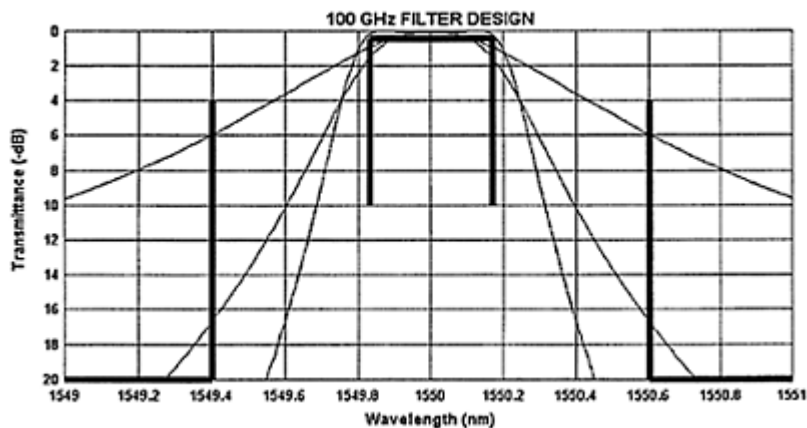


Fig. 1.27 Extended “skirts” of filters and typical specification limits.

in Fig. 1.28. There are “rabbit ears” (as they are called) which hang down, in this case, from the desired 0.0 dB level (or 100% T). Macleod⁴⁷ points out that: “The ears are a consequence of a deterioration of the matching towards the edges of the passband because there is no allowance for the variation of equivalent admittance that is very steep there.” These can be reduced or eliminated by changing the number of layer pairs in the outer cavities and/or changing the number of half waves in the spacer layers of the cavities. Cushing⁵⁰ addresses this technique also. Figure 1.29 shows such a design which has very little ripple in the pass band. This design is as follows:

(1H 1L)8 4H (1L 1H)8 1L (1H 1L)9 4H (1L 1H)9 1L (1H 1L)9 4H (1L 1H)9 1L (1H 1L)8 4H (1L 1H)7 1L .64691H .68219L.

Five and higher numbers of cavities can be dealt with by the same techniques. The antireflection coatings may vary from design to design, but are all essentially a “V-coat” from the final cavity mirror stack to the medium (air or vacuum). More than two layer ARs have been found to be of no advantage in the case of these narrow bandpass filters (except on the back side of the substrates).

Note that if the back side of a thin substrate is not coated with a proper AR, there can be ripples in the passband which depend on the substrate thickness and constitute an interference effect where the substrate is the spacer layer!

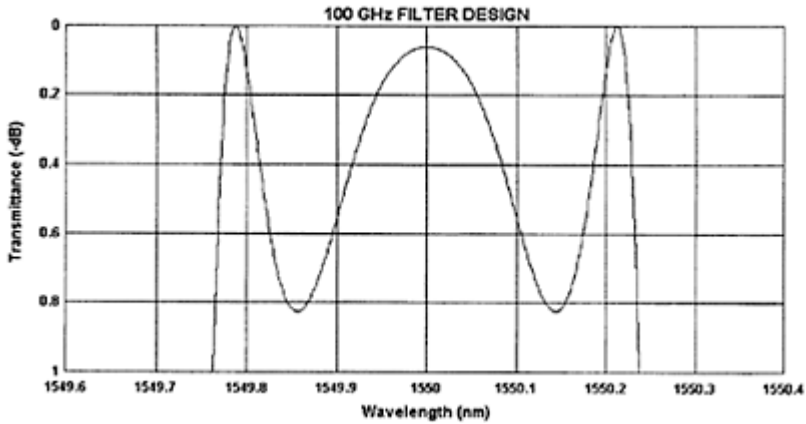


Fig. 1.28 Four-cavity NBP filter for 100 GHz with “rabbit ears”.

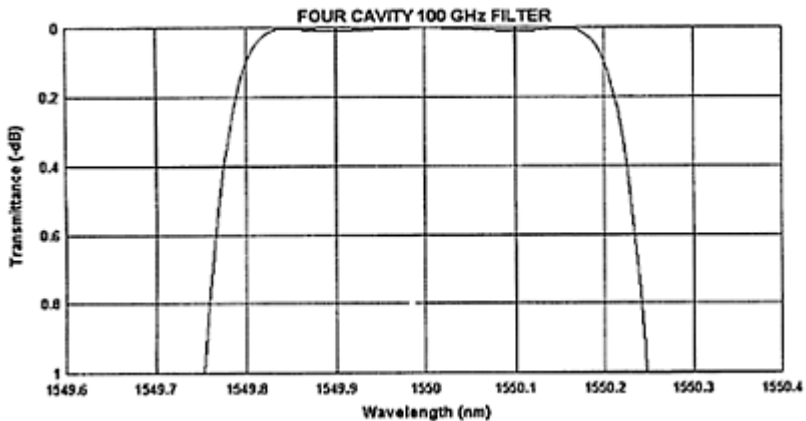


Fig. 1.29 Four-cavity NBP filter for 100 GHz without “rabbit ears”.

These filters are generally well suited to DWDM applications as “Add” or “Drop” filters. In the Add case, all other wavelengths are reflected by the filter with low losses, while the Add wavelength passes through the filter from the local station in the direction of the reflected wavelengths and is joined with them. The Drop case allows the selected wavelength to pass through the filter to the local station while all the others are reflected and travel on. The current requirements of 0.3 dB (6.7% R) maximum loss in the transmitted band are acceptable for a drop filter. However, if one were attempting to use such a filter to simultaneously perform an Add *and* a Drop, this 0.3 dB from the Dropped signal would combine with the Added signal and could never be eliminated. The requirement for a filter to do both Add and Drop simultaneously and also give 20 dB

isolation is that the in-band reflection would be less than 0.0044 dB (1% R). This is far from the current production capabilities at this time.

1.3.3.3 Fiber Bragg Gratings

There has been a great deal of work done in recent years in fiber Bragg gratings (FBG) for DWDM applications such as described in the book by Kashyap.²⁸ The concept is that a fiber can be made with a core composition which will gain a permanent change in index of refraction when illuminated with the appropriate wavelength and intensity of light. Interfering laser beams can be set up to record a periodic structure along the length of the fiber which resemble the index variations in a QWOT stack. When light of the QWOT wavelength travels in the fiber and encounters this photographically altered section, it will be reflected by this “dielectric mirror stack.” Such mirrors can be spaced by effective half wave “layers” to produce NBP filters of one or more cavities. Therefore, most of the ideas we have discussed are applicable to the FBG field also. However, the difference in index from high to low in the current FBGs is only on the order of .001! This means that the spectral width of the high reflecting area (as in Fig. 1.19) will be quite narrow. This would be like what Thelen⁴ calls a minus filter and is illustrated in Fig. 1.20. The attractive thing about the thin film DWDM filters is that the reflected band other than the NBP covers the whole C and L bands mentioned, whereas a FBG has no such general blocking capability. The FBG mirror, on the other hand, can be made to reflect only one narrow channel and pass the rest. This is the converse of the DWDM dielectric filters which transmit one channel and reflect the rest. It will be interesting to see how these technologies are mixed for best utility in the future.

1.3.4. Beamsplitters

Optical coatings with reflectance between that of high and low reflectors might be classified as beamsplitters. Combinations of the type of the first few layers shown in Fig. 1.19 could be starting points to make partial reflectors with all dielectric layers. Given some starting design, using any automatic optimizing thin film design program would probably result in a useful beamsplitter design.

1.3.4.1 Effects of Angle of Incidence

As the angle of incidence departs from normal incidence, the *s* and *p* polarizations behave as though they have different indices of refraction. The normal incidence index of refraction for the *s* polarization is multiplied by the cosine of the angle of incidence in the medium ($n_s = n \times \cos(i)$), and the index of refraction for the *p* polarization is divided by the cosine of the angle of incidence in the medium ($n_p = n / \cos(i)$).

It now also becomes necessary to know the angle of the light ray in the medium. This is found using Snell’s Law: $n_1 \sin(i) = n_2 \sin(r)$, where *i* and *r* (in this case only) are the angles of the incident and refracted rays and n_1 and n_2 are the indices of the media of the incident and refracted rays.

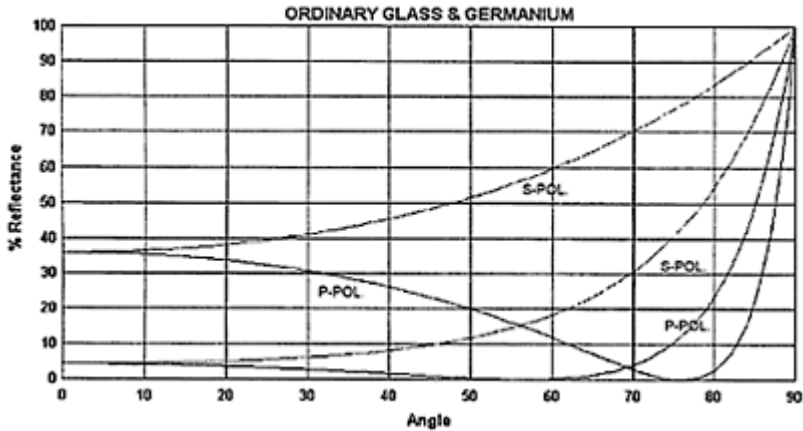


Fig. 1.30 Reflectance versus angle of glass and germanium surfaces.

Figure 1.30 shows these effects on uncoated substrates of ordinary glass (index 1.52) and germanium (index 4.0). Note that the p polarization reflectance goes to zero at angles of about 57° and 76° respectively. This is known as the Brewster Angle, and a window tilted at this angle will transmit all of the p polarized light while reflecting more s polarized light than at normal incidence. Such a window inside a laser beam will cause the laser to be polarized in the P orientation since the s polarization is much more “lossy”.

It can be seen that the polarization effects are small up to 10 or 20 degrees, so we generally can consider this region to be “near normal incidence” and ignore the angular effects. However, some special cases like DWDM filters need more careful attention to polarization. Figure 1.31 shows the effects of a 15° tilt on the 100 GHz filter shown in Fig. 1.26. The s polarization bandwidth gets narrower and has shifted from 1550 to 1533.65 nm. The p polarization has gotten wider and shifted further. This is probably not acceptable in the DWDM environment. Figure 1.32 shows the case of the filter tilted 9.9° and thereby shifted 7.2 nm to 1542.8 nm. This is a shift that covers 10 channels of 100 GHz bandwidth. If such distortion of the band were acceptable, the basic filter could be tuned by an appropriate tilt to any of these wavelengths in the range. To shift the band from

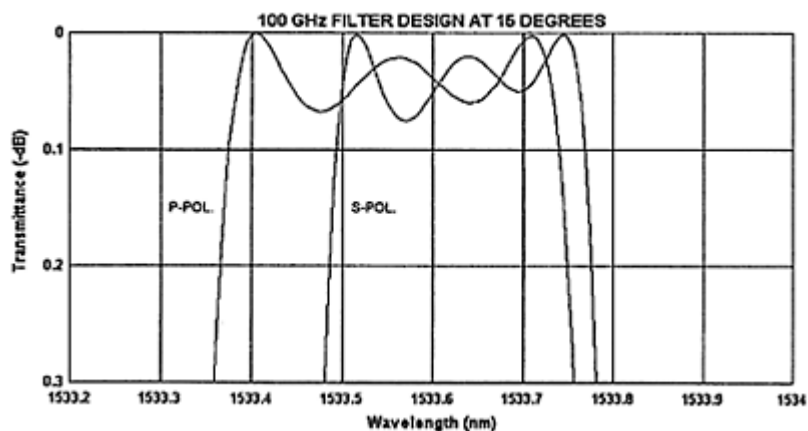


Fig. 1.31 Effects of a 15° tilt on a 3-cavity filter for 100 GHz DWDM applications.

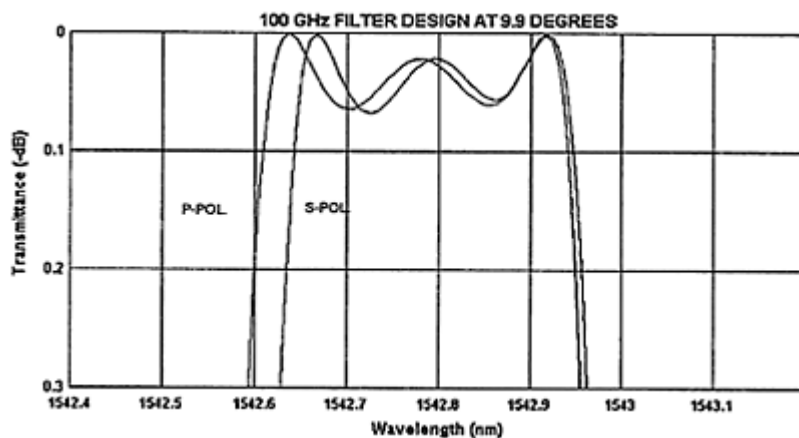


Fig. 1.32 Effects of a 9.9° tilt on a 3-cavity filter for 100 GHz DWDM applications.

1550 to 1549.2 nm or one channel, it takes a tilt of 3.3° . The wavelength shift is a function of the square of the angle of incidence, so the 9.9° ($3 \times 3.3^\circ$) shifts nine channels (3^2). This can give some indication of the alignment tolerances needed for such filters.

1.3.4.2 Polarizing Beamsplitters

The different behavior of the s and p polarizations at angles of incidence significantly different from normal incidence can be used to make polarizing beamsplitters. The situation is usually most favorable and useful if the beamsplitter coating is immersed in glass at 45° as in a beamsplitter cube. Figure 1.33 shows the results of a simple QWOT stack of 2.1/1.45 (16 layers) cemented in glass of index 1.52. In general, there is a region on both sides of the narrowed p polarization reflectance band where the broadened s polarization band can have very high reflectance and the p is quite low. It is therefore easy to make a polarizing beamsplitter for a narrow band like a laser on either the SWP or LWP side of the reflector. These are narrower if the coating is on a slab rather than immersed, but they still can be useful. This basic starting design can then be adjusted by optimization for the particular needs at hand.

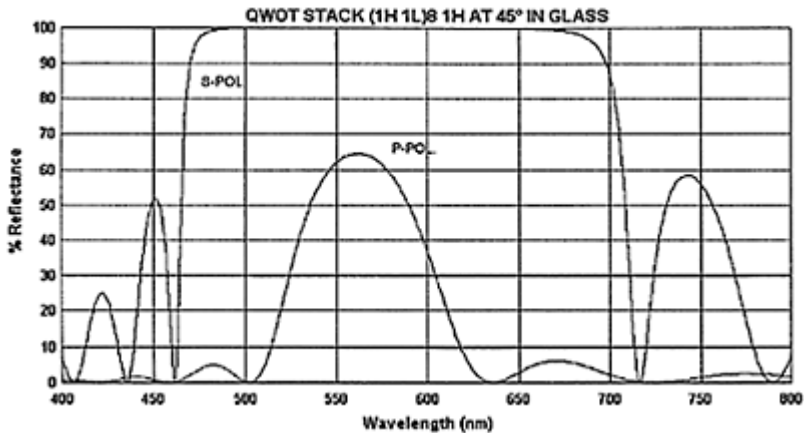


Fig. 1.33 QWOT stack (1H 1L)8 of indices 2.1/1.45 on glass at 45°

The ultimate broad band polarization beamsplitter is where the p polarization reflection in the middle of Fig. 1.33 can be reduced to near zero! This is the case of the MacNeille polarizer described in some detail by Macleod² on pages 328–332. Basically, conditions are satisfied to make the P polarization at the Brewster angle so that none is reflected and all is transmitted. The s polarization reflection is then boosted by an appropriate number of layers in the stack.

Cushing⁴⁸ described the design of a filter for use at 45° that had extremely low polarization. His approach was related to that of Baumeister⁴⁹ in using multiple indices to have less severe changes in index than just the highest to lowest.

Polarizing effects and coatings and any significant angle of incidence are best handled in thin film design software, but we will show a few more effects on admittance diagrams in Sect. 1.6.

1.3.5. Three-Layer AR Coating on Germanium, Example

Let us now apply the principles described above to the preliminary design of a three layer AR coating for a germanium substrate. We will use thorium fluoride at an index of about 1.38 as in Fig. 1.14 and germanium as a coating material at about 4.0 index as in Fig. 1.15.

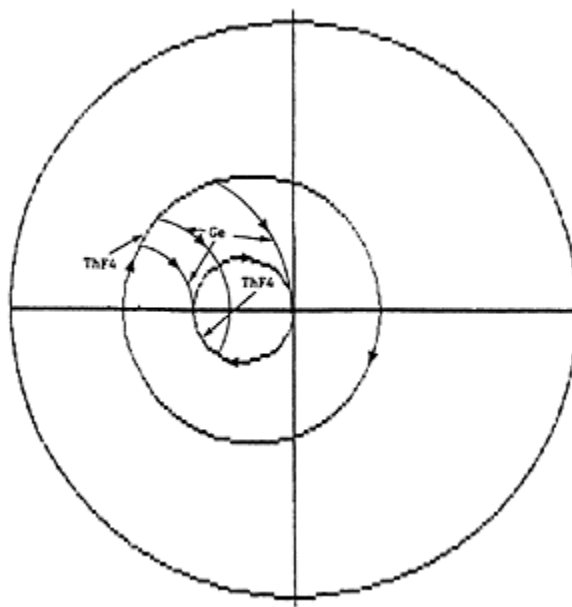


Fig. 1.34 Reflectance diagram of a preliminary design for a three-layer AR on germanium.

We know that the first layer (on the substrate) in this case would have to be thorium fluoride, because a germanium layer would have no effect on the germanium substrate. This means that the second layer would have to be germanium and the third thorium fluoride. We also know that to get zero reflectance the circle of the last thorium fluoride layer must pass through the origin of a reflectance diagram. We further know that the circle of the first layer must pass through the substrate index point of 4.0. We can use tracing paper over graphs like Figs. 1.14, 1.15, and 1.16 to draw these two circles as in Fig. 1.34. We can deduce that the germanium layer must lie in a specific range shown in Fig. 1.34 (or half wave additions thereto). By picking any of the germanium layers from the range and the corresponding thorium fluoride layers, the design can be quickly optimized over a specific (although limited) spectral range with any reasonable thin film design optimization program. It turns out that such a design optimized for a band from 8 to 12 μm will have the shortest first layer that allows the second layer to reach the left end of the circle for the third layer at the horizontal axis. This third (last) layer will be a QWOT. When we get to Fig. 1.80, it will be seen that such a solution is consistent with the principles that will become more apparent at that time.

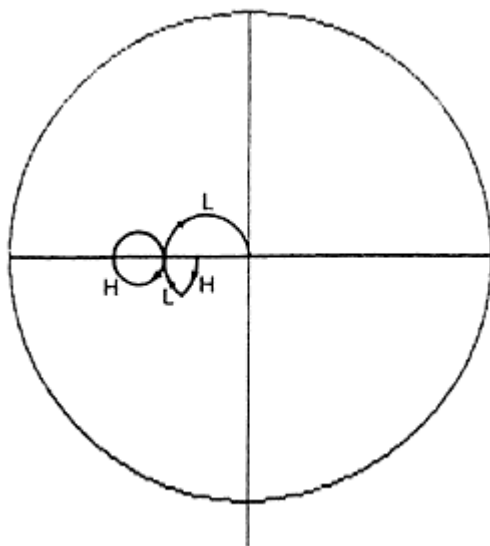


Fig. 1.35 Reflectance diagram of the construction of a two material, four-layer replacement for the three material, three layer AR in Fig. 1.11.

1.3.6. Example Four-Layer Broad Band AR Coating in the Visible

The index of 1.7 needed for the design of Fig. 1.11 mentioned in section 1.3.1 may not be easily available. We could think of using only the 2.25 and 1.38 indices and somehow simulate or approximate the 1.7 index with these materials. We know that the reflectance of the simulated index layer should start and end at the same points as for the QWOT layer with the desired 1.7 index. The only logical solution that comes to mind is a high index layer circle that starts at the substrate and goes clockwise, and a low layer circle that passes through the endpoint of the would-be 1.7 index layer as shown in Fig. 1.35. Where these two circles intersect is the end of the first layer and the beginning of the second layer. The third circle represents the half wave layer of high index, and the last circle the QWOT layer of low index from the three layer design of Fig. 1.11. This approximation of the intermediate layer by a combination of two layers of higher and lower index is attributed to Rock⁵. We discuss further approximations in more detail below.

1.3.7. Physical Thickness versus Optical Thickness

We have referred to the optical thickness of layers and the QWOT. It is sometimes necessary to know the physical thickness of layers. It is also necessary to keep in mind that an optical thickness (OT) is only a given value such as a QWOT at a specific wavelength. For example, a QWOT at 800 nm is two QWOTs or a half wave optical

thickness at 400 nm (neglecting dispersion effects). The physical thickness (t), on the other hand, is independent of wavelength. The relationship between the optical and physical thickness is $OT=n \times t$, where n is the index at the wavelength of interest. Therefore, if the index were 2.0 for a QWOT of high index material at 1550 nm, the physical thickness, t , would be 193.75 nm (or 1937.5 Ångstrom units or 1.9375 kiloÅngstroms (kÅ)).

1.4. ADMITTANCE DIAGRAMS

The admittance diagram gives essentially the same information as the reflectance diagram, as they are conformal mappings of each other. Macleod² describes their derivation in some detail. Figure 1.36 is the admittance diagram for the same design shown in the reflectance diagram of Fig. 1.11. Figure 1.38 is the admittance diagram for the high reflector stack shown in the reflectance diagram of Fig. 1.18. The goal for an AR coating to be used in air or vacuum is to bring the admittance at the top of the coating to $1.0-i0.0$ which is the normalized admittance of free space. It can be seen that either representation, reflectance or

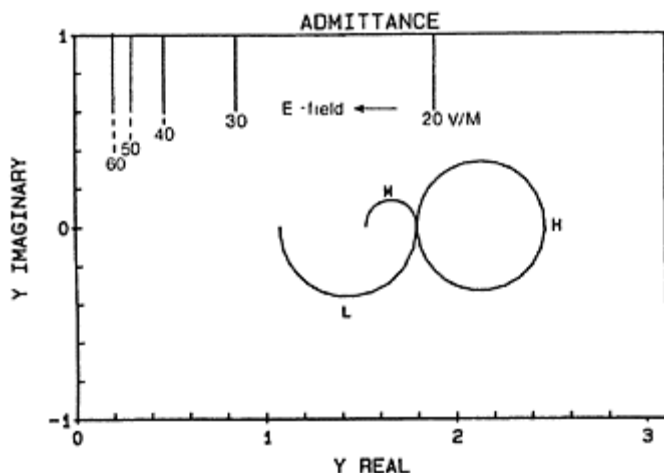


Fig. 1.36 Admittance diagram of the three layer AR in Fig. 1.11 for comparison and also showing the lines of equal electric field.

admittance, works well for designs like AR coatings which stay close to the origin. However, high reflectors quickly move out in admittance “orbits” that go off the page, whereas they remain always inside the unit circle of the reflectance diagram. One advantage of the admittance diagram is that the electric field at any point within the layer system can be determined simply from the distance from the imaginary axis. Macleod⁶

gives the derivation of the electric field as a function of the admittance in free space units, i.e., $Y=1.0$ in a vacuum. The resulting electric field in volts per meter as a function of the real part of the admittance units is

$$E=27.46/[Re(Y)]^{0.5} \quad (1.6)$$

where the incident power density is one watt per square meter. This generates lines of equal field on the admittance diagram which are parallel to the imaginary axis as shown on Fig. 1.36. On a reflectance circle diagram, the lines of equal volts per meter are as shown in Fig. 1.37. Here it is apparent that a design must stay away from the right outer portion of the diagram to avoid high fields. Apfel⁷ describes how to design optical coatings for reduced electric field at any particular level in the stack. This is useful in designing laser damage resistant coatings. As we will see in Sect. 1.5, one might want the converse of high fields in absorbing materials to maximize absorption. Smith⁸ gave an interesting comparison of admittance diagrams, reflectance diagrams, Smith charts (no relation), and other graphical representations. We use admittance diagrams for the balance of this chapter after the next section.

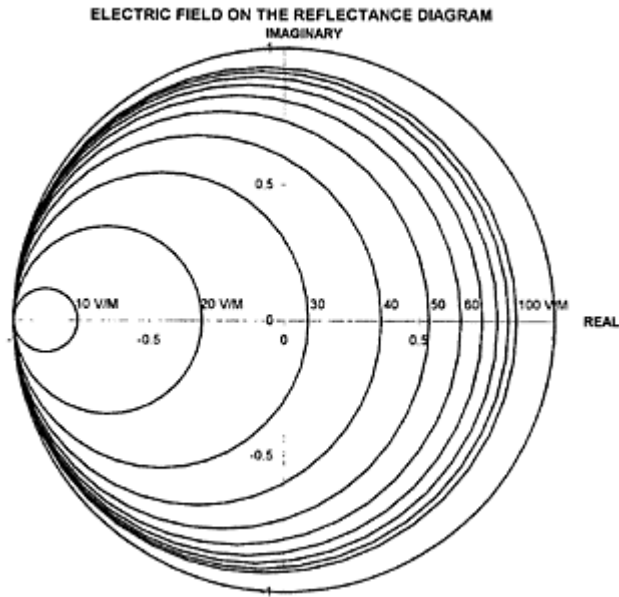


Fig. 1.37 Lines of equal volts per meter on a reflectance circle diagram.

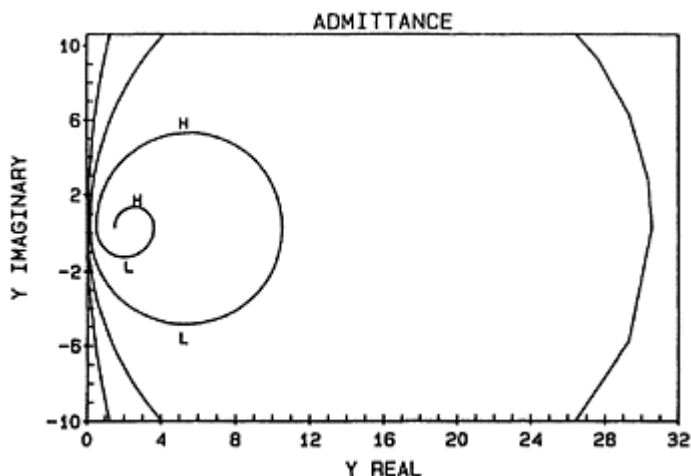


Fig. 1.38 Admittance of the QWOT high reflector stack shown in Fig. 1.18 for comparison.

1.5. TRIANGLE DIAGRAMS

Up to this point, we have dealt only with non-absorbing coating materials. Generally, the full description of index of refraction is a complex quantity, $N = n - ik$. Here the n is the index of refraction for a purely dielectric material that we have been considering thus far, and the k is the extinction coefficient needed to describe absorption in a material. The absorption coefficient, a , is defined as: $a = 4\pi k/\lambda$, where λ is the wavelength. The ratio of the transmitted intensity, I , to the initial intensity, I_0 , through an absorbing medium of thickness x can be found by: $I/I_0 = e^{-ax}$.

Apfel⁹ describes another useful graphical tool, the triangle diagram. These are useful in seeing at a glance what a semi-transparent film will reflect, transmit, and absorb, and what transmittance can be achieved if the layer were coated to reduce its reflectance. We will describe it here and go through some examples. Figure 1.39, taken from Apfel's work, shows the characteristics of twelve semitransparent metallic films when used as single layers of varying thickness. The three corners of the triangle diagram represent 100% transmittance (T), reflectance (R), and absorption (A). The sum of the three must equal unity ($A + R + T = 1$).

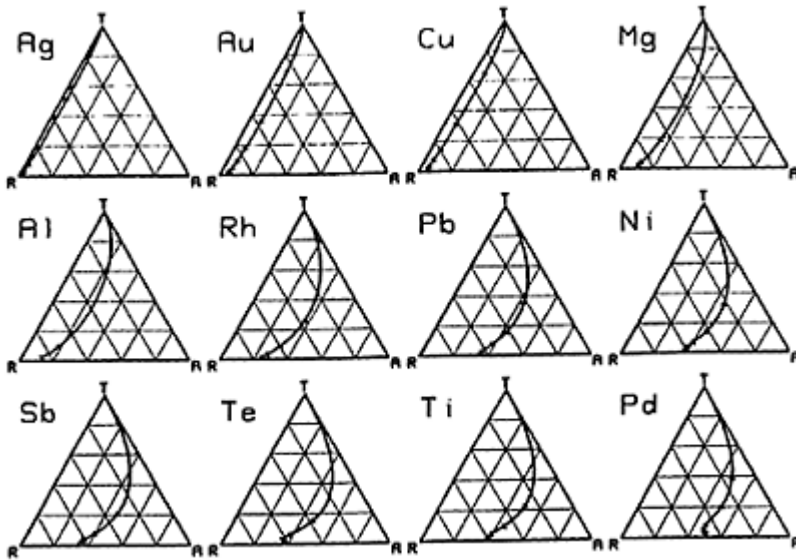


Fig. 1.39 Triangle diagrams plotting R, T, and A for various semitransparent metallic films versus thickness from zero to an opaque film (from Apfel^{1,5}).

At the start of deposition with zero thickness (as a freestanding film with no substrate), T equals 100%, and R and A equal 0%. As the thickness increases, the reflectance and absorption increase as a function of the real and imaginary parts of the refractive index. The transmittance moves from the 100% apex to the 0% baseline. Silver, for example, is seen to quickly reach a high reflectance with low absorption losses, while nickel moves quickly to much higher absorption values before reaching a peak $R=60\%$ and $A=40\%$ as the transmittance goes to zero.

Apfel shows that the potential transmittance described by Berning et al.¹⁰ can be found by a straight line from the $R=100\%$ point through the particular point representing the thickness of the metallic film. That is to say that an AR coating on the metallic film could reduce the reflectance to zero and move the T and A to the point where the line of equal potential transmittance intersects the triangle side T to A. The triangle diagram is a relatively new and useful tool for working with metallic (or other absorbing) thin films.

In the balance of the book other than this section, we confine the discussions to non-absorbing materials unless otherwise stated.

1.5.1. Designing Coatings with Absorbing Materials

In much of the literature, A is taken to be zero for most dielectric film studies, or at most a perturbation. However, when metals are used, A becomes an equal partner with R and T. Dobrowolski, et al.²⁹ have shown several examples of how and where metal layers can

be employed to take advantage of the possibility to reduce the unwanted reflections which occur from the rejected bands of wide and narrow bandpass filters. They also show that such benefits can only be obtained at the expense of some reduction of transmittance in the passbands due to the inherent A. A reflection filter by Tan, et al.³⁰ will also be shown below.

The graphic tools promoted by Apfel for this work are Reflectance Diagrams¹ (RD) and Triangle Diagrams⁹ (TD). The RD is illustrated in Fig. 1.40 where a 3 layer coating of chromium (Cr), dielectric, Cr is deposited on a glass substrate. For the wavelength depicted, this gives a very low reflection; it would be a “black mirror” for that wavelength. With all dielectric materials and no A, we could know the transmittance of a coating also from this diagram as $T=1-R$. However, when there is A involved, we cannot readily determine this from a RD. However, Apfel⁹ introduced the TD to help visualize absorption effects. Fig. 1.41 is such a diagram for the same coating as in Fig. 1.40. The first layer starts near the top at about 96% T from the substrate with 4% R and 0% A. The second dielectric layer can only move on a locus which radiates from the 100% R point through the point where that layer starts. With the proper combination of dielectric layers, it is possible to move on such a locus to the 0% R line (on the

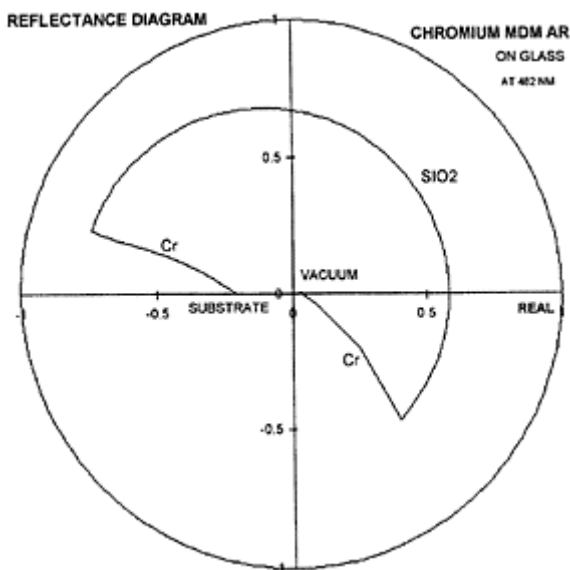


Fig. 1.40 Reflectance diagram (RD) of a MDM (Cr, SiO₂, Cr) AR on glass at 482 nm.

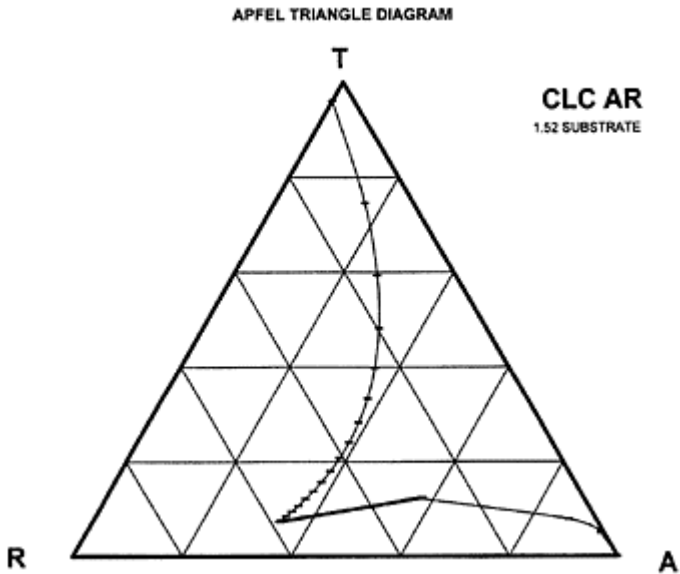


Fig. 1.41 Triangle diagram (TD) of the same coating as shown in Fig. 1.40.

right edge between T and A). This would be an antireflection coating on the chromium layer in this case. The point of intersection of this locus with the 0% R line would show the Potential Transmittance (PT) of the first chromium layer as about 17%. This same coating is illustrated on an Admittance Diagram (AD) in Fig. 1.42.

1.5.1.1 Absorbing Materials

Most of the material properties of n and k used here are derived from the compilations of Palik^{31,32}. Figure 1.43 shows that Aluminum (Al) is generally less absorbing than Cr. The Silver (Ag) shown in Fig. 1.44 is even less absorbing. The reflectance of a bulk material can be plotted on a RD from the Fresnel equation. Figure 1.45 shows lines of equal n and k on a RD. On an AD, these lines would form a rectangular grid. Figure 1.46 shows the opaque point or bulk reflectance of Cr as a function of wavelength. It can be seen that the distance from the origin and thereby the reflectance is fairly constant over the near UV and visible wavelengths. It then starts to climb from 2 to 3 microns and becomes a relatively high reflector at longer wavelengths. Cr has a generally neutral gray color with a little less reflectance in the blue. Ni has a similar locus but has a bit more reflection toward the blue. This is probably why Nichrome is favored for neutral density filters, the proper mixture of Ni and Cr should best compensate each other to give nearly constant R and T over the visible spectrum.

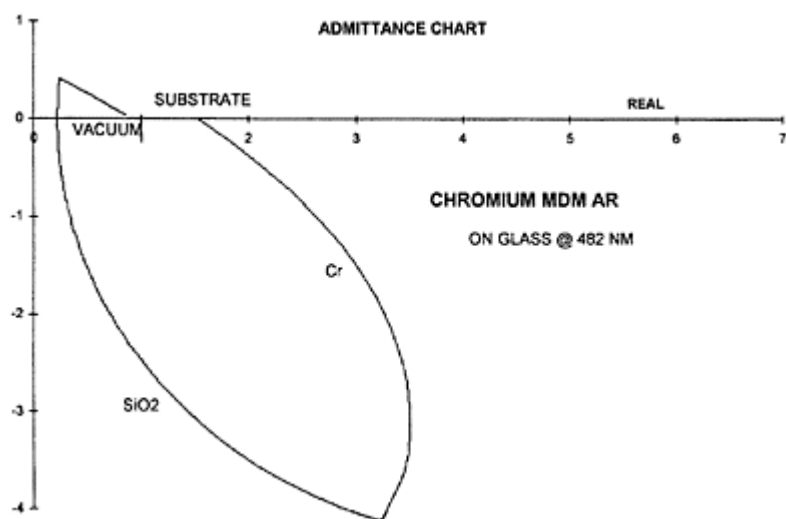


Fig. 1.42 Admittance diagram of the same coating as in Fig. 1.40.

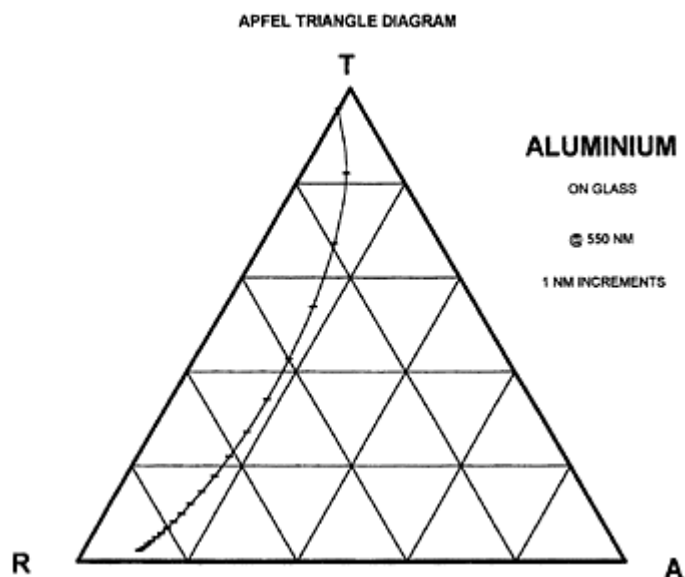


Fig. 1.43 Triangle diagram of a thick layer of aluminum (Al).

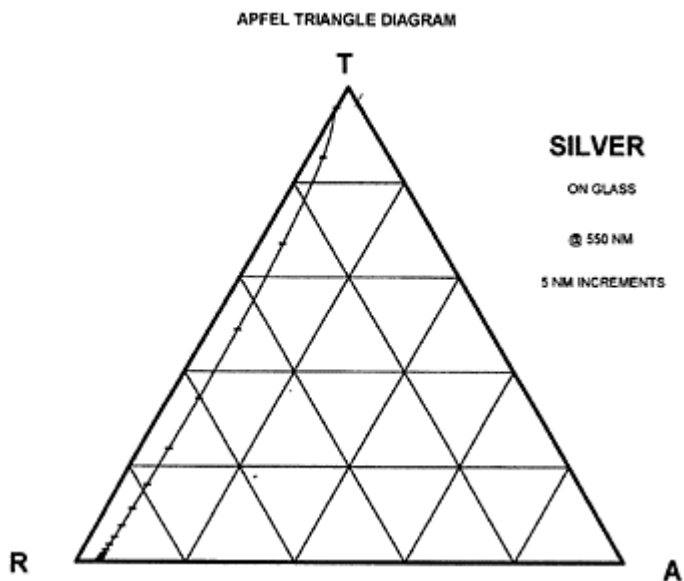


Fig. 1.44 Triangle diagram of s thick layer of silver (Ag).

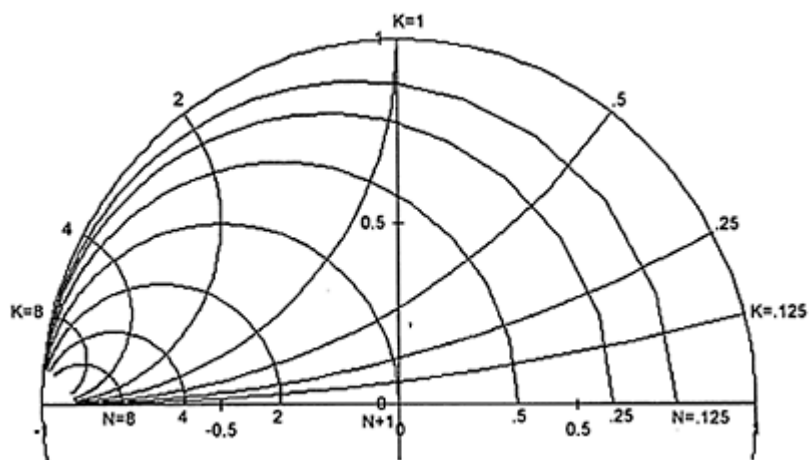


Fig. 1.45 Constant n and k lines on a reflectance diagram.

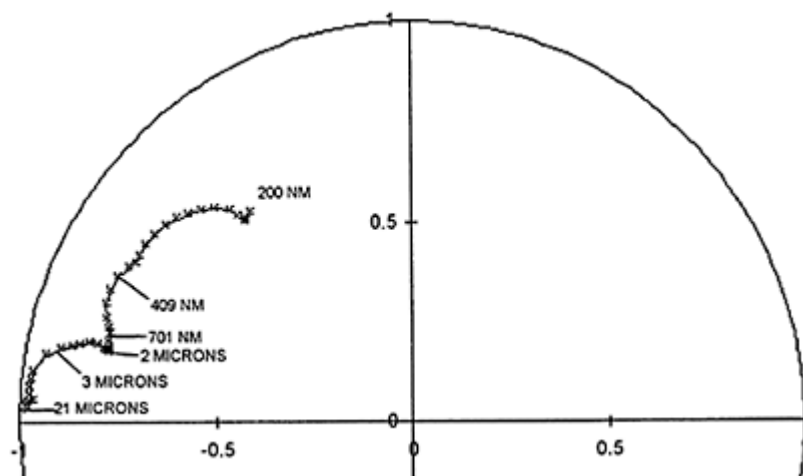


Fig. 1.46 Reflectance of the chromium (Cr) opaque point versus wavelength.

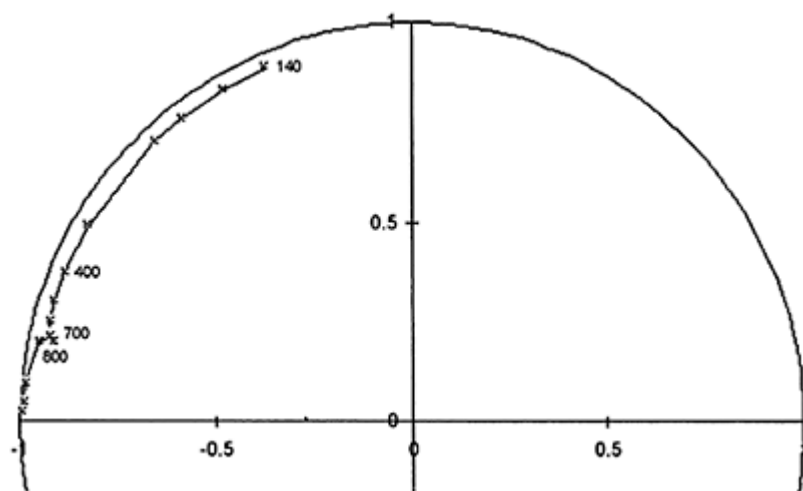


Fig. 1.47 Reflectance of the aluminum (Al) opaque point versus wavelength.

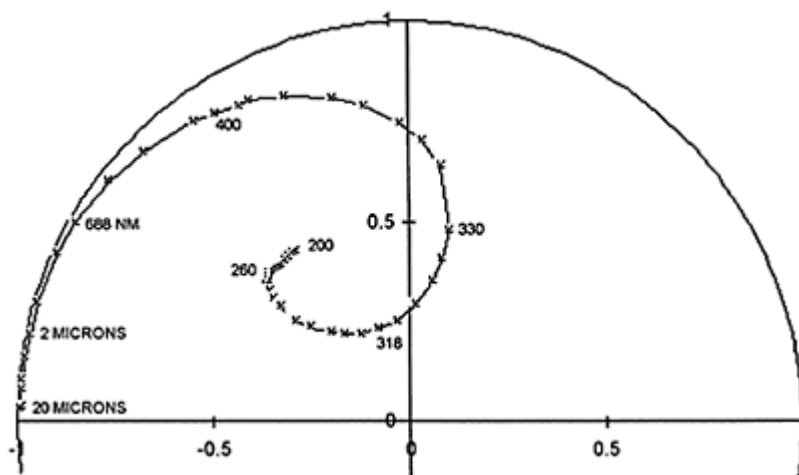


Fig. 1.48 Reflectance of the silver (Ag) opaque point versus wavelength.

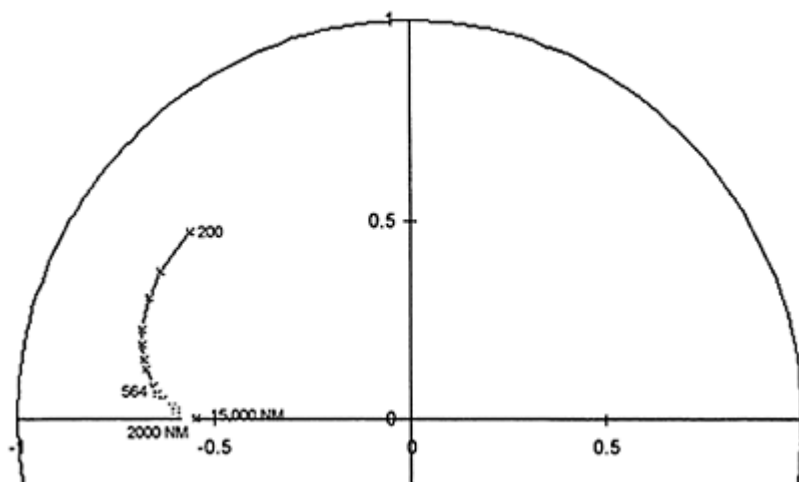


Fig. 1.49 Reflectance of the silicon (Si) opaque point versus wavelength.

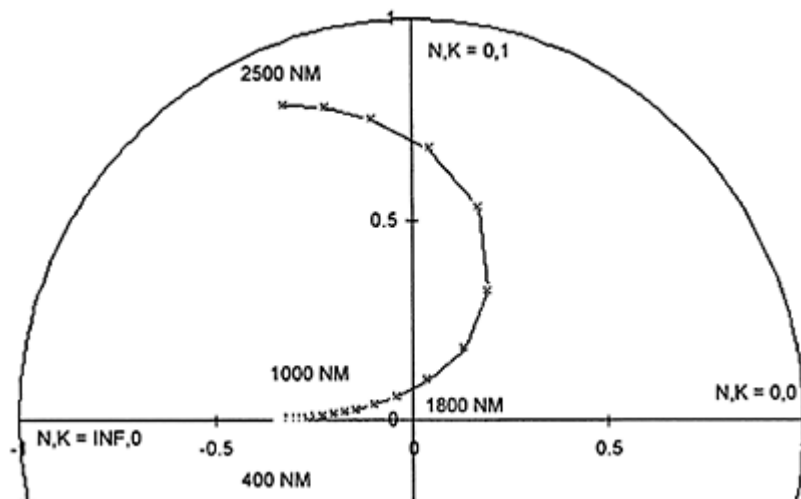


Fig. 1.50 Reflectance of the indium oxide (InO_x) opaque point versus wavelength.

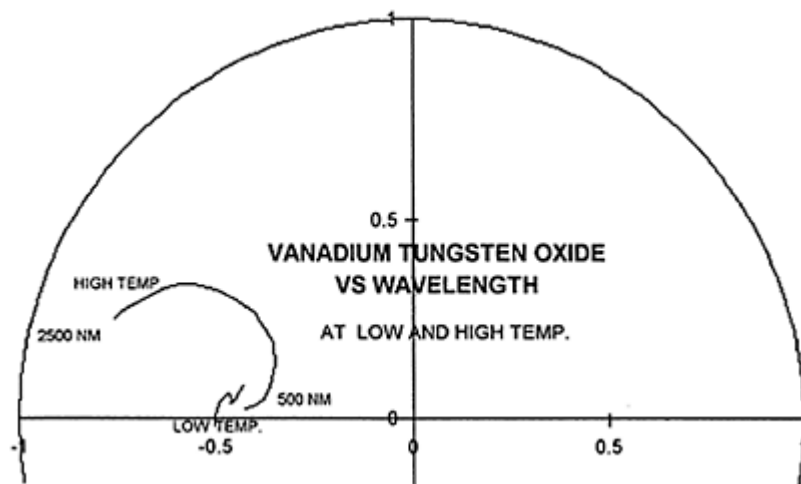


Fig. 1.51 Reflectance of the vanadium tungsten oxide opaque point versus wavelength.

Figure 1.47 shows that Al has a fairly flat spectral reflectance except around the well known dip near 800nm. Figure 1.48 shows that Ag has a minimum reflectance at around 318 nm and then it increases steadily with wavelength. The semiconductor materials of

silicon (Si) and indium oxide (InO_x) are shown in Figs. 1.49 and 1.50. The Si has high absorptance in the UV decreasing to none at about 1 micron. Conversely, InO_x has none in the visible and starts increasing from 1 micron to longer wavelengths where it becomes a high reflector.

Figure 1.51 shows the interesting thermochromic properties of tungsten doped vanadium oxide ($\text{V}_{1-x}\text{W}_x\text{O}_2$) as reported by Tazawa, et al.³³ At temperatures below the critical value of 47°C , the material has low absorptance and looks mostly like a high index dielectric. Above this temperature, it behaves more like a metal with increasing reflectance with wavelength. This type of behavior could be useful to make a passive window which would transmit sunlight and infrared warmth when the window was cool and block the heat when the window was hot.

1.5.1.2 Application Examples of Designing With Metals

When viewed on a RD, the locus of the reflection versus film thickness changes predictably from the starting point reflectance and phase. Figure 1.52, 1.53, and 1.54 show such loci for Cr, Al, and Ag. The fact that the metal loci

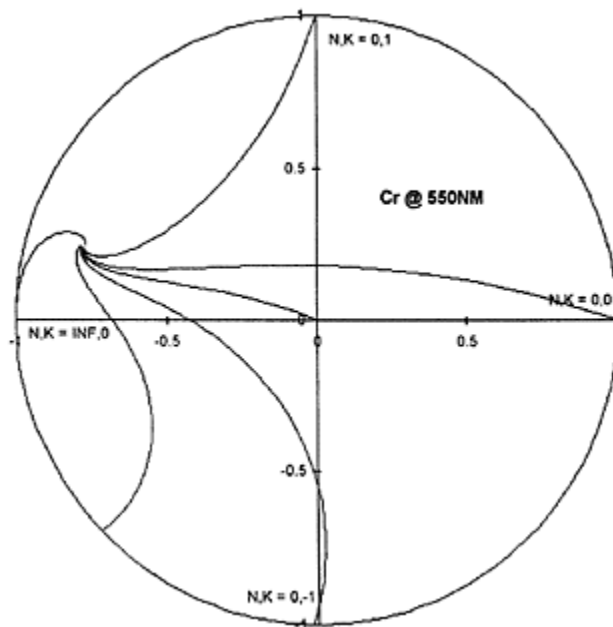


Fig. 1.52 RD showing loci of chromium (Cr) from various starting points.

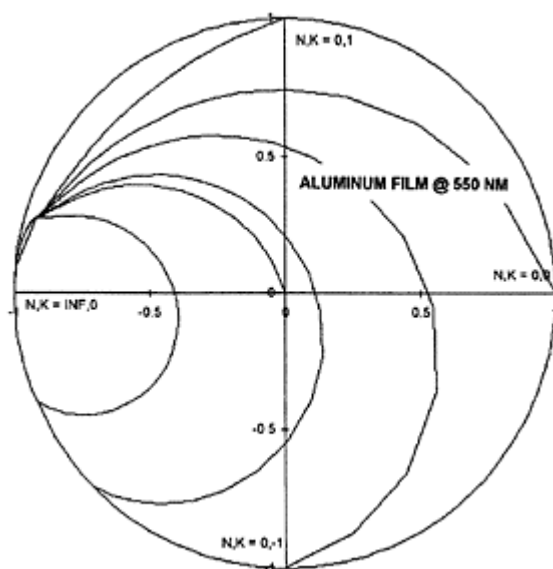


Fig. 1.53 RD showing loci of aluminum (Al) from various starting points.

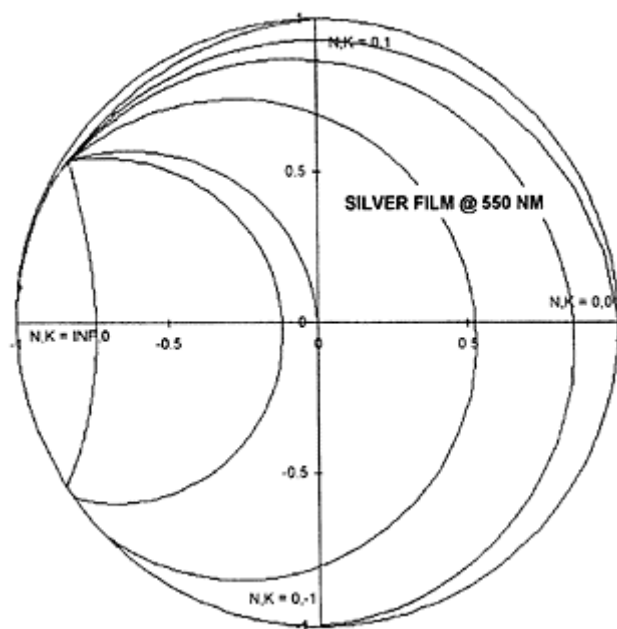


Fig. 1.54 RD showing loci of silver (Ag) from various starting points.

can in some cases move almost in the opposite direction from a dielectric locus may prove useful in future designs where some A can be tolerated. These type of charts can be used in conjunction with those like Figs. 1.14 and 1.15 to design films that combine metals and dielectric materials.

Induced Transmission Filter Example

An example of an induced transmission filter is taken from Fig. 6 of the work by Dobrowolski, et al.²⁹ Figure 1.55 shows the transmittance of a similar design with and without suppression of the unwanted reflections shown in Fig. 1.56. The higher reflection curve in Fig. 1.56 is the unwanted reflection and the lower curve is the result of the reflection suppression. Figure 1.57 shows how this would look on a TD; the unsuppressed design at 570 nm in the passband ends near the top of the TD with about 90% T and 10% R. The suppression brings the locus to about 50% T and 2% R.

The TD viewpoint as seen in Fig. 1.57 is not very revealing with respect to what is happening to the R and T when the dielectric layers are deposited, just as the RD does not show what is happening to the A and T when there is A present. To overcome this limitations, we have modified the TD to make a Prism Diagram

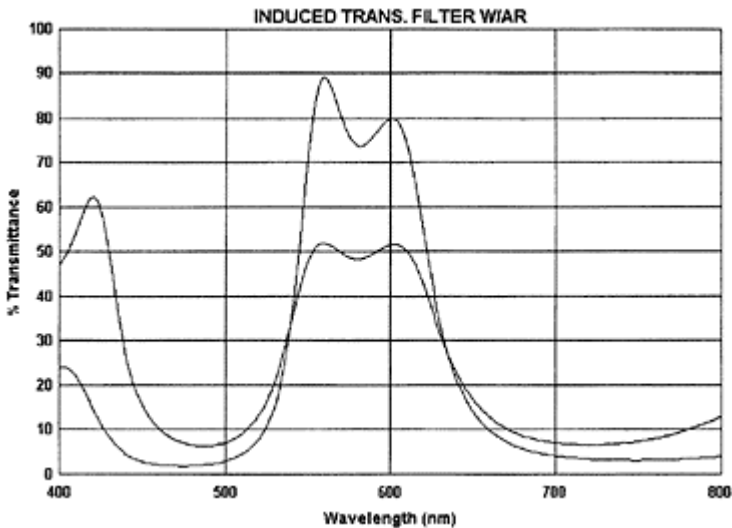


Fig. 1.55 Transmittance spectrum of Dobrowolski, et al.²⁹ induced transmission design with and without suppression AR at 570 nm.

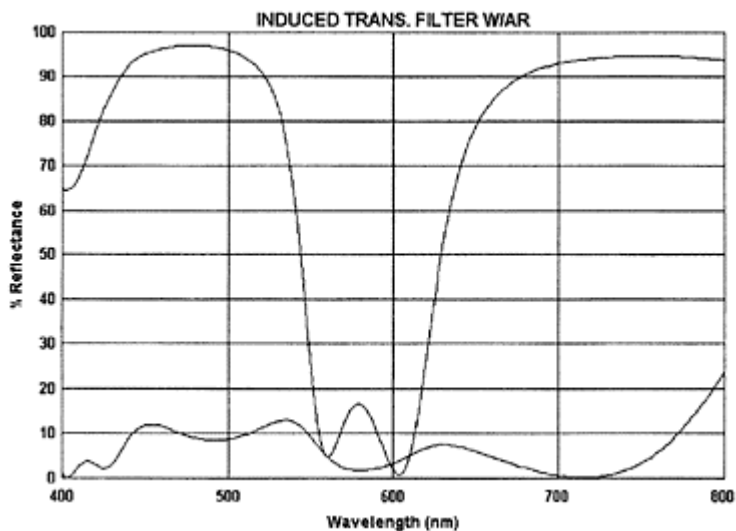


Fig. 1.56 Reflectance spectrum of Fig. 1.55 design with and without AR.

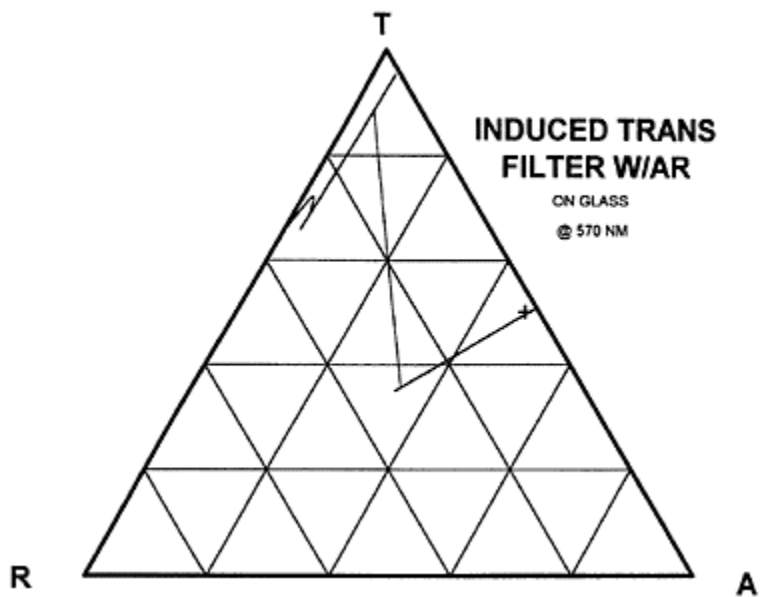


Fig. 1.57 Triangle diagram of Fig. 1.55 design with AR.

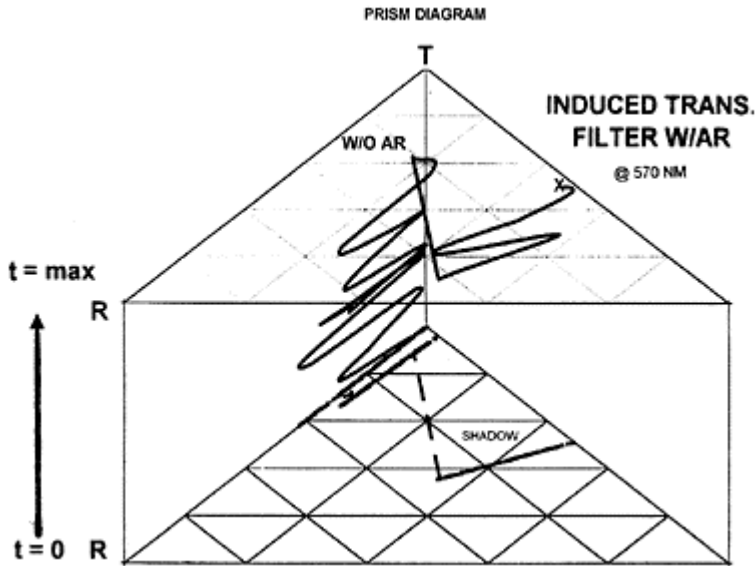


Fig. 1.58 Prism diagram (PD) of Fig. 1.55 design with AR at 570 nm.

(PD) as shown in Fig. 1.58 wherein the same TD as Fig. 1.57 forms the base, but the values are also offset vertically with increasing thickness of the coating up to the maximum thickness at the top plane. Here, we can now see that the first five dielectric layers in the first half of the thickness oscillate in R and T in the 0% A plane. The Ag layer then adds a small amount of A at this 570 nm wavelength and the rest of the dielectrics without the AR brings the transmittance to a high %T and low %A. When the reflection suppression coating is added, the Cr layer moves the locus about half way toward the 0% T edge and then the dielectric layers antireflect this to the maximum PT near the 0% R edge at the top. This point is marked with an "X" in the figure. The shadow of this three-dimensional path is seen on the base plane of the Fig. 1.58 PD and is the same as seen in the Fig. 1.57 TD. Figure 1.59 shows the same display for 700 nm wherein the coating without AR has a very high R, but with the AR it is reduced to virtually zero with a small residual %T.

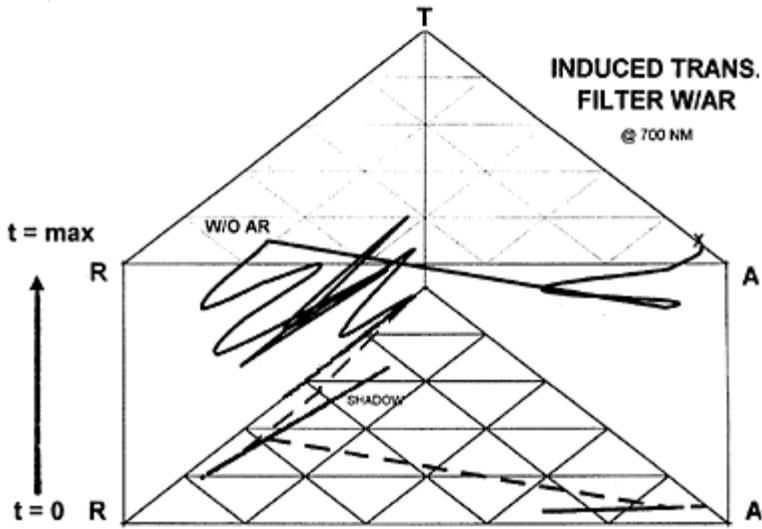


Fig. 1.59 Prism diagram (PD) of Fig. 1.55 design with AR at 700 nm.

NBP Reflection Filter Example

Another example is from Fig. 1 of the work of Tan, et al.³⁰ where totally opaque narrow bandpass reflection filters were studied. Figure 1.60 shows the reflection of the filter where there is 0% T at all wavelengths. There is basically an opaque Ag layer, dielectric stack, and a final thin layer of Cr. The principal used here is

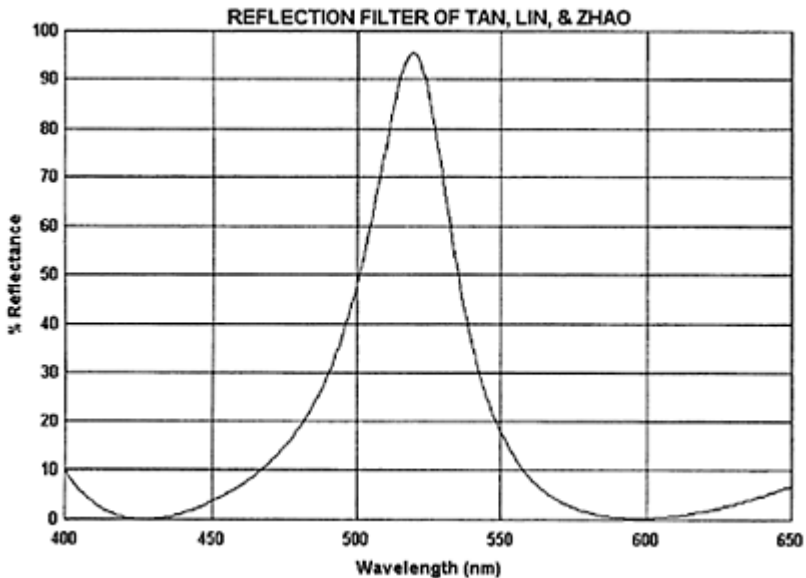


Fig. 1.60 Reflectance spectrum of Tan et al.³⁰ filter.

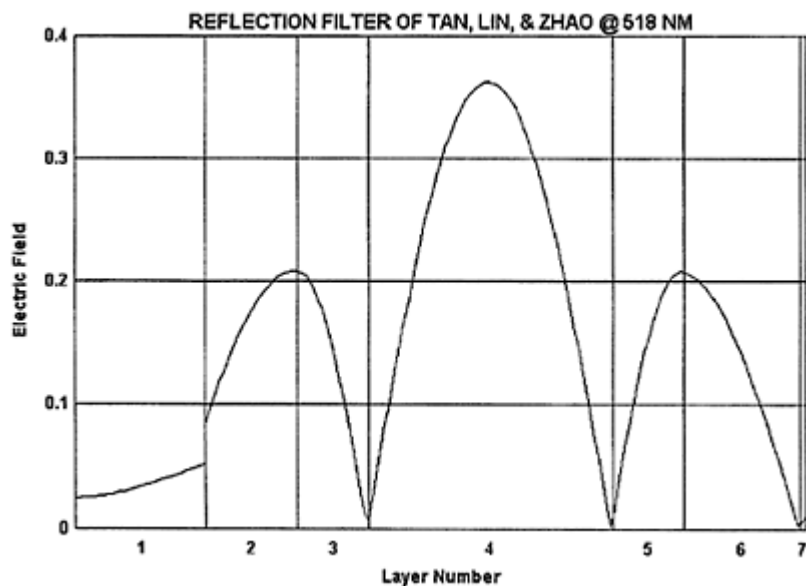


Fig. 1.61 Electric field in coating of Fig. 1.60 at 518 nm.

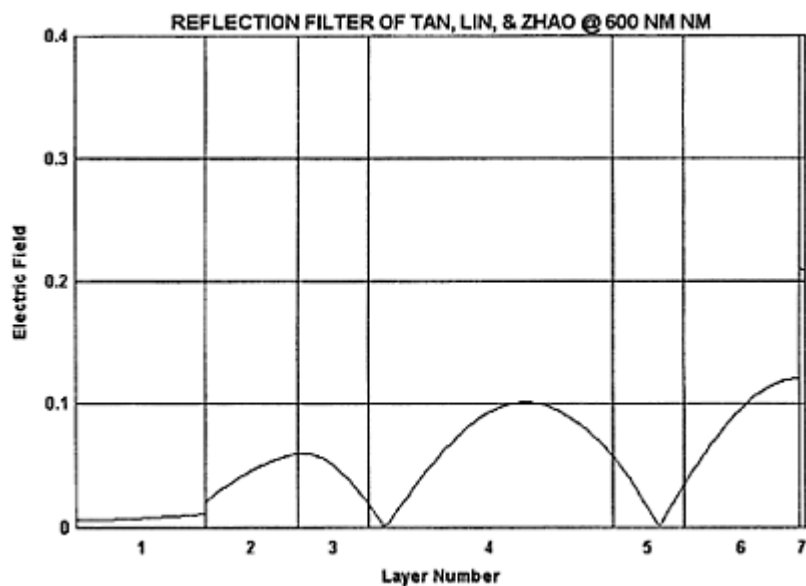


Fig. 1.62 Electric field in coating of Fig. 1.60 at 600 nm.

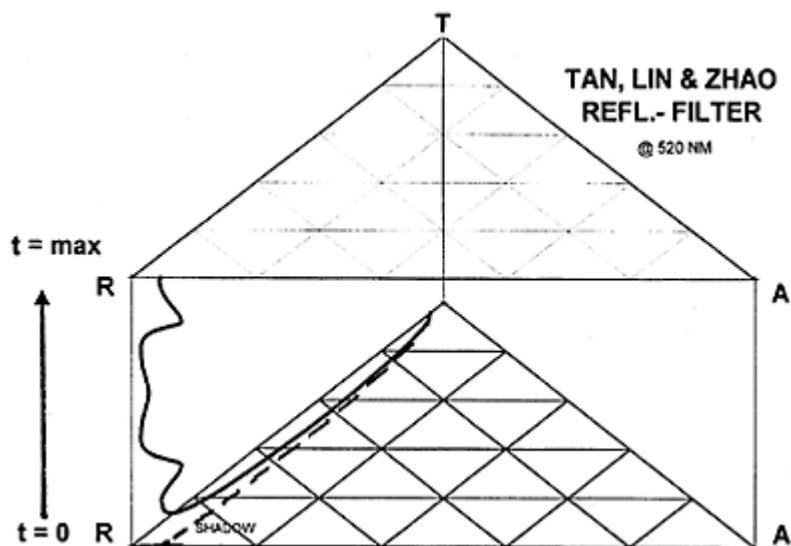


Fig. 1.63 Prism diagram for coating of Fig. 1.60 at 520 nm.

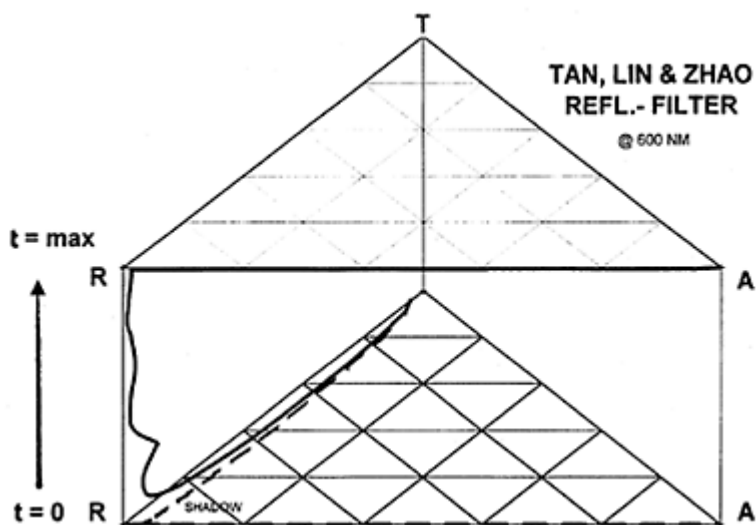


Fig. 1.64 Prism diagram for coating of Fig. 1.60 at 600 nm.

that the stack creates a spacing such that the electric field is near zero at the thin Cr layer for the passband wavelength as seen in Fig. 1.61. Under these conditions, the Cr layer absorbs very little. As the wavelength moves either way from this, the electric field increases rapidly and causes great A in the Cr layer as shown in Fig. 1.62 and thereby low reflection. Figures 1.63 and 1.64 show the PD for this filter in the passband at 520 nm and in the blockband at 600 nm. It can be seen in Fig. 1.60 for 520 nm that the coating ends at very high %R and low %A where there is no %T. In Fig. 1.64 for 600 nm, we see that A goes to 100% and thereby T and R are zero.

AR Coated Variable Neutral Density Filter Example

Neutral density (ND) filters can be made of a variety of metals deposited to a thickness that produces the desired optical density. Aluminum will transmit more in the blue than in the red. Chromium is more neutral than aluminum, but nickel is more neutral than chromium. An alloy of nickel and chromium, “nichrome”, is the most neutral of these in the visible spectrum.

Variable ND filters have been produced for decades with either a linear variation along a straight line or a circular filter which varies with rotational position. If two otherwise identical linear or circular filters are positioned with their gradients of density in opposing directions, a nominally uniform density over an area can be achieved which can be varied by moving the filters in opposing directions.

A problem can occur in such a case and also when two or more uniform ND filters are placed in sequence along and at near normal incidence to an optical path. The reflections from the second ND filter return to the first and are again reflected to the second, etc., etc. These “ghost” reflections can cause two problems: one is that the effect of two ND filters in sequence is no longer the sum of their optical densities, and the other is that multiple ghost images are formed in the transmitted beam. To eliminate these effects, it is necessary for the ND filters to be AR coated. The general nature and principles of such a coating would be the metal-dielectric-metal (MDM) stack shown in Fig. 1.40. However, this must be of different thicknesses for each different optical density. A few principles can be found operative in these cases.

One principle is that the substrate can have an AR for the spectral band of interest on it before the MDM is applied (it can even be on both sides). The AR might typically be a four layer broadband AR, or it could be a V-coat if only a narrow wavelength band was needed. The substrate then acts as if it were not there. This means that the MDM can be designed to be the same on both the substrate and air side of the main metal film or as though it were a self supporting and free standing film. Note that the result would then be MDMDM on top of the AR. The reflection amplitude from the top M layer essentially needs to be equal to the reflectance from the M layer below the D spacer which reaches the top and out of phase with it at the center of the AR band. The two reflections will then cancel each other and create the AR effect. The D spacer then can be the same thickness for all densities in the variable ND filter because it is only required to maintain the phase between the reflections from the top and next lower M layers. This is the second principle of interest. The D spacer could be either high or low index, but low index layers seem to give better design results. We believe this to be related to the fact that the low index locus on a circle diagram is more nearly concentric with the origin than a high index locus. From a circle diagram viewpoint, the first metal layer would move out from the origin

instead of the substrate index point as in Fig. 1.40. The D layer would travel in approximately a semicircle (QWOT) toward the origin as in Fig. 1.65. The second M layer would pass near the origin from lower right to upper left and terminate at a maximum reflectance amplitude. The second D layer would travel on another larger semicircle. The third M layer would bring the reflectance amplitude essentially to the origin again (the place of no reflection). The higher density filters would have longer metal lines that the lower density filters because this is what causes the absorption for the optical density.

Figure 1.66 shows the reflectance (upper curve) and the transmittance (lower curve) for an 2.0 OD filter (10% T) as shown in the circle diagram of Fig. 1.65.

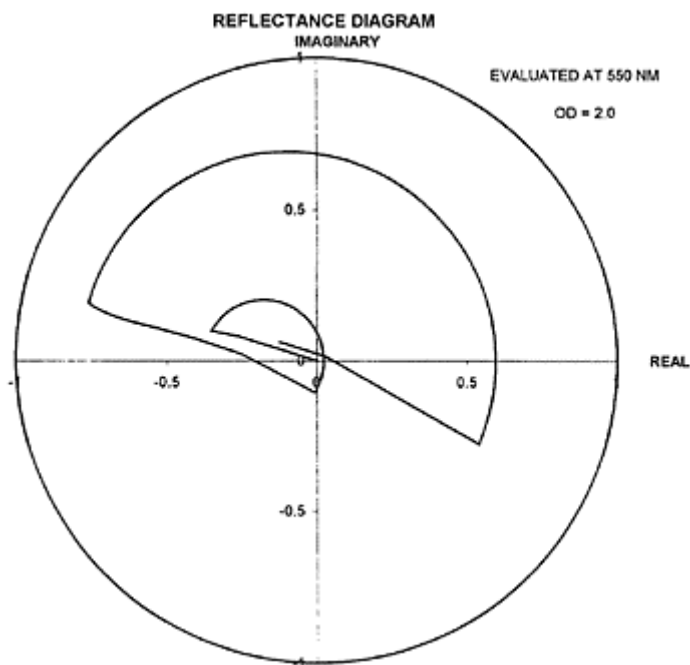


Fig. 1.65 Reflectance diagram of an antireflected neutral density filter at 2.0 OD.

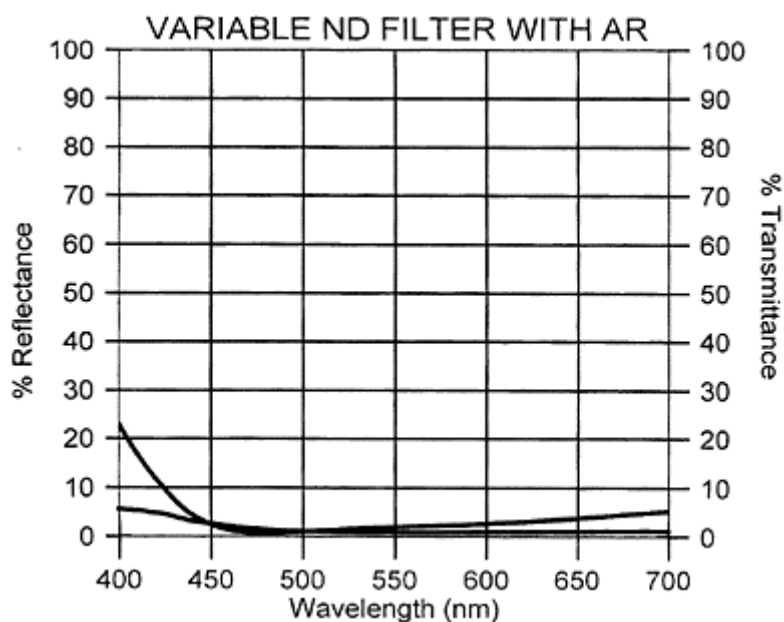


Fig. 1.66 Reflectance and transmittance spectrum of an antireflected filter at 2.0 OD.

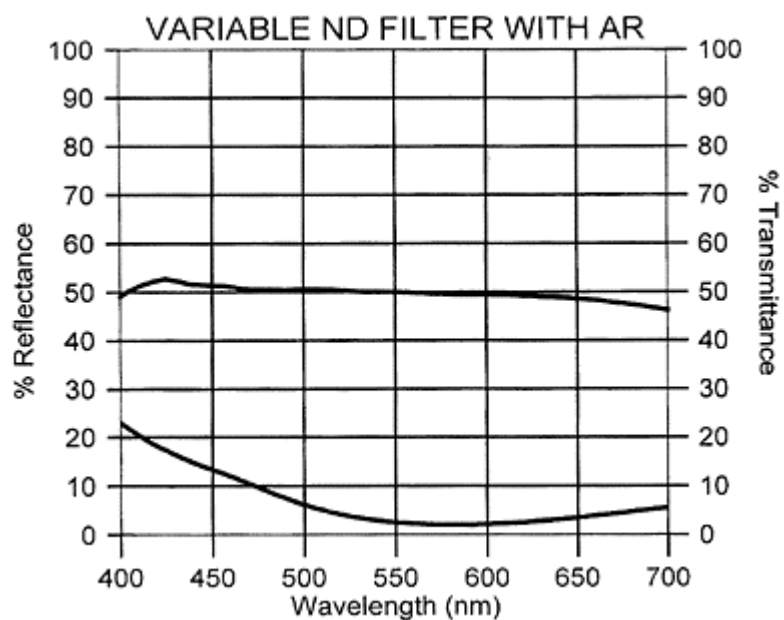


Fig. 1.67 Reflectance and transmittance spectrum of an antireflected filter at 0.3 OD.

Figure 1.67 shows the reflectance (upper curve) and the transmittance (lower curve) for an 0.3 OD filter (50% T). The reflectance of the AR at the blue end of the spectrum is higher than might be desired, but it is a reasonable compromise considering that the AR is only two layers (on each side of the metal). The properties of the metal also pass more of the blue in transmittance. However, this approach will generally overcome the problems of ghost images and non-additive optical densities of filters. Nickel instead of chromium might be a better choice, and nichrome might be even better yet. However, also recognize that the properties of the metal films deposited may not be exactly the same as those reported in the literature or used here.

Conclusions of Designing With Metals

The design of coatings with absorbing layers adds the potential to accomplish characteristics which are not possible with only dielectric materials. Some compromise is made with the inherent energy loss due to absorption in such coatings, but the effects of these limitations can be minimized and balanced. Several ways of visualizing the behavior of such coatings have been shown which include: reflectance diagrams (RD), triangle diagrams (TD), admittance diagrams (AD), and a new prism diagram (PD). These can aid in the understanding of certain types of coatings. We have added, in the Appendix, a selection of TDs and RDs for various materials of possible interest and usefulness that have not yet been shown here.

1.5.1.3 Additional Graphics for Visualization

It is also possible to make reflectance diagrams and prism diagrams, using a spreadsheet program, which are stereoscopic and show what is happening in three dimensions (3D). In some cases, this has been helpful to the author in gaining further visualization and insight concerning the behavior of some coating designs. Figure 1.68 shows a stereoscopic reflectance diagram (SRD) of a high-low stack which starts on glass and spirals clockwise toward the observer with increasing thickness.

Figure 1.69 shows a stereoscopic prism diagram (SPD) of the growth of an opaque film of a metal like Ni starting on a glass substrate.

For many decades, photogrammetric journals have published stereoscopic photo pairs which are printed with a separation less than the typical interpupillary distance of the average observer. Their subscribers have become comfortable with an ability to view the left image with the left eye and the right image with the right eye.

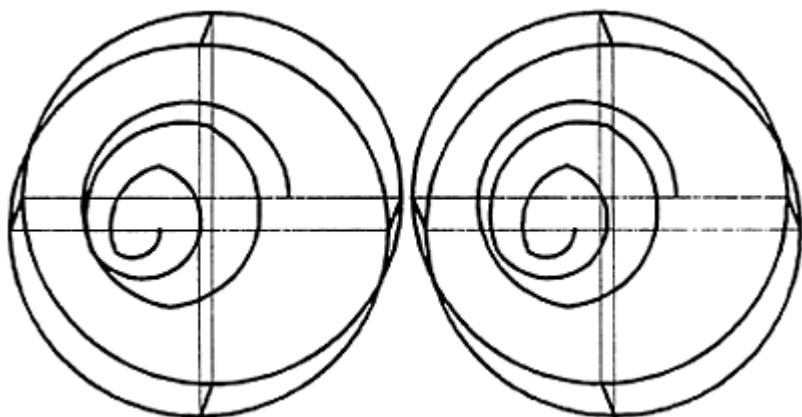


Fig. 1.68 Stereoscopic reflectance diagram (SRD) of a high-low stack which starts on glass and spirals clockwise toward the observer with increasing thickness.

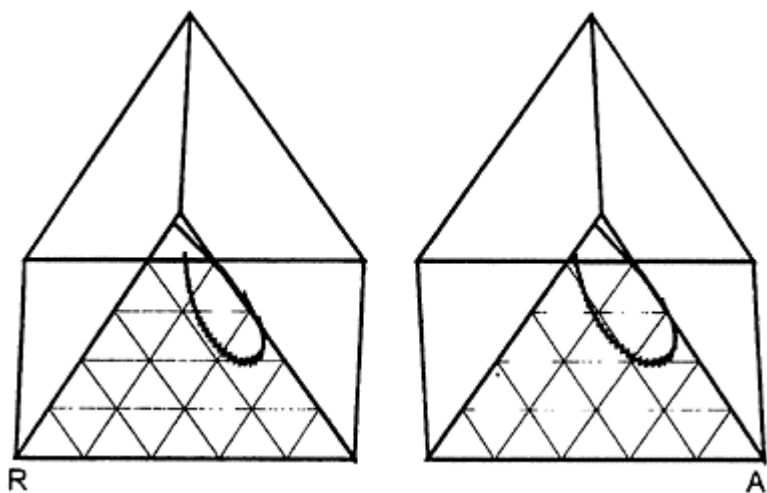


Fig. 1.69 Stereoscopic prism diagram (SPD) of the growth of an opaque film of a metal like Ni starting on a glass substrate.

If you have not done this before, try the following:

- 1) Position the stereo pair as close as possible in front of your eyes.

2) Slowly move the pair further in front of you until you are able to focus sharply on the images, and keep the line between the centers of the images parallel to the line between the pupils of your eyes.

3) With some practice, it is usually possible to use this technique to keep the left eye on the left image and the right eye on the right image.

4) The stereoscopic or 3D picture should then become apparent as the middle image in a set of three images!

1.6. APPROXIMATIONS OF INDICES AND DESIGNS

We will now touch on the concepts of Herpin equivalent index and Epstein periods. Herpin¹¹ showed that any film combination is equivalent at one wavelength and angle of incidence to a two film combination. Epstein¹² carried this further to show that a symmetrical combination of layers was equivalent to a single film at one wavelength and angle. We will illustrate those graphically and expand on the concept of approximations of the ideal when the ideal is not directly achievable. We described¹³ the possibilities of equivalent index approximations which we will review here for the convenience of the reader.

We discussed above the case of approximating the desired solution shown in Fig. 1.11 and 1.36 by the solution in Fig. 1.35 with only two available real materials. Rabinovitch et al.¹⁴ and others have pointed out that these approximations are only fully equivalent at one wavelength and one angle of incidence. There are various ways to approximate the layer with a combination of two indices that are higher and lower than the index to be simulated. Figure 1.70 shows the principle on the first layer of a three-layer design in graphic form. When a QWOT of medium index 1.65 is deposited on a substrate of index 1.52, the admittance curve follows a semicircular path M from point A to Z in the figure. Any set of layers which bring the admittance from point A to point Z will have the same performance as M at this one wavelength and angle of incidence, but not necessarily at others. The Epstein period is defined as having a central layer of high or low index and two equal and symmetric outer layers of the opposite index.

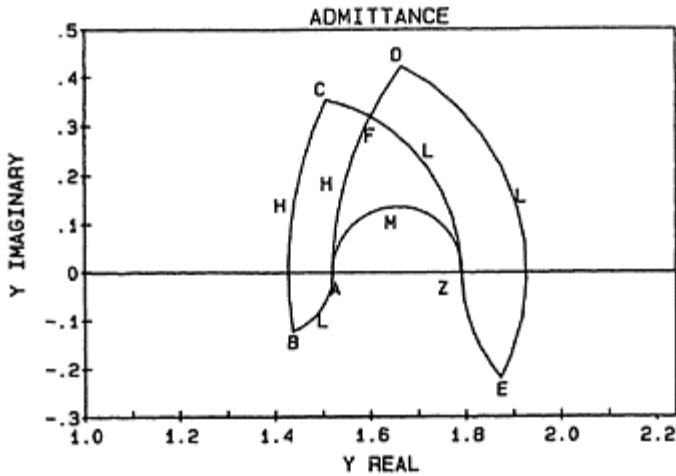


Fig. 1.70 Admittance of a QWOT of index 1.65 (curve M from A to Z), Epstein LHL period from ABCZ, Epstein HLH from ADEZ, and Rock 2-layer Herpin HL from AFZ.

For any given available high and low index materials, there are two solutions to the Epstein period for a given case. These are shown in Fig. 1.19 by the path ABCZ for the LHL period and by ADEZ for the HLH period. We have already mentioned the two layer solution of Rock⁵ which is represented by the path AFZ and seen in Fig. 1.35. Herrmann¹⁵ discussed a non-symmetric three-layer solution which was an improvement on the Epstein periods over a broader wavelength region. We showed¹³ that the closer the admittance path of the simulation approaches the path of the system to be approximated, the more nearly the same will be the performance of the simulation to the design approximated at all wavelengths and angles, as one would justifiably expect. This is a key point to keep in mind for later sections of this chapter.

By dividing the simulated layer into an arbitrary number of thin layers, the resulting admittance locus can be made to follow the admittance curve to any arbitrary degree of closeness. Figure 1.71 illustrates a 20 layer approximation of the medium 1.65 index first layer. Figures 1.72 and 1.73 show the differences in performance between the approximated design and the various approximations over a broad wavelength range at an angle of incidence of 45 degrees. The simulations were designed to match the 1.65 index at 524nm and normal incidence. It can be observed that the above key point is well demonstrated. Figures 1.74 and 1.75 illustrate how the admittance diagrams change from Fig. 1.70 for the *s* and *p* polarizations for each of the approximations at 45 degrees angle of incidence. The Herpin and Epstein concepts are valuable tools as long as their limitations are kept in mind. The concept of the optimum degree of approximation is what we want to emphasize here, however.

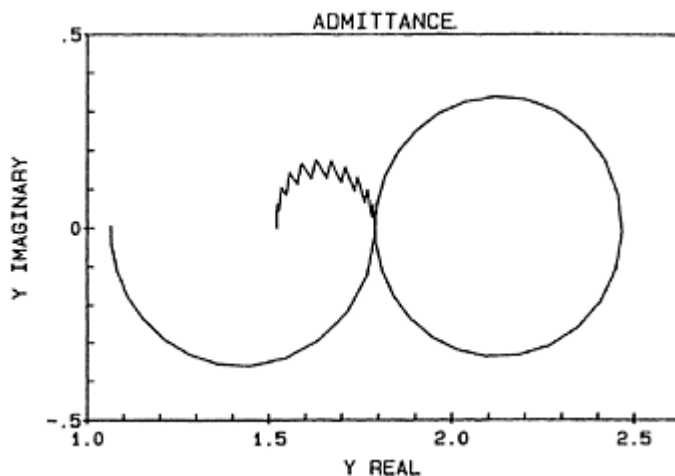


Fig. 1.71 Admittance diagram at 524 nm and normal incidence of the three-layer AR with the 1.65 index layer replaced by and optimized 20-layer Herpin equivalent.

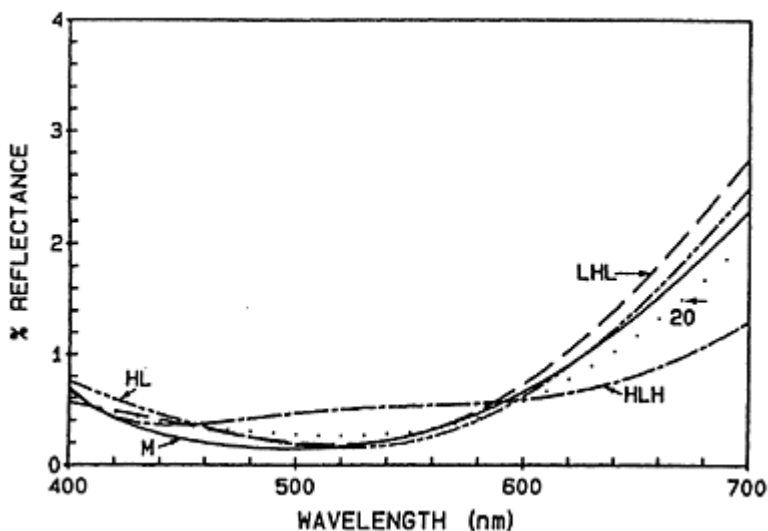


Fig. 1.72 Reflectance at 45° in *p*-polarization of the three-layer AR and the various approximations of Fig. 1.70.

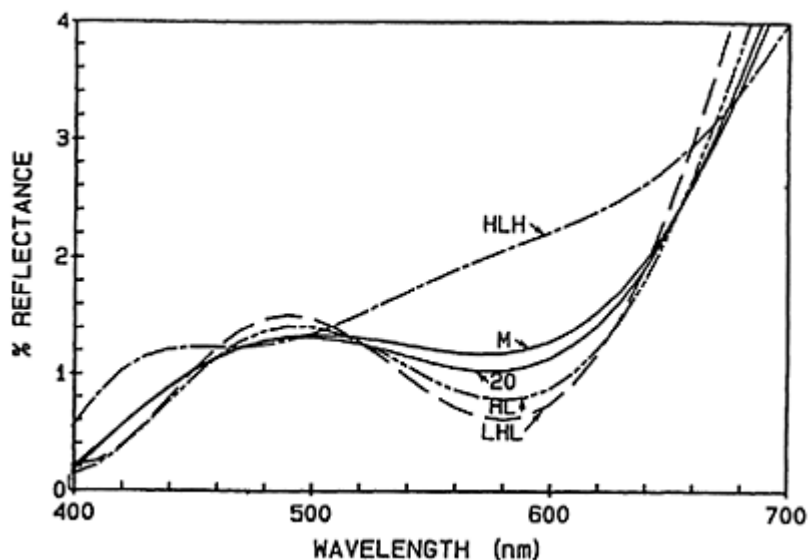


Fig. 1.73 Reflectance at 45° in s -polarization of the three-layer AR and the various approximations of Fig. 1.70.

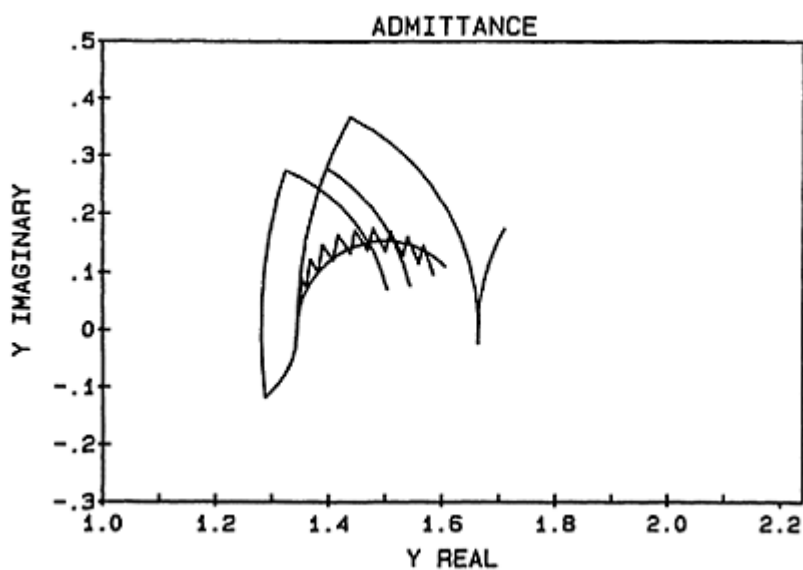


Fig. 1.74 Admittance diagram at 600 nm and 45° incidence in s -polarization

first layer of the three-layer AR and its approximations in Fig. 1.70.

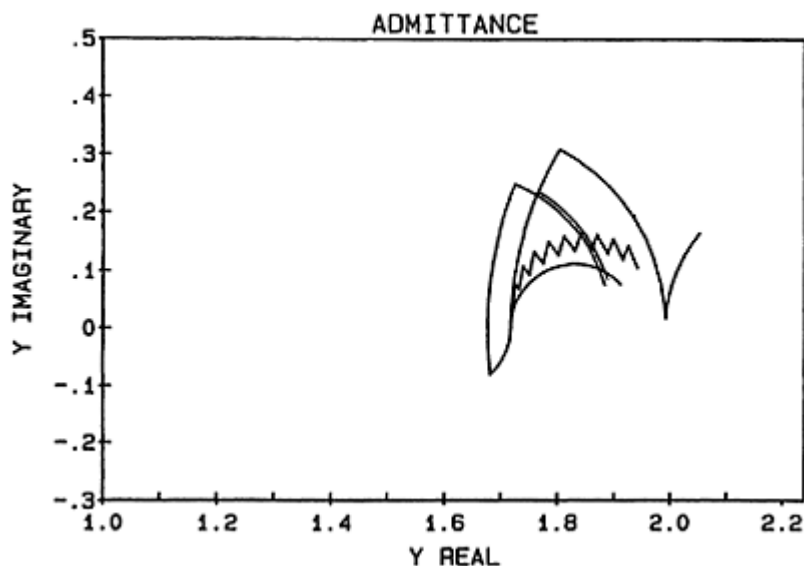


Fig. 1.75 Admittance diagram at 600 nm and 45° incidence in p -polarization first layer of the three-layer AR and its approximations in Fig. 1.70.

1.7. INHOMOGENEOUS INDEX FUNCTIONS

The question of what are the underlying principles and “ideal” design for the ultimate broad band AR coating has intrigued us for almost a decade. The question of how to design it, and how to fabricate it has followed us through this period until recent years when the probable answers seem to have finally come together.

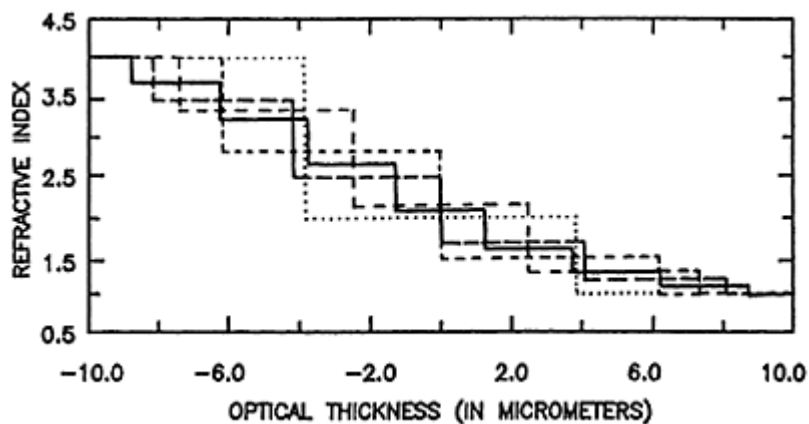


Fig. 1.76 Index profile of step-down coatings of 1, 2, 3, 4, and 7 layers from index 4.0 to 1.0, all giving the same long wavelength limit to the AR band.

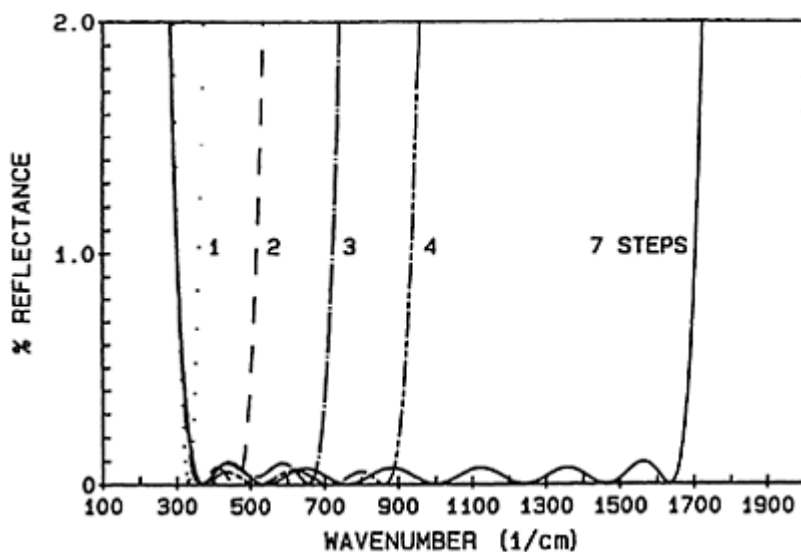


Fig. 1.77 Spectral reflectance of the step-down ARs in Fig. 1.76.

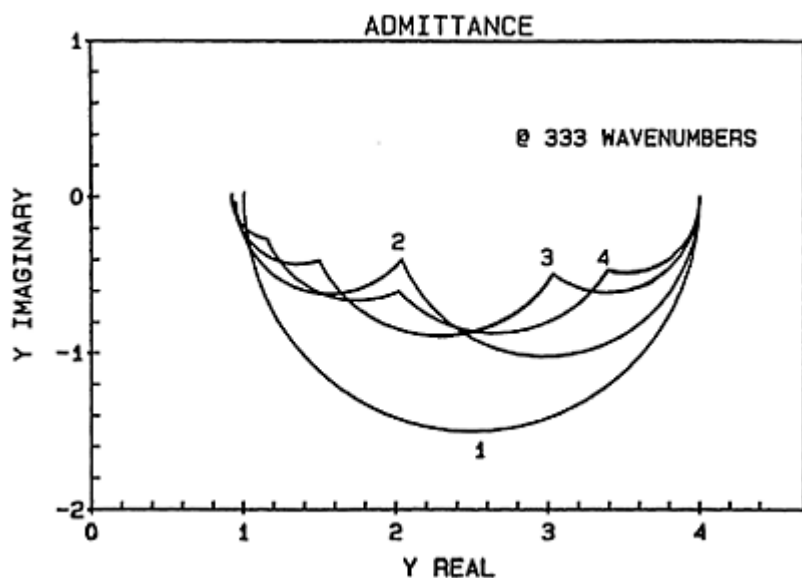


Fig. 1.78 Admittance diagram of 1-, 2-, 3-, and 4-layer step-down ARs in Figs. 1.76 and 1.77 at 333 wavenumbers or 30 micrometers.

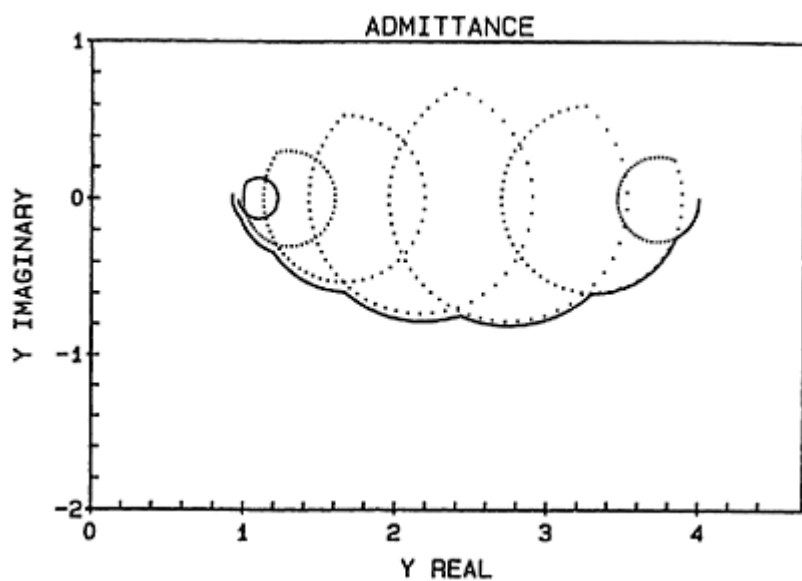


Fig. 1.79 Admittance diagram of the seven-layer step-down ARs in Figs. 1.76 and 1.77 at 333 and 1667 wavenumbers or 6 micrometers.

Jacobsson et al.¹⁶ and Dobrowolski et al.¹⁷ described the merits of step-down layers where the space between the substrate and the ambient medium is divided into approximately equal steps of decreasing index of refraction from the substrate to the medium. The author investigated this in terms of admittance diagrams and reported some of the results in 1989¹⁸. Figure 1.76 illustrates such a step-down layer from germanium to air with 1, 2, 3, 4, and 7 steps. Note that the overall thicknesses have been adjusted to give the same long wavelength or low frequency limit at 300 1/cm (for 1% reflectance). We will discuss this thickness difference later. Note also that the origin of zero thickness is near the center of Fig. 1.76 and later plots. This causes the profiles in Fig. 1.76 to be more clearly seen as approximations of the ideal shown later in Fig. 1.85. The choice of this origin is also a characteristic of the Fourier transform technique in Sect. 1.7.2. Figure 1.77 shows the performance of each profile in Fig. 1.76. We can observe that the AR band gets wider with an increasing number of steps as the short wave limit (high frequency) moves to the right while the long wave limit is fixed. Figure 1.78 shows the admittance diagrams resulting for the 1-, 2-, 3-, and 4-layer versions at the long wavelength end of the AR band. Figure 1.79 shows the admittance diagram of the seven layer at the longest and shortest wavelengths in the AR band. This figure was something of a Rosetta Stone to deciphering what might be the basics of the AR coating. At the long wave limit of the AR band, the admittance of the “ideal” AR appears to be of an approximately catenary form which seems to osculate the admittance locus of the short wave or high frequency end of the AR band. The admittance locus of the shortwave end has a coiled spring appearance. As the number of layers or steps is increased, the loci become more smooth and the AR bandwidth becomes progressively wider. Figure 1.80 shows this for 24 layers at the long wave end of the band. Figure 1.81 shows the admittance at the high frequency end of the band, and it seems to fit “inside” the admittance of the low frequency end (Fig. 1.80). We shall return to this concept later.

A coincident study of the paper by DeBell¹⁹ and a plot of the index versus thickness of some of his solutions led to Figs. 1.82, 1.83, and 1.84. DeBell looked at the possibilities of inserting additional half waves of alternating high and low index layers between the quarter wave start and end layers of the classical QHQ in order to improve the bandwidth and minimize the reflectance of broadband ARs. He optimized the indices of each layer while holding the thicknesses constant. The index profiles of some of his solutions are shown as broken lines in Figs. 1.82, 1.83, and 1.84. Looking at the index profiles from DeBeirsworckwith homogeneous layers we noticed that these might be taken as approximations of smooth periodic variations of index with thickness, as shown in solid lines. Southwell²⁰ stated, “Any arbitrary generalized gradient-index interference coating (including homogeneous and inhomogeneous layers) possesses a digital configuration (sequence of thin high- or low-index layers), which is spectrally

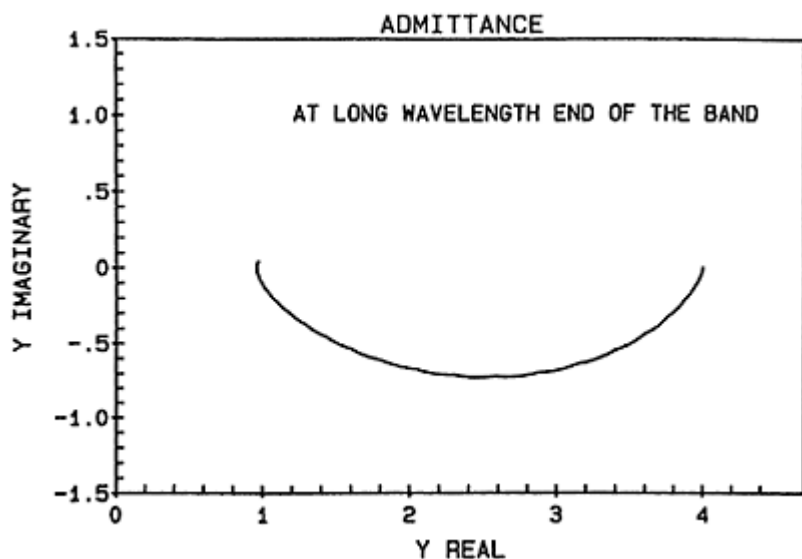


Fig. 1.80 Admittance diagram of 24-layer step-down, near “ideal” AR at the longest wavelength of the AR band.

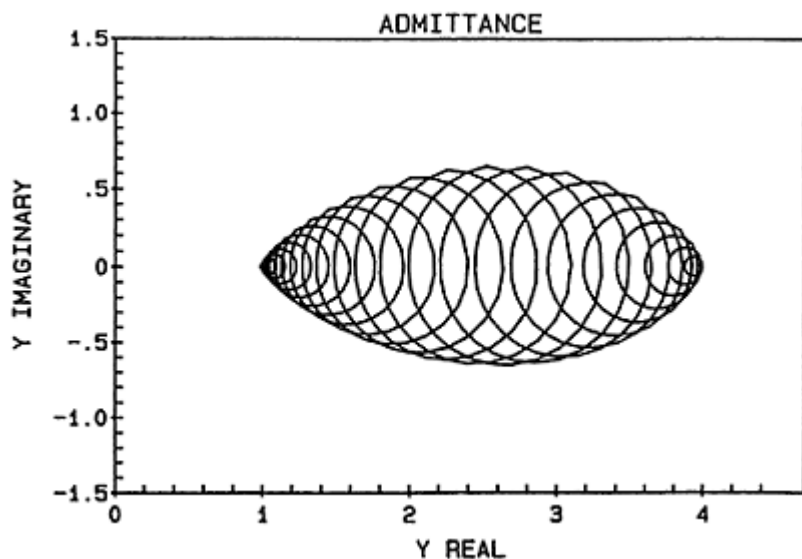


Fig. 1.81 Admittance diagram of 24-layer step-down, near “ideal” AR at the shortest wavelength of the AR band.

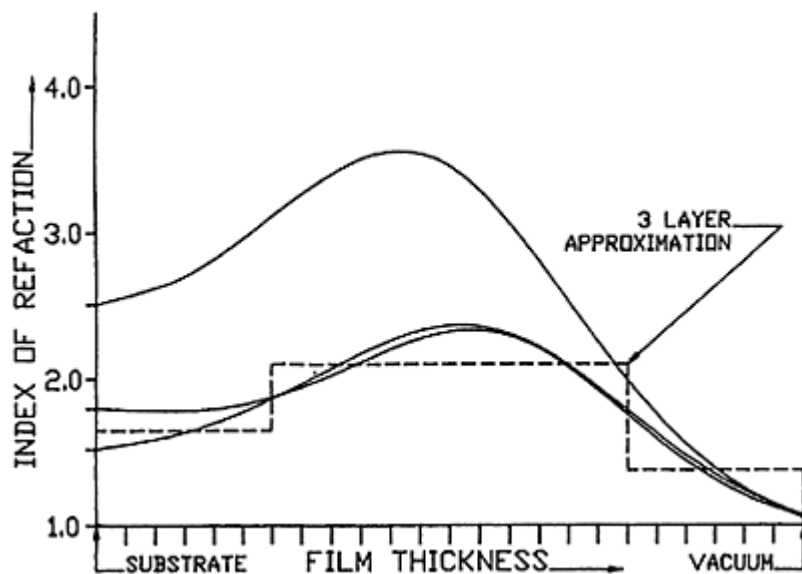


Fig. 1.82 DeBell's QHQ homogeneous index three-layer AR (dashed line) and one-cycle inhomogeneous index functions for three different index substrates.

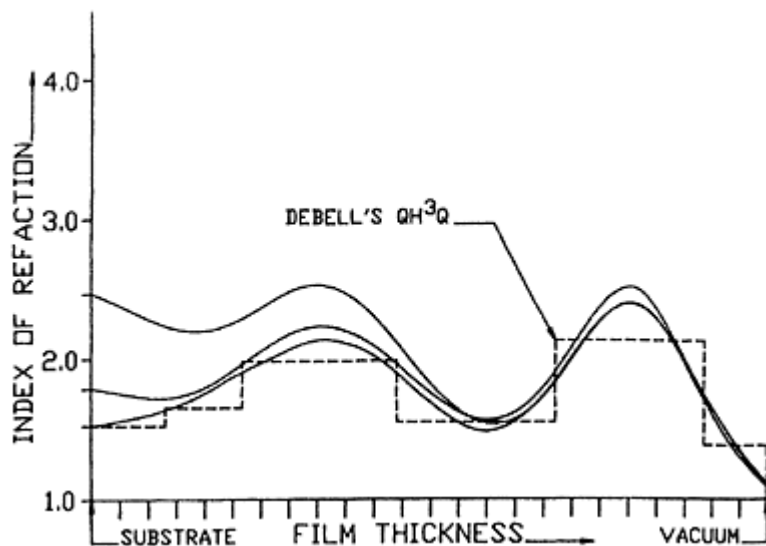


Fig. 1.83 DeBell's QHHHQ
homogeneous index five-layer AR
(dashed line) and two-cycle
inhomogeneous index functions for
three different index substrates.

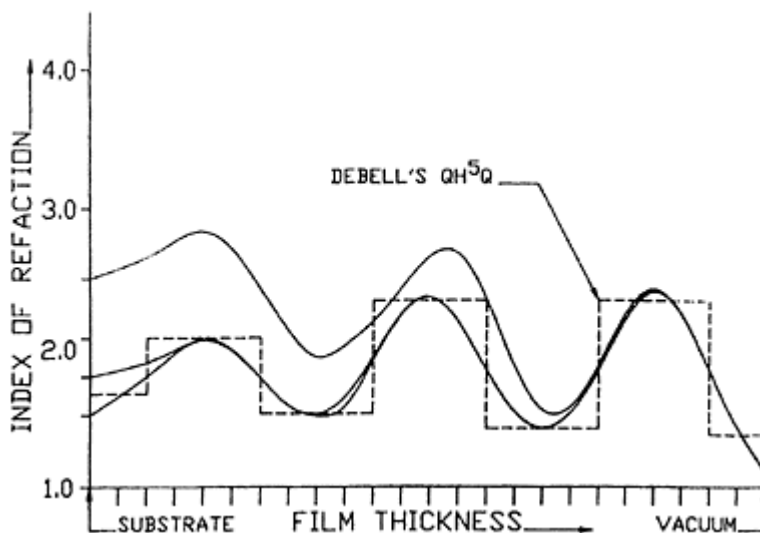


Fig. 1.84 DeBell's QHHHHHQ
homogeneous index seven-layer AR
(dashed line) and three-cycle
inhomogeneous index functions for
three different index substrates.

equivalent at all wavelengths. Such digital configurations are found directly from arbitrary index profiles by using a prescribed two-layer high-low equivalent to a thin layer of arbitrary index." Using this basic principle with some modification, we divided the overall thickness of our quasi-periodic functions into 24 layers of equal thickness and optimized the index of each layer with respect to the broadest AR band practical. This included the "step-down" of Fig. 1.76. In that case, the smooth curve in Fig. 1.85 resulted. We call this the zeroth order or 1/2 cycle function. Its admittance is seen in Curve A of Fig. 1.86 when applied to a 1.52 index substrate. The normal SLAR coating of magnesium fluoride on glass would be as in Curve C, and a homogeneous layer of index 1.233 (the ideal value for a SLAR coating on glass) would produce Curve B.

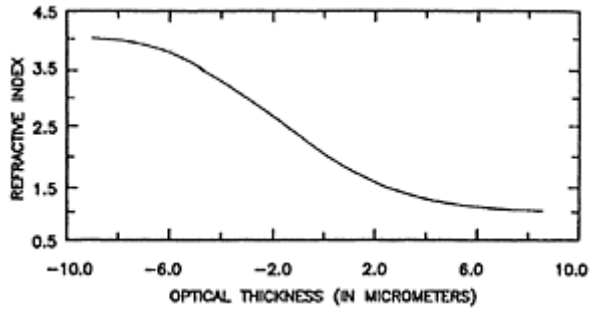


Fig. 1.85 Half cycle inhomogeneous index function from Fig. 1.80 and also generated by the Fourier method.

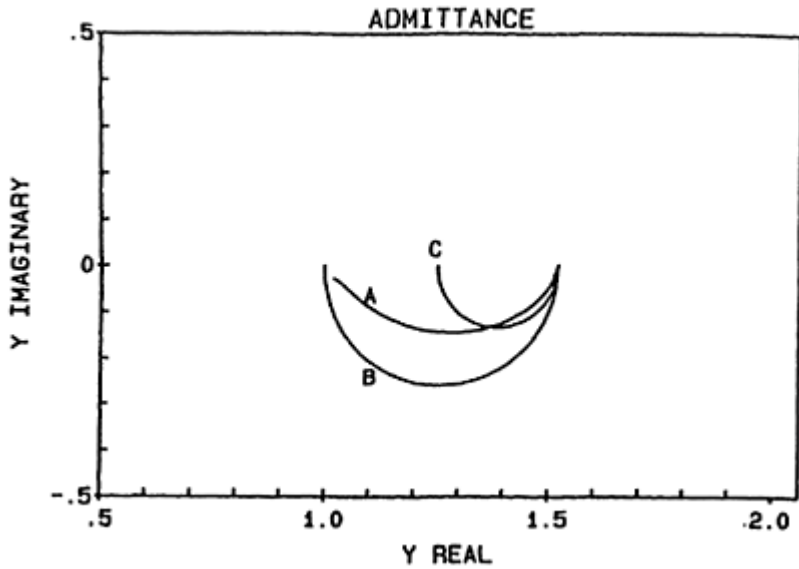


Fig. 1.86 Admittance: curve A is for the “ideal” inhomogeneous function on 1.52 index substrate, curve B is a homogeneous SLAR of index 1.233, and curve C is a homogeneous SLAR of index 1.38.

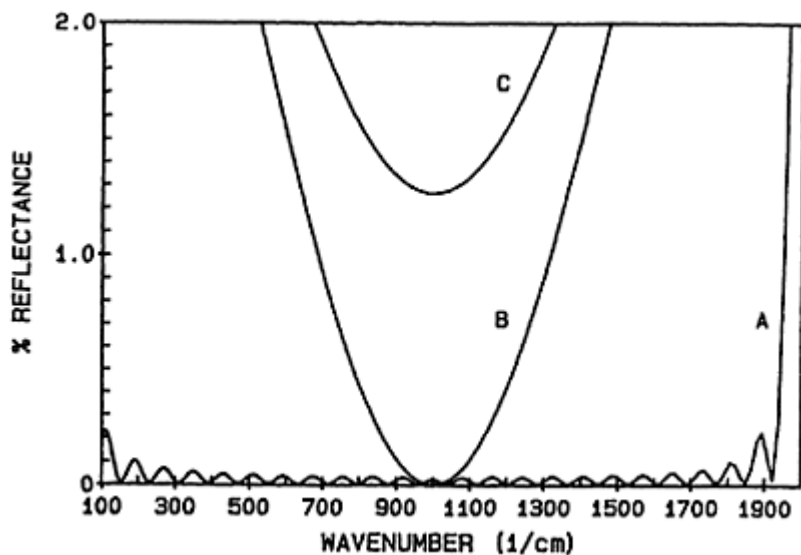


Fig. 1.87 Spectral reflectance of the three coatings in Fig. 1.86.

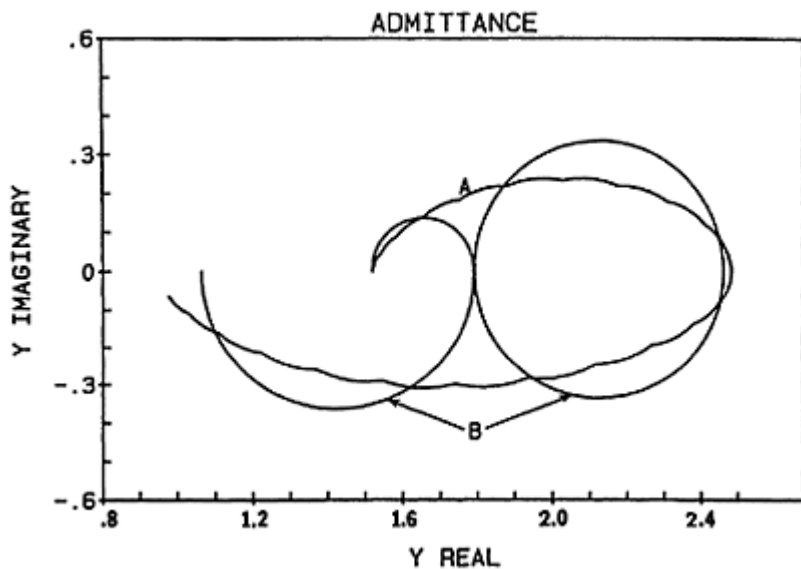


Fig. 1.88 Admittance diagram of the classic three-layer homogeneous layer AR (curve B) compared to the “ideal”

one-cycle inhomogeneous function of
Fig. 1.82.

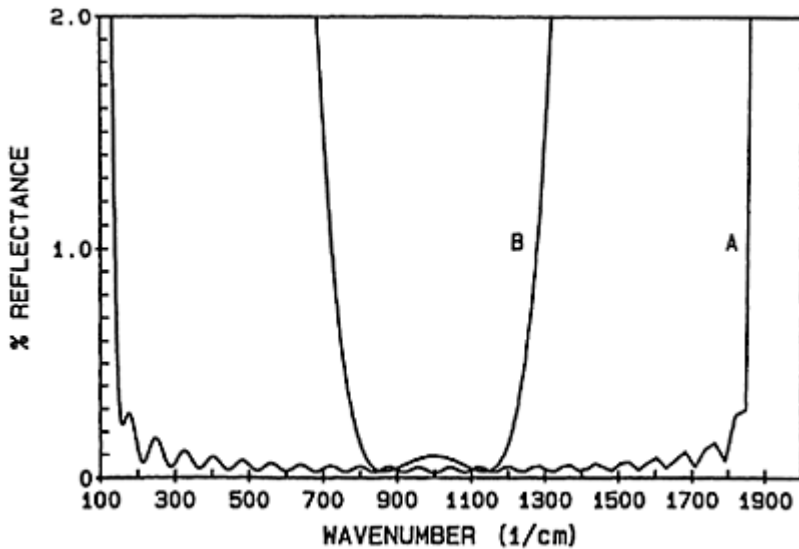


Fig. 1.89 Spectral reflectance of the two coatings in Fig. 1.88.

Fig. 1.87 shows the spectral performance calculated for these single layer approximations of the “ideal” inhomogeneous $1/2$ cycle index profile function. The ripples in Curve A and the rise in reflection at the short wave or high frequency end are due to the limitations of 24 layers. The rise occurs when the individual layers approach $1/2$ wave optical thickness, and thus become ineffective absentee layers. If we used twice as many layers in the same overall stack thickness, the bandwidth would be twice as wide. This is because the frequency (wavenumber) at which each layer becomes a half wave would be twice as high because the layers would be half as thick. The low frequency (long wavelength) end of the band would remain the same because the overall thickness would be unchanged. Note by comparison of Figs. 1.86 and 1.87 that the closer Curves B and C approximate the admittance of the “ideal” Curve A, the better the result match the “ideal”. This was a major point made in Sect. 1.6.

We saw earlier that changes of the angle of incidence from normal will cause the spectral properties to generally shift toward shorter wavelengths or higher frequencies. If a broadband AR coating is needed which will function at increasing angles of incidence, the AR band at normal incidence needs to be “flat” to longer wavelengths (or lower frequencies). This will then allow the coating to be tilted to some greater angle before the long wavelength end of the band encroaches on the AR band of interest. The basic principle for the design of a broadband AR coating to handle large angles is, therefore, to approach a design like that of Fig. 1.85 and curve A of Figs. 1.86 and 1.87. Here the long

wavelength end of the AR band must be sufficiently longer than the band of interest at angle.

The performance of any of these “ideal” ARs is ultimately limited by the lowest real index of refraction available as we will discuss in more detail in Sects. 1.7.1, 2.2.3, and 3.2.4.1. The current limits on the indices of real homogeneous materials is approximately the 1.38 of MgF_2 . However, a new possibility has appeared, at least as a laboratory curiosity. The photolithography techniques have etched and produced “moth eye” surfaces which have a microscopic texture like “pyramids” which taper from contiguous solid bases to “sharp” points. This then approximates the change in effective index from the substrate to air/vacuum as seen in Fig. 1.85. The index of 1.0 is approximated to the degree that the pyramids come to sharp points. It will also be noted that the pyramids cannot be flat sided, but must curve more like a bullet shape in order to obtain the correct density as a function of thickness from the substrate.

At the time of this writing, we had attempted to contract to have an actual 21 to 67° beamsplitter “coated” in this way for a real application (which cannot currently be done with ordinary coating technology). However, it appears that the technology is not economically viable at this time, and the communications and semiconductor industries offer more commercial interest for related technologies.

Back to the earlier discussions, the techniques used for Figs. 1.80 and 1.81 were applied to the QHQ solution by DeBell for different substrate indices generated the one-cycle function curves in Fig. 1.82. The dotted line is the homogeneous index profile for the classical QHQ broad band AR on a 1.52 substrate previously seen in Figs. 1.11 and 1.36. Figure 1.88 (Curve A) shows the admittance loci (24 layer approximation) for the inhomogeneous profile on a 1.52 index substrate of Fig. 1.82 and the homogeneous (Curve B) solution. Figure 1.89 illustrates the resulting spectral performance of the two. Note in the case of Figs. 1.88 and 1.89 that the three homogeneous layers are an approximation of the “ideal” one cycle function.

Carrying these concepts further to multiple periods, after DeBell’s QHHHQ and QHHHHHQ as seen in Figs. 1.83 and 1.84, led us to the reduction of these to similar approximations with thin homogeneous layers^{21–23} according to the technique of Southwell²⁰.

The term “rugate” is defined in *The American College Dictionary* as: “*adj.* wrinkled; rugose.” Rugose is defined as: “*adj.* having wrinkles; wrinkled, ridged.” The term has come into use in the optical coating community in the last two decades. In this case, it generally is referring to inhomogeneous film structures of which Figs. 1.82 to 1.85 are related examples of “Rugate ARs.” The more common references to rugate filters, however, are to structures where there are many cycles of index variation with thickness that have been designed to reflect selected bands much like minus filters. The FBG is also an example of a rugate filter. We will touch on this more in subsequent chapters.

1.7.1. Low Index Limitations

There is a serious limitation placed by indices of real and available materials on what can be achieved by an AR coating over a broad band. This author²², Aguilera et al.²⁴, and others have shown that for a given bandwidth, the lowest achievable average reflectance of an AR coating over the low reflectance band is limited by the lowest index available.

As we can see in Fig. 1.76, the broader step-down solutions require very low (and unavailable) indices. If 1.35 or 1.45 or such is the lowest index available, the discontinuity from the ideal index profile causes a ripple perturbation in the resulting AR band. There is also another factor in the case of approximating an “ideal” design with a limited number of layers. If the band is made wider when optimizing a given homogeneous layer design, the residual reflectance in the AR band goes higher as seen in Fig. 1.90. Adding more cycles to the ideal inhomogeneous function does not significantly improve the achievable bandwidth. This observation raises the question, “What advantage would a multiple period function have?” It was shown that the higher order “ideal” inhomogeneous index functions tend to spiral into the desired admittance

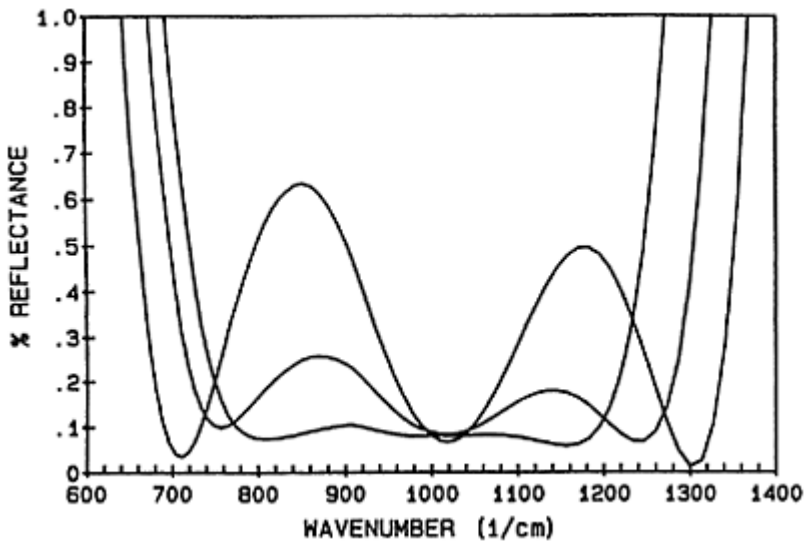


Fig. 1.90 Change in the minimum average reflectance of an AR as the bandwidth of a given homogeneous layer design is changed and the design is reoptimized.

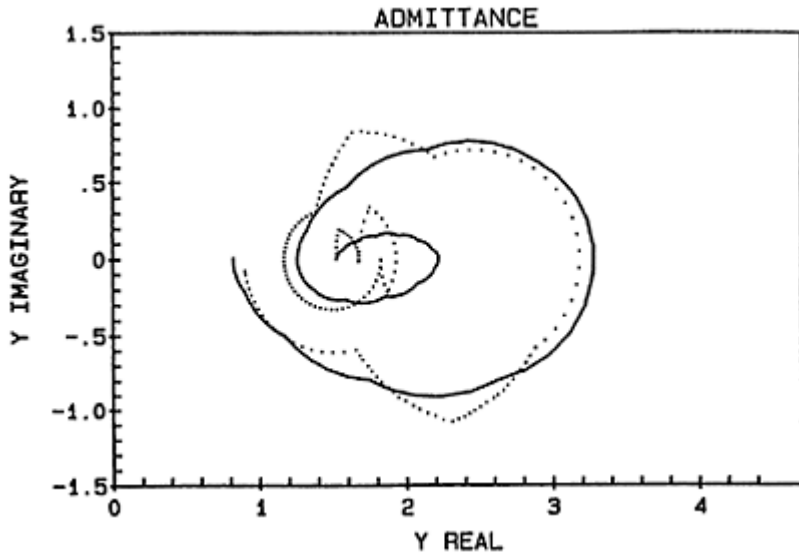


Fig. 1.91 Admittance of the two-cycle inhomogeneous index function (solid curve) and a 12-layer approximation using 2.35, 1.46, and 1.38 index homogeneous layers (dotted line).

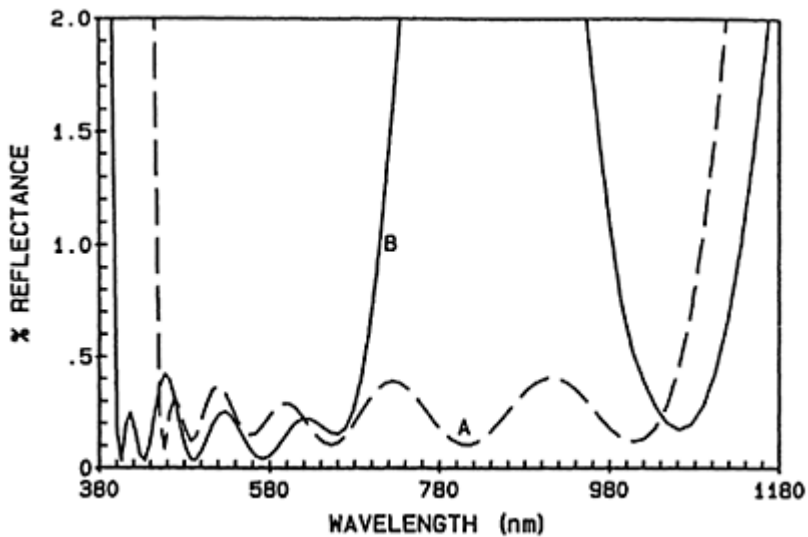


Fig. 1.92 Spectral reflectance of the 12-layer design in Fig. 1.91 (dashed

curve) and an 11-layer design optimized for 400 to 700 and 1064 nm.

of 1.0 at a steeper slope. For example, the solid line in Fig. 1.91 corresponds to the 2 cycle function of Fig. 1.83 and is to be compared to Curve A in Fig. 1.88 for the one cycle function. This makes it easier to approximate the low index termination of the AR layer with a real index. This was demonstrated in Refs. 1.22 and 1.23 on a practical BBAR coating from 400 to 1064 nm. Here the inhomogeneous admittance profile was approximated by alternating layers of 2.35 and 1.46 index except for the last layer of 1.38 index (because we already know of the desirability of having the lowest index possible in the last layer). The thicknesses of the homogeneous layers were then optimized to minimize the reflectance in the desired wavelength band. The result of this 12-layer approximation of the “ideal” 2-cycle AR using titanium dioxide, silicon dioxide, and magnesium fluoride is seen as Curve A in Fig. 1.92 and the dotted line in Fig. 1.91. This basic design was then modified to give the best practical AR coating from 400 to 700 nm and at 1064 nm using titanium dioxide and magnesium fluoride. The realization was done with an approximation of only 11 layers. The design result is shown as Curve B in Fig. 1.92. This will be discussed in more detail later in Sec. 7.7.3.

1.7.2. A Fourier Approach

Observing the periodic nature of DeBell’s multiple half-wave solutions, as seen in Figs. 1.82 to 1.84, and the subsequent discovery of a family of inhomogeneous index functions for AR coatings were the trigger for the following idea. The Fourier synthesis techniques described by Dobrowolski et al.²⁵ might lead to these types of periodic inhomogeneous solutions and lend further insight into these matters. The Fourier technique transforms the desired reflectance/transmittance function of frequency (wavenumber) to an inhomogeneous (continuous) index of refraction profile as a function of thickness which will generate a similar R/T function. Although this currently has some limitations, it has the advantage of being a direct synthesis technique requiring no prior knowledge of what the index profile might look like. A collaborative work²⁶ added to the understanding of the possibilities and limitations, as follows.

Figure 1.85 also represents the “ideal” AR coating which resulted from our studies. Such a smooth function generates an AR range for all frequencies higher than the low frequency (long wavelength) limit. The long wavelength limit is directly proportional to the overall optical thickness of the transition profile, and it is twice that length. That is to say that the profiles of Figs. 1.76 and 1.85 with an optical thickness of 17.5 micrometers provide an AR band starting at about 35 micrometers or 286 1/cm. Note two things about Fig. 1.76 with respect to Fig. 1.85. First all of the homogeneous layers approximate the smooth inhomogeneous function as closely as possible. Second, the effective width of all the profiles is the same, even though the single layer approximation is only 7.6 micrometers in optical thickness. In combination with a portion of the substrate and the medium, it is the best single layer approximation of this “ideal” inhomogeneous function for the 35 micrometer AR band.

The Fourier point of view explains the high frequency limits seen in Fig. 1.77 as follows. The steps in the homogeneous layer approximations are high frequency

perturbations in the smooth low frequency “ideal” profile. These “ripples” in the profile cause a reflection peak at a corresponding high frequency point in the spectral reflectance curve, and thus we get the high frequency limit of the AR band. Note that, to avoid confusion in Fig.1.77, the higher order AR bands which occur at $3x$, $5x$, etc. of the center band frequency are omitted.

Verly²⁶ tried to achieve a long wavelength limit for an AR coating with less than the minimum thickness indicated above. Figure 1.93 shows such a solution compared to a three layer step-down. Inference yields the “ideal” profile in this case to be in excess of 8 micrometers, whereas the “too-thin” profile is only about 6.5 micrometers thick. The figure indicates that the “thin layer” solution has approximated the longer smooth profile by adding the spikes on the ends of the layer. These make the average effect over a few micrometers on each end similar

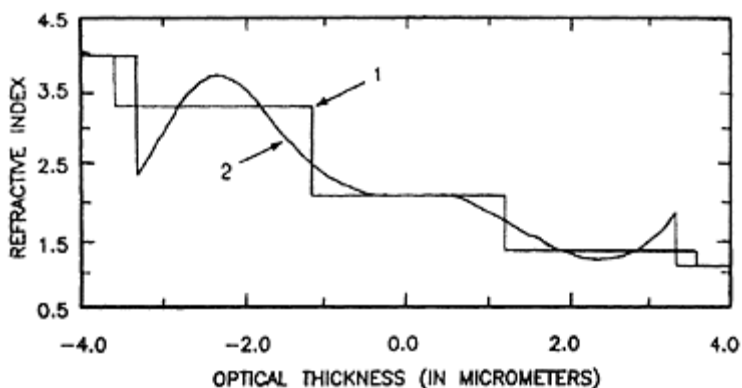


Fig. 1.93 Index profile of three homogeneous layer step-down (curve 1) and a profile of a “too thin” inhomogeneous layer by the Fourier method (curve 2).

to the “ideal”. Figure 1.94 and 1.95 show the admittance at the long and short wavelength ends of the AR band for the three layer and the Fourier “too-thin” solutions. The striking similarity of these figures is consistent with all of the foregoing discussion.

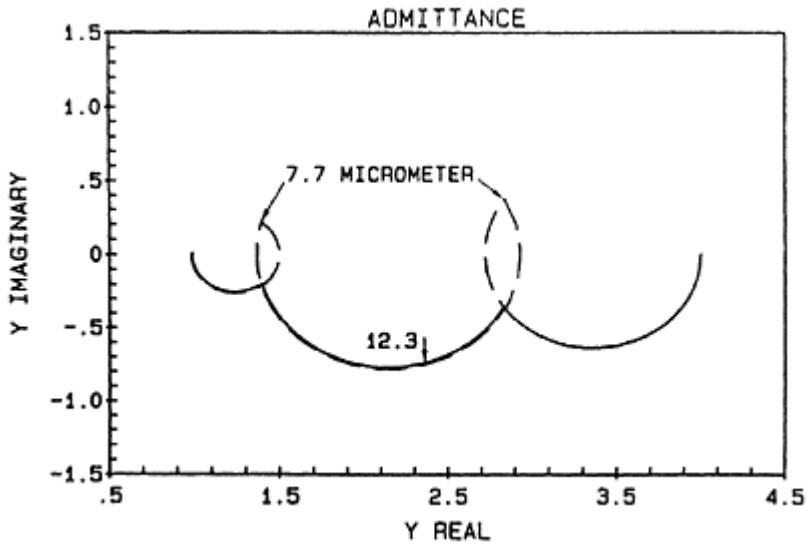


Fig. 1.94 Admittance diagram of the homogeneous layer design of curve 1 in Fig. 1.93 at the longest and shortest wavelengths in the AR band.

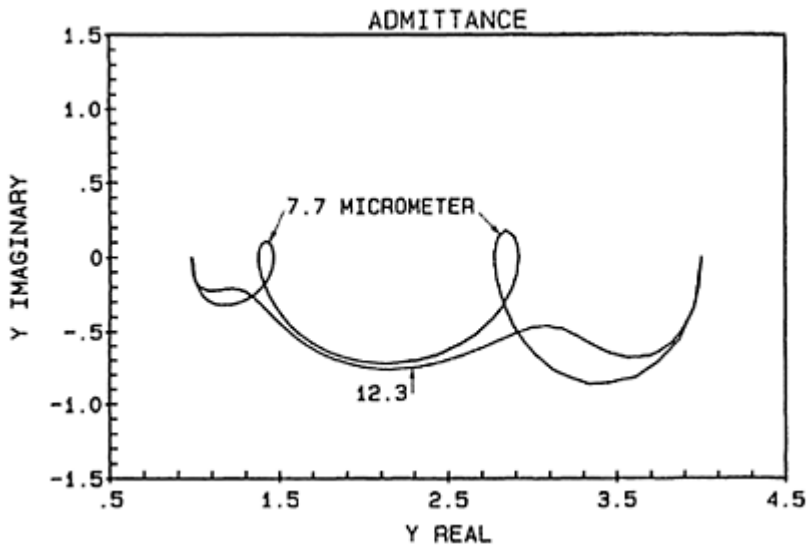


Fig. 1.95 Admittance diagram of the inhomogeneous index profile of curve

2 in Fig. 1.93 at the longest and shortest wavelengths in the AR band.

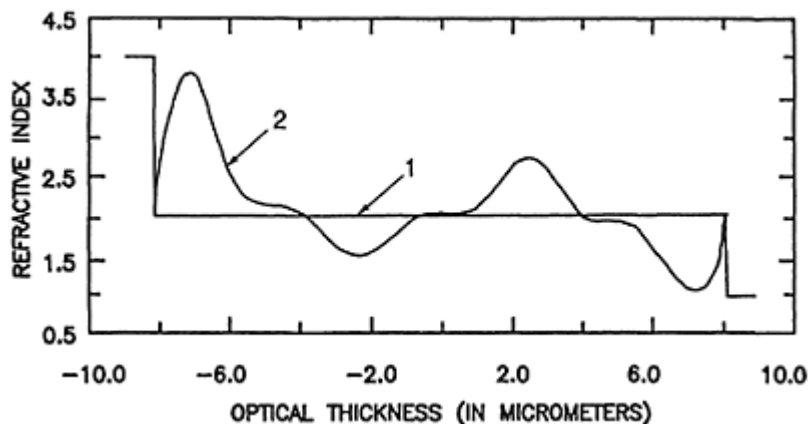


Fig. 1.96 Index profile of a “too-thick” inhomogeneous AR design by the Fourier method (curve 2) and its starting design (curve 1).

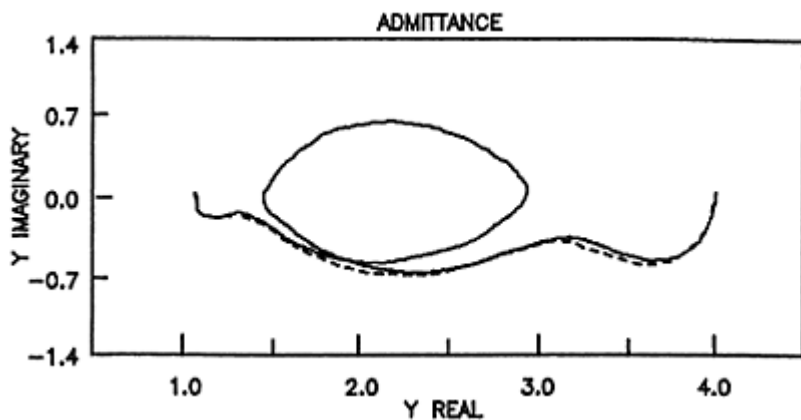


Fig. 1.97 Admittance of the design in Fig. 1.96 (solid curve) and the design of Fig. 1.95 (dashed curve) at the longest wavelength in the AR band.

What happens if the profile is thicker than indicated by the long wavelength limit? Figure 1.96 shows a profile generated by the Fourier method which was intentionally started

from a base profile which was approximately twice as thick as it needed to be from the long wavelength limit as discussed above.

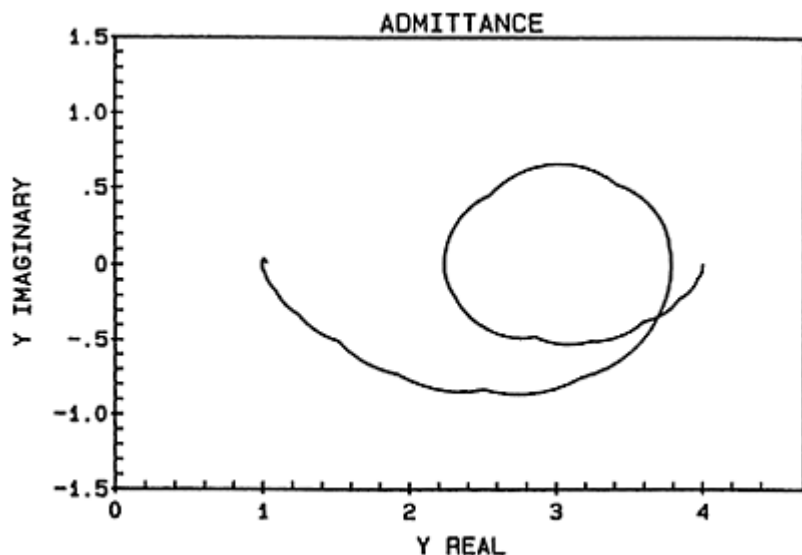


Fig. 1.98 Admittance of a “too-thick” design by the method of varying the index of 24 layers of equal optical thickness.

Figure 1.97 shows the resulting admittance at the long wavelength end of the AR band and in dashed line the same curve from Fig. 1.95 (the too-thin profile). The difference between the two admittances is a loop which has “grown” on the too-thick profile’s admittance locus. We can also see in Fig. 1.96 that the portion of the profile between -3 and $+3$ micrometers might be removed to give a result much like the profile in Fig. 1.93. Figure 1.98 shows a case generated by the author’s version of the Southwell method which demonstrates the same effect when compared to Fig. 1.80.

We may draw three conclusions from these observations. The first is that excess thickness is “used up” by the design in the form of admittance loops as seen in Figs. 1.97 and 1.98. The second is that the excess is of no apparent value except to help in overcoming the low index limitations mentioned previously. The third is that the excess loop could be anywhere on the admittance loci and may be more than one loop. This accounts for the findings of Verly²⁶ that there are an unlimited number of solutions to an AR band requirement if there is excess thickness. These conclusions are not rigorously shown here, but are rather the result of empirical examination of many other related cases. These observations aid significantly in the understanding of the basic phenomena of AR coating designs.

We have expended some effort to date in search of the ideal profile of the index of refraction as a function of thickness. We have referred above to the “ideal” in quotes,

because we do not yet know if an ideal profile exists or what it is exactly. However, we do think that we have the “ideal” design narrowed down closer than any current ability to actually fabricate such a coating. The following few specific cases approximate the ideal function to a sufficient degree for all present practical purposes: Verly (private communication) has preferred an Exponential Sine Function of the form, where x is optical thickness,

$$N(x) = A e^{-C \sin(x - x_0)} \quad (1.7)$$

The author has started many designs with a Gaussian profile of the form

$$N(x) = A e^{x^2} - B \quad (1.8)$$

with the offset B to bring it to the index of the medium in a finite thickness. The

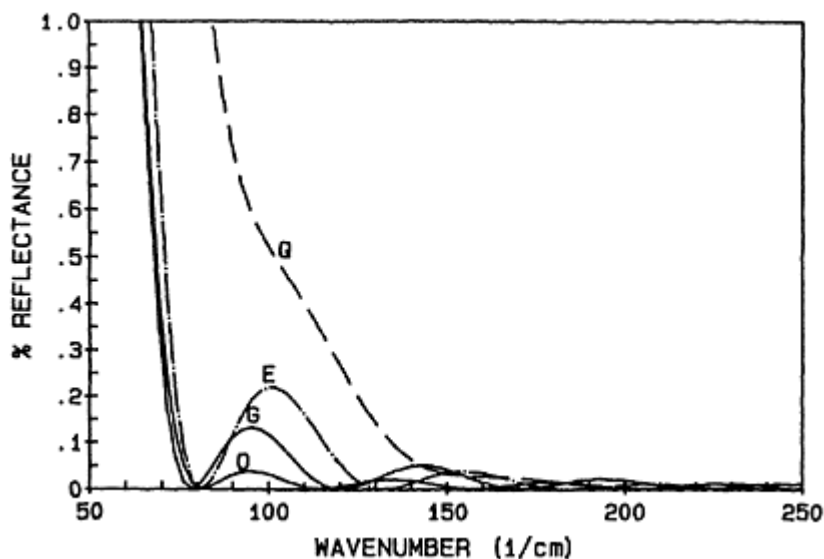


Fig. 1.99 Reflectance at the long wavelength end of the band for the “ideal” index profiles: Q is the Quintic function, E is the Exponential sine function, G is the truncated Gaussian function, and O is the optimized Gaussian function.

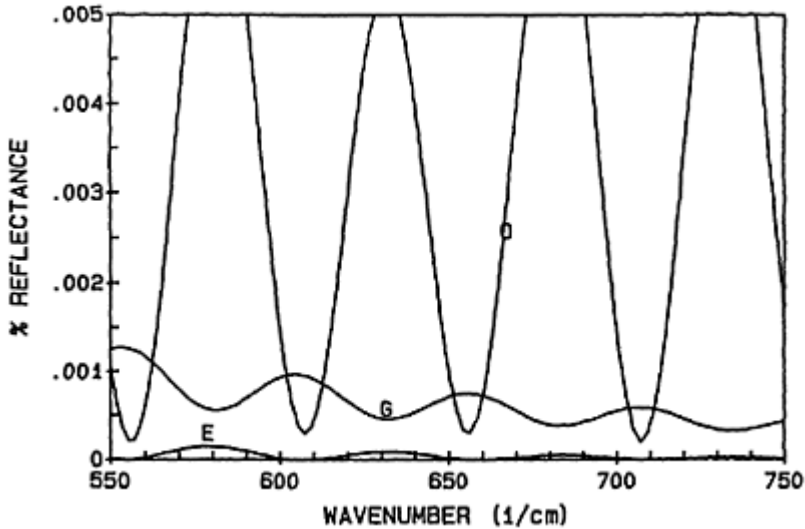


Fig. 1.100 Reflectance as in Fig. 1.99 but toward the center of the AR band and on a highly expanded scale. The reflectance of the Quintic is so low that it is lost on the baseline.

profile given in Eqn. 1.8 also served as a starting point for its optimization over a given bandwidth. We will refer to the result as the Optimized Gaussian.

Southwell²⁷ reported work with another interesting refractive index profile following the Quintic Function where n_s is the substrate index and n_j is the index of the final (or incident) medium

$$N(x) = n_i + (n_s - n_i)(10x^3 - 15x^4 + 6x^5) \quad (1.9)$$

All of these functions look just like Fig. 1.85 to within a pen width or so, and are hardly worth plotting on that scale to show the differences. We compare the reflectance of each of these four “ideal” layers at the same profile thickness in Figs. 1.99 and 1.100. At the long wavelength end of the band, the Optimized Gaussian has the lowest reflectance because the optimization had been weighted to make it so. The Gaussian and Exponential Sine have intermediate values, and the Quintic rises more rapidly. However, in the middle of the band, the reverse is the case. The Quintic is so low it is hidden in the baseline of Fig. 1.100, whereas the Optimized Gaussian exceeds .005%. Not that .005% is necessarily bad, but this is just to illustrate the differences of the principles employed. What we can observe here is that the usual tradeoffs seem to exist between band width and band depth. It is like a tube of toothpaste, if you squeeze it in one place, it will bulge in another. It may be that the area under the total reflectance band is constant or approximately so. The pursuit of the ideal profile is really somewhat academic and

esoteric at this point, but the author believes that (if it exists) it will be totally symmetric when seen in admittance as in Figs. 1.80 and 1.81. The pursuit thus far has been well worth the effort for the author in terms of gaining a better understanding of the underlying principles of AR coatings.

1.8. OPTIMIZATION

We have mentioned the use of optimization many times above without addressing any of its details. One can generally think of an “optimizer” as a black box which takes for its input a starting design, a set of parameters which may be varied, and a set of performance goals, weightings, and constraints. The desired output of this black box is a design which is the best fit to the weighted performance goals that can be achieved with the variables and within the constraints given. The optimizers usually deal with a merit function that is actually a *demerit* function which is reduced to a minimum by the process. The “merit” function might be computed as the sum of the squares of the difference between the targets and the actual spectrum. The program finds the effect of each variable on the merit function and predicts where the best design would be in variable space. This process iterates for a given number of times or until the goals are met. Such “black boxes” are provided as software packages for personal computers by organizations such as lead by Goldstein³⁴, Noe³⁵, and Macleod³⁶. The starting design, such as might have been obtained through the methods discussed above, would be represented by the index of refraction and the thickness of a given number of layers plus the substrate and media indices. We will discuss some of the other factors and consequences below.

1.8.1. Performance Goals and Weightings

If the goal were to design a four-layer AR for the visible spectrum, one must specify the performance goals as 0.0% reflectance at 50 nm intervals from 400 to 700 nm. This should lead to a good solution such as shown in Fig. 1.8 curve B. In the software which we use³⁴ (and most likely others), the weighting on each goal or target value is determined by the reciprocal of the tolerance on that target. For example, the above targets of 0.0% for wavelengths of 500, 550, and 600 nm might have tolerances of 0.5% while 450 and 650 nm have 1.0% and 400 and 700 nm have 2.0%. This would give an approximation of weighting for the visual or photopic response of the eye putting the most “pressure” on wavelengths centered near 555 nm. The same design might also result from leaving out the 400 and 700 nm targets entirely. In former times of slower and more expensive computers, keeping the number of targets smaller helped speed up and reduce the costs of the optimization process. These considerations are no longer as much of a concern.

The situation changes when we look at the case of the design in Fig. 2.1 labeled “B=2.5”. This design was optimized from 420 to 1100 nm. If we only used seven targets as mentioned above, we would encounter a problem. The “optimizer” would tend to move the reflection toward 0.0% as much as it could at the targets (only), but the spaces in between the targets would be free to move to much higher values of reflectance. It is advisable to have at least twice as many targets across the band of interest as there are

cycles in the ripple pattern, and four times is our usual choice. In this case, that would be about 32 targets. The ripple patterns are more evenly spaced on a frequency or wavenumber (cm^{-1}) scale, therefore it is better to have the targets equally spaced in frequency also.

The art of designing, after the indices and number of layers have been determined (which could be aided by the concepts discussed in Chapters 1, 2, and 3), involves the choices of the magnitude and position of the targets and their weightings in order to achieve the desired results. The author has sometimes found that the original choices need to be modified to apply more or less pressure on some target points to achieve the best balance.

1.8.2. Constraints

There are sometimes occasions where it is desirable to constrain certain aspects of a design such as minimum and maximum layer thicknesses. Some software packages³⁴ allow great flexibility in the definition of constraints. This is done by bringing design variable and results into a spreadsheet where they can be manipulated and fed back to the optimization part of the program.

We recently used this approach⁴⁶ in an investigation of which classes (if any) of AR designs might be improved by design techniques which considered the effects of probable production errors. We further reported on the influence of different error distribution assumptions such as: random errors uniformly distributed within a tolerance range, worst case error distributions, and various sensitivities to errors which might realistically represent those in actual practice. We were not able to discover any class of coatings which gains a significant benefit from the optimization with respect to the expected random errors of production. It was not surprising to find that normal designs result in each layer thickness lying at a minimum of the (de)merit function wherein a movement from the nominal design thickness in any direction will increase the (de)merit. Since this is also the point of the zero first derivative of the (de)merit with respect to thickness, the sensitivity to small errors is at a minimum.

We will discuss the use of this spreadsheet and constrained optimization feature in more detail in Chapter 7 in connection with achieving matching layers for a QWOT stack edge filter which also give specific optical monitoring characteristics.

1.8.3. Global versus Local Minima

We have seen in Fig. 1.7 and 1.8 that there are two solutions to the typical V-coating problem. The optimizer will usually converge on the solution closest to the starting design, or the “Zlocal minimum.” The other solution in this case might actually be better than the first, depending on the targets and weightings, but the typical optimizer might not find it. The best of all possible solutions is the “global” minimum. One can find discussions of global optimizers in the literature, but they of course must systematically examine the whole universe of parameter space in enough detail to not miss the global minimum.

The analogy would be that a search for the lowest point starting at Salt Lake City would find it to be Salt Lake. A search starting at Barstow, California should find Death

Valley, which is lower. However, if one started at Jerusalem, they might find the true global minimum at the Dead Sea.

Therefore, it is well to be alert to the possibility of local minima when using automatic optimizers. This is where some knowledge and understanding can be helpful in evaluating the results of an optimization. It is like the fact that we can know that the Dead Sea is the lowest point on the globe from the collective experience of others, even if we have not been there yet ourselves.

1.8.4. Some Optimizing Concepts

There are a variety of concepts for optimizers which have been used for decades and new ones are being developed all of the time. We will comment briefly on some of the most commonly used that we are aware of, and we will not discuss the “Simplex” or Parabolic Approximation³⁷ and other methods which have appeared over the years in some applications such as lens design.

Optimization is like letting a ball slide into a bathtub and roll until it rests on/in the drain hole. Its progress will not generally be in a straight line to the drain.

1.8.4.1. Damped Least Squares Optimization

There are many descriptions in the literature such as by Meiron^{38,39} which describe the concept of least squares and damped least squares (DLS) optimization. The derivative of the merit function with respect to each variable is calculated. These are used to compute the solution including each variable which would minimize the merit function. The solution is not likely to be the actual minimum because the merit function is not apt to be linear in all of the variables. However, this should move the design in the direction of the minimum (local). There is a risk, particularly as the actual minimum is approached, that the predicted change will overshoot the actual minimum. Therefore, damping is added so that the step taken is shortened to avoid overshooting, if that has been detected. Conversely, the step may be lengthened if the movement continues in the same direction without passing over a minimum. This all done automatically in the software, of course.

1.8.4.2. Flip-Flop Optimization

Southwell²⁰ introduced a simple but effective optimization scheme which he called “flip-flop” optimization. It stems from the concepts which we discussed earlier concerning index of refraction approximations and also inhomogeneous index functions. As will become more clear in Chapter 2, there is generally an advantage to using the highest and lowest indices practical in a design. Since we have seen that we can approximate any index in between the highest and lowest, there is usually little justification for using more than two indices and the attendant complications.

Southwell’s proposed synthesis (design) algorithm is as follows:

(1) Select a total physical thickness for the coating. Divide this thickness into thin layers of equal thickness.

(2) Assign some initial index, either high or low, to each layer. Usually the convergence of iterative solutions depends on starting values, so this step may be important. Here are four suggestions:

- (a) Start with all high-index layers.
- (b) Start with all low-index layers.
- (c) Start with alternating high- and low-index layers.
- (d) Start from some known approximate solution.

The first three require no knowledge of thin-film theory, and the fourth attempts to utilize such experience.

(3) Evaluate a merit function based on the desired spectral response. One example is the least-squares sum: square the difference between calculated reflectivity and the desired reflectivity at various wavelengths across the band of interest and add them together. Use the characteristic matrix theory to evaluate the calculated response.

(4) Change the state of each layer (from low to high index or from high to low) one at a time and reevaluate the merit function. If the performance is better in the flipped state, retain the change; otherwise restore it.

(5) If, after testing all the layers (a single pass), the merit function has improved, go to (4) for another pass; otherwise end.

He shows several examples of how well this works.

1.8.4.3. Needle Optimization

We were first introduced by Tikhonravov⁴⁰ to “needle optimization” in 1989. DeBell⁴¹ described the method as: “The key idea of this method is to introduce a needle like variation in the refractive index somewhere in the coating design in such a way that it will optimally decrease the value of the merit function. Once this variation, really it is just a very thin (needle like) layer, has been inserted into the existing design the resulting assembly of layers is refined to further decrease the value of the merit function by adjusting the thickness of all the layers. This process may be successively iterated until the introduction of a thin layer of material no longer effects a decrease in the merit function.”

Our understanding is that the influence of a very thin layer on the merit function is evaluated through the whole thickness of the design (“scanned” through it). The point of greatest beneficial effect is then chosen to insert the thin layer and reoptimize. From what we will show in Chapter 2, there is likely to be an optimum number of layers for a given overall thickness of coating. This is particularly true for AR coatings. More layers would add more reflection to be dealt with and eliminated; less would reduce design potency. We have seen published cases of needle designs that could be slightly improved by removing one layer and reoptimizing.

There have been more details published about the application of the needle method by Baumeister⁴², Sullivan and Dobrowolski⁴³, Tikhonravov et al.⁴⁴, and Furman and Tikhonravov.⁴⁵

1.9. SUMMARY

We have reviewed the principles and shown applications of various useful graphical tools (or methods) for optical coating design: Reflectance Diagram, Admittance Diagram, Triangle Diagram, Prism Diagram, and stereoscopic versions of these. We have shown graphically how unavailable indices can be approximated by two available indices of higher and lower values than the one to be approximated. The basis of ideal antireflection coating design has been shown empirically. The practical approximation of these inhomogeneous index profiles has been demonstrated. Much of the discussions have centered on AR coatings, but most other types have been seen in the perspective of the same graphics and underlying principles. We will show in subsequent chapters that *reflection control* is the basis of essentially all dielectric optical coatings and that transmittance, optical density, etc., are byproducts of reflection (and absorption). We believe the best insight is gained by the study of reflectance. We will also show further that AR coatings, high reflectors, and edge filters are all in the same family of designs. The author has found the graphical tools described in this chapter to be very useful as an aid to understanding and insight, and hopes that they will prove helpful to the reader also.

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